



Noise Hotspot Identification and Deeper Analysis in West Bengal

B.Stat. 3rd Year Statistics Comprehensive Group Project

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Research Questions Addressed

In this project, we address several statistical and environmental questions related to large-scale noise monitoring data across West Bengal. The study focuses on hotspot detection, temporal variation in environmental noise, festival-related acoustic behaviour, and scalable methodologies for analysing high-dimensional cyclic noise patterns.

- Analysis of seasonal variation in environmental noise characteristics across different regions and time periods.
- Identification of environmental noise hotspots across the Greater Kolkata Region.
- Study of noise behaviour during major festive occasions such as Diwali, Christmas, New Year, and related public events.
- Segregation and comparative analysis of noise contributions arising from fireworks and DJ-based sound systems during festive periods, with emphasis on peak-noise characteristics and temporal intensity patterns.
- **Additional Challenge:** Development of statistically meaningful summaries for large collections of daily and annual noise cycles. Since reporting hundreds of temporal cycles individually is impractical, the study explores clustering of similar cyclic profiles and Functional Principal Component Analysis (FPCA) for compact representation and interpretation.

Environmental Noise Analysis: Sound Metrics, Weighting Filters, and Time-Series Preprocessing

Abstract

Densely populated environments are characterised by complex, time-varying acoustic landscapes arising from traffic, commercial activity, and human settlement. Quantifying this noise in a meaningful and reproducible manner requires a rigorous framework of acoustic metrics, frequency-weighting schemes, and robust preprocessing of the measured data. This report introduces the fundamental concepts of sound and sound levels, outlines the standard noise metrics employed in environmental monitoring — including the equivalent continuous sound level L_{eq} and its A-, C-, and Z-weighted variants — and describes the preprocessing pipeline applied to noise time-series data collected at 311 monitoring stations across West Bengal. The analysis uses data from 2021 onwards and evaluates the discrete computation of daily equivalent noise levels and the suitability of an additive decomposition model for the resulting series.

Introduction

Sound is a mechanical wave that propagates through a medium — typically air — as a result of pressure variations caused by vibrating sources. In the context of environmental science and public health, the characterisation of *noise* (unwanted or harmful sound) has become increasingly important, particularly in areas with high population density and diverse land-use patterns.

Environmental noise arises from a multitude of overlapping sources: vehicular traffic on arterial roads, loudspeakers at commercial and religious gatherings, bursting firecrackers during festivals, industrial operations, and ambient crowd activity. The resulting acoustic signal is highly non-stationary — its level, frequency content, and statistical properties change substantially across the day, across seasons, and across different land-use zones.

Effective monitoring and regulation of environmental noise demands agreed-upon metrics that reduce a continuously fluctuating signal to interpretable summary statistics. This report is structured as follows. Section 2 introduces the physical basis of sound and the instantaneous sound level. Section 3 presents the equivalent continuous sound level L_{eq} . Section 4 reviews standardised frequency-weighting filters. Section 5 defines the weighted metrics L_{Aeq} , L_{Ceq} , L_{Zeq} , and L_{Apeak} . Sections 6 and 7 describe the data preprocessing pipeline and the discrete computation of daily L_{eq} values. Section 8 examines the suitability of an additive decomposition model for the noise time series.

Sound and Sound Levels

Physical Basis of Sound

Sound is produced whenever a mechanical disturbance causes rapid variations in the local pressure of a medium. These pressure fluctuations propagate outward as longitudinal waves. The human auditory system is sensitive to

root-mean-square (RMS) pressure fluctuations spanning approximately six orders of magnitude — from $20 \mu\text{Pa}$ (threshold of hearing) to 20 Pa (threshold of pain) — across frequencies from roughly 20 Hz to 20 kHz . Because of this enormous dynamic range, acoustic quantities are invariably expressed on a logarithmic (decibel) scale.

Sound Pressure Level

The **sound pressure level** (SPL) at a given location is defined as

$$L_p = 20 \log_{10} \left(\frac{p_{\text{rms}}}{p_0} \right) \quad [\text{dB}] \quad (1)$$

where p_{rms} is the measured RMS sound pressure and $p_0 = 20 \mu\text{Pa}$ is the internationally agreed reference pressure (ISO 1683).

Instantaneous Sound Level

The **instantaneous sound level** $L(t)$ represents the SPL measured at a particular instant in time. Because environmental noise is driven by many independent, transient sources, $L(t)$ is highly non-stationary and can exhibit sharp spikes (e.g., a passing motorcycle or firecracker burst) as well as sustained elevated periods (e.g., peak-hour traffic). Its key characteristics are:

- Highly time-varying; statistical properties change across diurnal and seasonal cycles.
- May contain sharp spikes and sudden drops.
- Difficult to interpret directly over long durations without suitable summarisation.

These properties motivate the use of energy-equivalent time-averaged metrics for regulatory and comparative purposes.

Equivalent Continuous Sound Level (L_{eq})

Definition

The **equivalent continuous sound level** L_{eq} is the level of a hypothetical *steady* sound that would deliver the same total acoustic energy as the actual fluctuating sound over the same observation period T . Formally,

$$L_{eq} = 10 \log_{10} \left(\frac{1}{T} \int_0^T 10^{L(t)/10} dt \right) \quad (2)$$

where $L(t)$ is the instantaneous sound level (dB) at time t and T is the total observation duration.

Physical Interpretation

Equation (2) converts the dB-scale signal $L(t)$ to its linear intensity equivalent $10^{L(t)/10}$, time-averages this quantity, and converts back to decibels. The result is an *energy-equivalent* average: brief high-level events contribute disproportionately compared to sustained moderate-level sound, consistent with the physiological mechanism of noise-induced hearing loss.

Importance

- L_{eq} is mandated by most national and international environmental noise standards (ISO 1996, EU Environmental Noise Directive 2002/49/EC).
- It provides a single, interpretable summary over any chosen integration period (1 h, 8 h, 24 h, etc.).
- Day–night composites such as L_{dn} and L_{den} are derived directly from hourly L_{eq} values.

Frequency Weighting in Noise Measurement

The human auditory system does not respond with equal sensitivity to all frequencies: sensitivity peaks in the 1–4 kHz range and falls sharply at both low and high frequencies. To make instrument readings correspond more closely to perceived loudness, standardised **frequency-weighting filters** are applied to the microphone signal before computing the sound level.

A-Weighting

A-weighting (IEC 61672) was derived by inverting equal-loudness contours at moderate sound levels. It:

- Approximates the frequency sensitivity of the human ear.
- Attenuates frequencies below ~ 500 Hz and above ~ 10 kHz significantly.
- Is the most widely used filter in environmental noise legislation worldwide.

C-Weighting

C-weighting applies a much flatter response than A-weighting. It:

- Retains sensitivity at low and high frequencies.
- Is more appropriate for impulsive and bass-heavy sounds (e.g., explosions, amplified music).

Z-Weighting

Z-weighting (“zero” weighting) applies no frequency correction. It:

- Represents a raw, flat-frequency measurement.
- Is used where physical sound energy is required without any perceptual correction.

Table 1 summarises the three standard filters.

Table 1: Comparison of standard frequency-weighting filters.

Property	A	C	Z
Low-freq. attenuation	Strong	Mild	None
High-freq. attenuation	Moderate	Mild	None
Hearing approximation	Yes	Partial	No
Primary regulatory use	Yes	Limited	No

Weighted Noise Metrics

L_{Aeq}

L_{Aeq} is the L_{eq} computed after applying A-weighting. It is the most commonly reported environmental noise metric and is appropriate for road and rail traffic, crowd noise, and general environmental exposure.

L_{Ceq}

L_{Ceq} uses C-weighting and is more sensitive to low-frequency content. It is the preferred metric for loud music, bass-heavy amplification systems, and firecracker and pyrotechnic activity.

L_{Zeq}

L_{Zeq} applies no weighting and represents total unweighted acoustic energy. It is used in research contexts where perceptual corrections are undesirable or where physical energy budgets are of primary interest.

L_{Apeak}

L_{Apeak} is the peak instantaneous A-weighted sound pressure level:

$$L_{Apeak(t)} = 20 \log_{10} \left(\frac{|p_A(t)|_{\max}}{p_0} \right) \quad (3)$$

where $p_A(t)$ is the A-weighted instantaneous pressure. L_{Apeak} is particularly important for impulsive noise events such as firecracker bursts, where a single short-duration spike may cause acoustic trauma even when L_{Aeq} remains moderate.

Table 2 summarises the four metrics and their principal applications.

Table 2: Weighted noise metrics and primary applications.

Metric	Primary Application
L_{Aeq}	Traffic, crowd, general environmental noise
L_{Ceq}	Loud music, bass amplification, firecrackers
L_{Zeq}	Raw acoustic energy; unweighted research use
L_{Apeak}	Impulsive events (firecrackers, explosions)

Preprocessing of Noise Time Series

Before computing summary noise metrics, raw data from each monitoring station underwent a structured quality-control pipeline to remove erroneous records and handle missing observations.

Preprocessing Steps

1. **Duplicate removal.** Records sharing identical timestamps were identified and removed, retaining the first occurrence.
2. **Period separation.** Measurements were split into daytime and nighttime sub-series for independent analysis (Table 3).
3. **Gap computation.** Consecutive timestamp differences were computed as

$$\Delta t_i = t_i - t_{i-1} \quad (4)$$

Gaps ≤ 15 min were considered acceptable sampling irregularity.

4. **Missing data quantification.** Total missing time per day was assessed using:

$$T_{\text{miss}} = \sum_i \max(\Delta t_i - 15, 0) + \max_i(\Delta t_i) \quad (5)$$

5. **Day rejection.** Days whose T_{miss} exceeded a site-specific threshold were discarded entirely (daytime threshold: $T_{\text{miss}} \leq 300$ min; nighttime threshold: $T_{\text{miss}} \leq 150$ min).
6. **Interpolation.** Short contiguous gaps of ≤ 2 days were filled by linear interpolation of the surrounding valid L_{eq} values.

Table 3: Daytime and nighttime period definitions.

Period	Time Range
Daytime	06:00 – 22:00
	00:01 – 06:00
Nighttime	22:00 – 23:59

Computation of Daily L_{eq}

Rationale for Discrete Formula

Because noise loggers record at irregular time intervals, the continuous integral in equation (2) must be replaced by a duration-weighted discrete sum. Only data from 2021 onwards was retained for this computation.

Discrete Formulation

$$L_{eq} = 10 \log_{10} \left(\frac{\sum_i \Delta t_i 10^{L_i/10}}{\sum_i \Delta t_i} \right) \quad (6)$$

where:

- L_i = observed sound level (dB) at observation i ,
- Δt_i = duration (min) attributed to observation i ,
- $\sum_i \Delta t_i 10^{L_i/10}$ = total accumulated sound energy,
- $\sum_i \Delta t_i$ = total valid observation duration.

Equation (6) reduces to the standard time-average when all Δt_i are equal, and naturally down-weights observations recorded over shorter intervals. Daily L_{eq} values were computed separately for the daytime and nighttime sub-series.

Checking Suitability of the Additive Model

A time-series decomposition model can be either *additive* or *multiplicative*. In an additive model the observed series Y_t is expressed as

$$Y_t = T_t + S_t + R_t \quad (7)$$

where T_t is the trend component, S_t the seasonal component, and R_t the residual. An additive structure is appropriate when the magnitude of seasonal fluctuations and residual variability remain roughly constant regardless of the level of the series. If variability instead scales proportionally with the mean, a multiplicative (or log-transformed additive) model is preferred.

To assess which structure is appropriate for the daily L_{eq} series, rolling-window(30 days) Mean vs. Standard Deviation (Mean-SD) plots were constructed for both monitoring stations and both time periods. Under an

additive model, the scatter of standard deviations should be approximately flat across the range of rolling means; a strong positive slope would indicate multiplicative behaviour.

WBPCB Barrackpore — Commercial Zone (Night)

The nighttime commercial series shows somewhat higher variability at larger mean levels compared to daytime observations. Although a mild positive relationship between the rolling mean and standard deviation is visible, the increase in variance is not excessively strong. The additive decomposition model can therefore still be considered acceptable for practical analysis of this series.

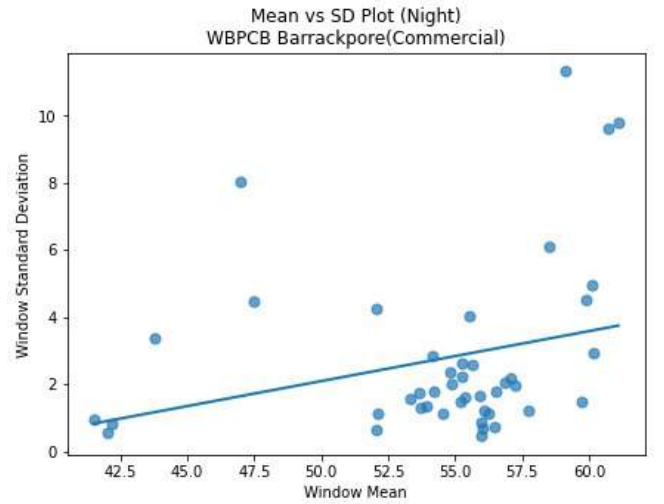


Figure 1: Rolling Mean vs. Standard Deviation plot for the nighttime L_{eq} series at WBPCB Barrackpore (Commercial Zone). The fitted trend line indicates only a mild positive association, supporting the additive model.

WBPCB Barrackpore — Commercial Zone (Day)

The daytime commercial scatter plot shows a relatively stable spread of standard deviation across varying mean levels. The fitted trend line is nearly flat with a slight negative slope, indicating weak dependence between mean and variability. This behaviour supports the suitability of an additive decomposition model for this series.

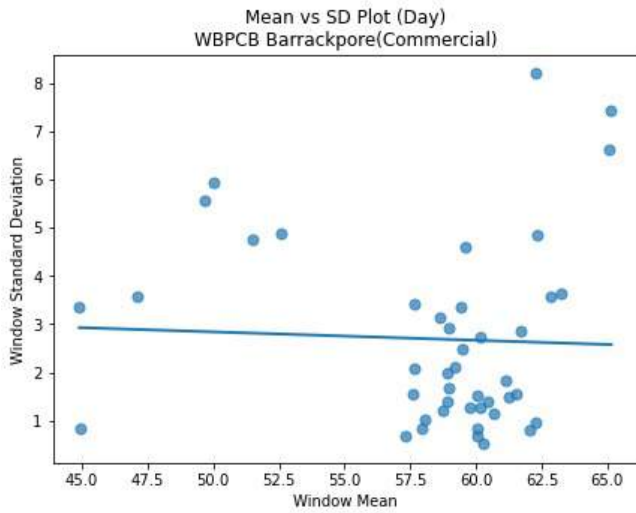


Figure 2: Rolling Mean vs. Standard Deviation plot for the daytime L_{eq} series at WBPCB Barrackpore (Commercial Zone). The nearly flat trend line confirms approximate variance stability consistent with an additive structure.

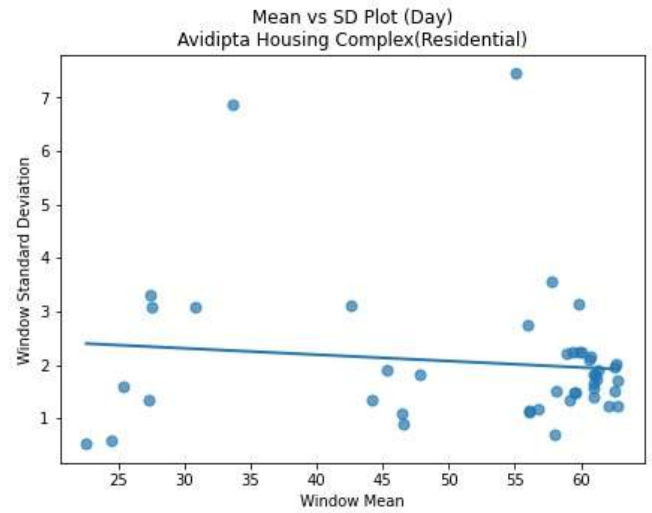


Figure 4: Rolling Mean vs. Standard Deviation plot for the daytime L_{eq} series at Avidipta Housing Complex (Residential Zone). The weak slope confirms approximate variance stability, consistent with an additive structure.

Avidipta Housing Complex — Residential Zone (Night)

The nighttime residential series exhibits moderate dispersion with no strong increasing variance pattern. Most observations cluster around similar standard deviation levels despite changes in the rolling mean. The nearly horizontal fitted line suggests approximate variance stability, supporting the additive assumption.

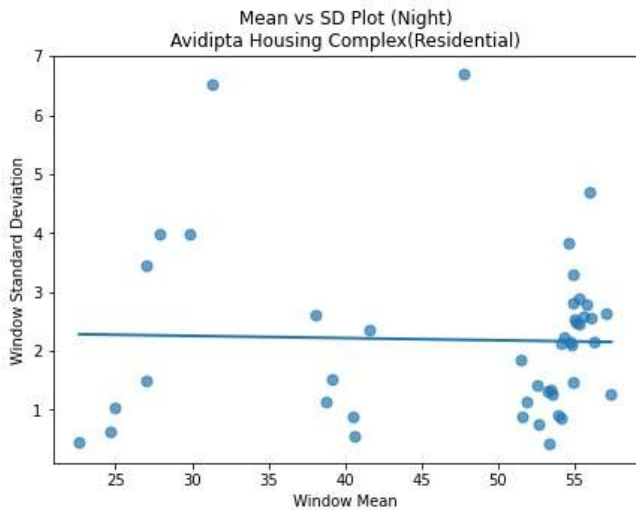


Figure 3: Rolling Mean vs. Standard Deviation plot for the nighttime L_{eq} series at Avidipta Housing Complex (Residential Zone). The approximately horizontal trend line supports the additive decomposition assumption.

Avidipta Housing Complex — Residential Zone (Day)

Daytime residential observations show slight variability in standard deviation across the range of means. However, the absence of a strong upward trend indicates that variance does not systematically increase with the mean, suggesting that the additive model remains reasonably appropriate for the daytime residential noise series.

Across all four series, the Mean-SD plots showed no strong proportional increase in variability with increasing mean levels. Although a weak positive association was observed in some cases, the variance remained approximately stable. The daily L_{eq} time series for the two shifts (Day and Night) at both stations were therefore treated as consistent with an **additive** decomposition structure for all subsequent analyses.

Time Series Analysis of Imputed Noise Data

Abstract

Urban environmental noise exhibits substantial temporal variation driven by traffic intensity, industrial activity and human movement. This study analyzes long-term environmental noise measurements obtained from monitoring stations maintained by the West Bengal Pollution Control Board (WBPCB) across multiple acoustic zones, including industrial, residential, commercial and institutional regions. The study investigates stationarity characteristics using the Augmented Dickey-Fuller (ADF) test and explores temporal behavior through seasonal decomposition techniques. Results indicate the coexistence of stationary and non-stationary behavior among monitoring stations, while seasonal decomposition reveals distinct long-term trends and recurrent seasonal structures across different acoustic zones.

1 Introduction

Environmental noise pollution has become an increasingly important urban issue due to rapid industrialization, population growth and transport expansion. Long-term exposure to elevated noise levels may adversely affect human health through sleep disturbance, stress, reduced productivity and cardiovascular effects. Environmental sound levels generally vary over time and are often influenced by recurring temporal patterns. Therefore, understanding long-term trends and seasonal characteristics of urban soundscapes is important for identifying persistent noise sources and evaluating mitigation strategies. The present study investigates longitudinal noise dynamics using time-series analysis techniques across selected monitoring stations representing different acoustic environments.

2 Data Description

Noise data used in this study were collected from the official monitoring network of the West Bengal Pollution Control Board (WBPCB). Monitoring stations were selected from four major acoustic categories:

- Industrial Zones
 - Bandel Thermal WBPDCB
 - Jalan Complex
 - Shyam Power LTD
- Commercial Zones
 - Leather Complex

- Howrah Municipal
- WBPCB Barrackpore

- Residential Zones
 - Avidipta Housing
 - Solaris Ideal Riverview
 - Urbana Housing
- Institutional / Silence Zones
 - Ballygunge CU Campus
 - Loreto College
 - Presidency University

3 Methodology

The analysis consists of two major stages:

3.1 Stationarity Assessment

The Augmented Dickey-Fuller (ADF) test was used to determine whether the observed time series contained unit roots. The hypotheses are:

$$H_0 : \text{Series is non-stationary} \quad (1)$$

$$H_1 : \text{Series is stationary} \quad (2)$$

The null hypothesis is rejected when:

$$p < 0.05 \quad (3)$$

indicating stationarity. The results of the ADF Test are presented in **Appendix A** and **Appendix B** of this report.

3.2 Seasonal Decomposition

Seasonal decomposition separates a time series into:

$$Y_t = T_t + S_t + R_t \quad (4)$$

where:

$$T_t : \text{Trend component} \quad (5)$$

$$S_t : \text{Seasonal component} \quad (6)$$

$$R_t : \text{Residual component} \quad (7)$$

The decomposition enables identification of long-term movement, periodic structure and irregular variation.

4 Stationarity Analysis

The ADF results indicate both stationary and non-stationary behavior among monitoring stations. Several stations exhibited strong evidence of stationarity:

- Loreto College
- Presidency University
- Leather Complex
- Shyam Power LTD

Several residential and commercial stations exhibited non-stationary characteristics indicating:

- Long-term trends
- Seasonal effects
- Temporal variability

Nighttime observations appeared relatively more stationary than daytime observations, possibly due to reduced human activity and traffic movement.

5 Seasonal Decomposition Analysis

Seasonal decomposition was performed separately for daytime and nighttime noise levels.

Distinct temporal behavior was observed across the four categories. All the Time Series plots are presented in

Appendix A of this report.

6 Results and Discussion

Table 1: ADF Test Results for Daytime Noise Data

Station	ADF Statistic	p-value	Conclusion
Avidipta Housing	0.6643	0.9891	Non-Stationary
Ballygunge CU Campus	-3.0224	0.0329	Stationary
Bandel Thermal WBPDCCL	-1.7865	0.3872	Non-Stationary
WBPCB Barrackpore	-1.6528	0.4556	Non-Stationary
Howrah Municipal	-2.5904	0.0950	Non-Stationary
Jalan Complex	-1.7740	0.3934	Non-Stationary
Leather Complex	-3.6370	0.0051	Stationary
Loreto College	-4.6218	0.0001	Stationary
Presidency University	-3.1489	0.0231	Stationary
Shyam Power LTD	-3.6963	0.0042	Stationary
Solaris Ideal Riverview	-2.1859	0.2114	Non-Stationary
Urbana Housing	-4.0074	0.0014	Stationary

Table 2: ADF Test Results for Nighttime Noise Data

Station	ADF Statistic	p-value	Conclusion
Avidipta Housing	0.1378	0.9685	Non-Stationary
Ballygunge CU Campus	-2.9546	0.0394	Stationary
Bandel Thermal WBPDCCL	-1.9093	0.3277	Non-Stationary
WBPCB Barrackpore	-1.8669	0.3478	Non-Stationary
Howrah Municipal	-2.4283	0.1339	Non-Stationary
Jalan Complex	-1.7563	0.4024	Non-Stationary
Leather Complex	-3.3374	0.0133	Stationary
Loreto College	-4.2094	0.0006	Stationary
Presidency University	-3.1484	0.0232	Stationary
Shyam Power LTD	-5.2125	0.0000	Stationary
Solaris Ideal Riverview	-3.0506	0.0304	Stationary
Urbana Housing	-5.3209	0.0000	Stationary

Industrial regions exhibited comparatively high and persistent noise levels due to continuous operational activities and freight movement. Commercial regions displayed elevated variability associated with traffic intensity and public movement. Residential regions showed

moderate noise fluctuations influenced by local transportation and neighborhood activity. Institutional and silence zones generally exhibited lower noise levels with reduced variability. Seasonal components revealed recurring periodic patterns across all categories.

7 Conclusion

This study examined long-term environmental noise dynamics using stationarity testing and seasonal decomposition techniques. The analysis demonstrated:

- Coexistence of stationary and non-stationary behavior
- Significant seasonal structures across acoustic zones
- Distinct temporal differences between daytime and nighttime observations
- Strong influence of anthropogenic activity on environmental sound dynamics

The framework can assist in long-term urban noise assessment and environmental planning.

Hotspot Identification and Noise Segregation from Equivalent Continuous Sound Level data

1 Introduction

We have chosen the Greater Kolkata region as our subject of study. The region is delimited by $[22.45^\circ, 22.75^\circ]$ N and $[88.20^\circ, 88.55^\circ]$ E, and has plenty of noise monitoring stations(104 stations), many of them being quite close to one another, which can allow for some good analysis subject to very few data being missing.

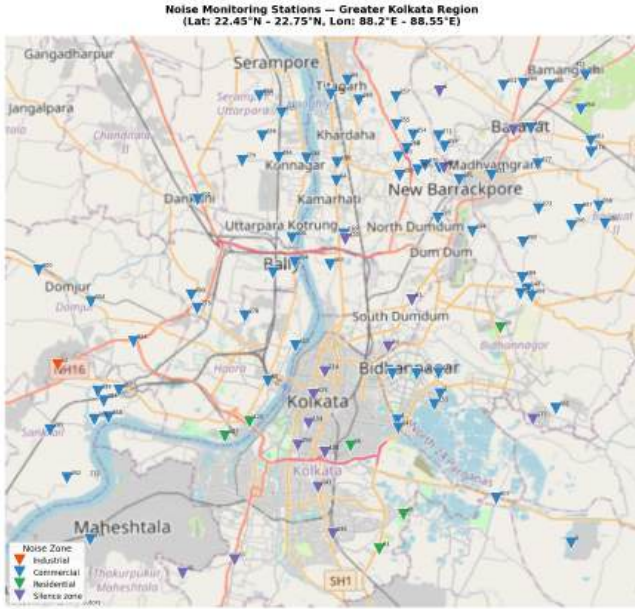


Figure 1: Greater Kolkata Region

2 Examining trends

Here, we inspect the behaviours of different types of sound levels($LAeq_T$, $LCeq_T$, $LAPeak_T$) across noise monitoring stations that are, owing to their location, have data that possibly have signature noise sources nearby (i.e. stations near a music venue, stations with major traffic roads nearby, stations that possibly record louder noise levels/impulsive noises during festive periods, likely arising from firecrackers ; the last type of stations may be obtained from hotspot identifications) .

2.1 Traffic based noise

We shortlisted 4 stations, which don't have much missing data(throughout 2025) and are within 40m distance of a major road, namely, Kamarhati Municipality(< 10m,

10.31% missing data), Karunamoye Bus Stand(< 20m, 6.92% missing), Panihati Municipality(< 10m, 0.53% missing), Office of The Assistant Commissioner of Police, Belghoria(< 20m, 17.2% missing). Polerhat Police Station(> 450m, 6.53% missing) has no major traffic roads within 400m. All of these monitoring stations fall in commercial zones.

2.1.1 Contrasting stations near major traffic roads against stations farther from ones

We choose a year(here, we chose 2025), and observe the behaviour of $LAeq_T$ values, during the day period for two stations over this period; one that's pretty farther away from the nearest major traffic road and one that's pretty close to a major traffic road. $LAeq_T$ values are the most suitable for this since traffic noise is mostly continuous in nature. Here, we take the day period to be from 9AM to 8PM, since that's when the traffic dominates. This may be affirmed by considering the plot of the hourly mean $LAeq$ values.

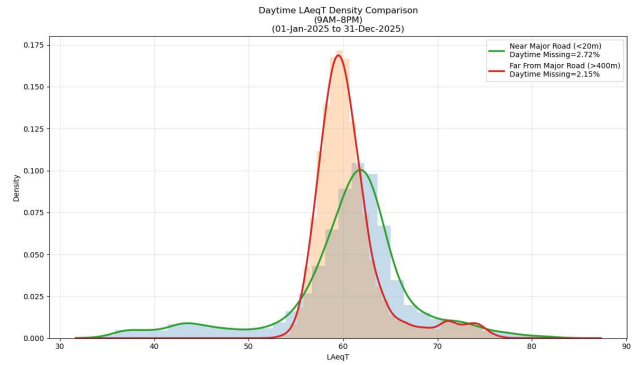


Figure 2: Karunamoye Bus Stand- Polerhat Police Station, 2025

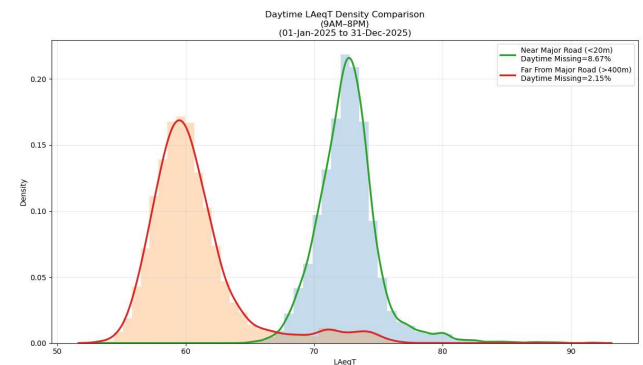


Figure 3: Kamarhati Municipality- Polerhat Police Station, 2025

We overlay the density plots(KDE with the Gaussian Kernel function and an appropriate bandwidth, the histogram shaded in a lighter shade underneath the curves) of the LAeqT values from the two stations, in 2025.

From the plots, it may be seen that the LAeqT profiles for the stations(which are near major traffic roads) have much heavier tails than the station which has none nearby. (plots for stations not nearby traffic roads show similar behaviour, some instances have been given in the appendix).

We may also perform a one sided Anderson-Darling test of the hypothesis:

$$\begin{aligned} &H_0 : F(x) = G(x) \\ \text{against } &H_1 : F(x) < G(x) \quad \text{for sufficiently large } x \end{aligned}$$

where, F, G are distribution functions of the LAeqT values from the station with a major road nearby and the station further from it's nearest major road, respectively.

The Anderson-Darling test has the following test statistic in the case of such an one sided alternative:

$$A_{m,n}^- = \frac{1}{N} \sum_{k=1}^{N-1} \frac{N[G_n(Z_{(k)}) - F_m(Z_{(k)})]}{k(N-k)/N}. \quad (1)$$

where, $Z_{(k)}$ is the k -th order statistics computed from the pooled data. Large positive values of $A_{m,n}^-$ are consistent with H_1 : if $G(x)$ consistently exceeds $F(x)$ in the upper tail, then $G_n(Z_{(k)}) - F_m(Z_{(k)})$ will be positive for large k , and those terms receive the largest weights.

The tests report p-values($<< 0.05$) in all cases, so we may conclude that LAeqT values are significantly higher for traffic related noises, and may use this as a base for classification. From the empirical survival probability plots, the LAeqT being greater than 60 may be used as a measure for classification.

2.1.2 Diurnal Profile

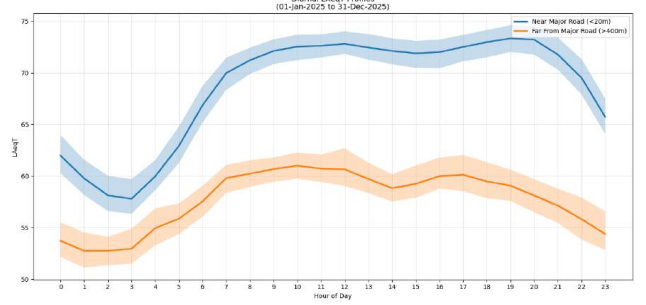


Figure 4: Kamarhati Municipality

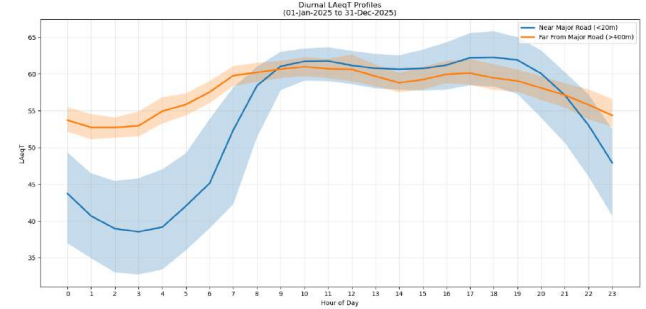


Figure 5: Karunamoye Bus Stand

From the plots of the hourly mean LAeq values, 9AM-8PM may be regarded as the traffic rushhour.

2.2 Loud Music based

In this regard, there weren't any known reference stations that could have frequent/known patterns of loud music sources nearby, from which a possible characteristic (equivalent continuous) sound level profile could otherwise be obtained. The presence of bass heavy, drum like sounds during music events indicates the presence of low frequency components in loud music sources, a major part of which falls towards the lower end of the human hearing range, and the human ear isn't as responsive to such frequencies. LAeq levels attenuate lower and higher frequencies to better imitate the human hearing, whereas LCeq allows many of the low frequency influences to pass through. From this, it appears that $LCeq_T - LAeq_T$ could serve as a possible indicator of the presence of low frequency components, which may include music sources. Also, music sources lack any sudden impulsive sounds, so the impulsiveness could be upper bounded.

We used $LCeq_T - LAeq_T > 10$ and impulsiveness < 15 , as being indicators of a possible music source.

2.3 Firecracker/impulsive noise based

Let impulsiveness during a time window, be defined as, $LAPeak_T - LAeq_T$; a high value of this might signify the presence of some instantaneously pretty high sound level which lasted very short, thereby not affecting the LAeqT value as much. Such impulsive noise could indicate towards firecracker based noise occurrences. We inspect it's behaviour across festive and non festive periods/months,

and across day and night periods during festive periods, to suggest bounds.

The stations possibly near such impulsive noise sources, may be obtained from hotspot identifications. Stations(having enough data in 2025) from regions that record more hotspots during the festive periods may be considered. Here, we consider 25th September, 2025 to 10th January, 2026, to be the festive period(hosting major festivals like Diwali, Durga Puja, Kali Puja, Christmas, New Year's Eve etc.)

2.3.1 Impulsiveness, LAPeakT Distribution between Festive and Nonfestive periods

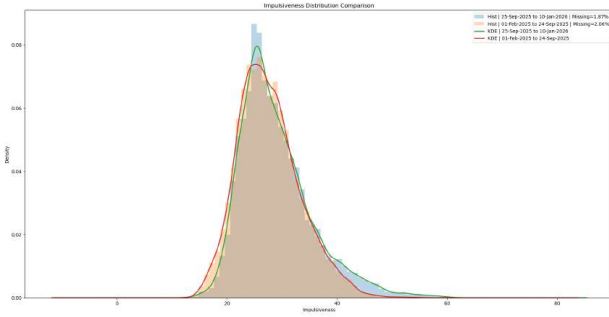


Figure 6: Impulsiveness Festive-Nonfestive(Rajarhat Police Station)

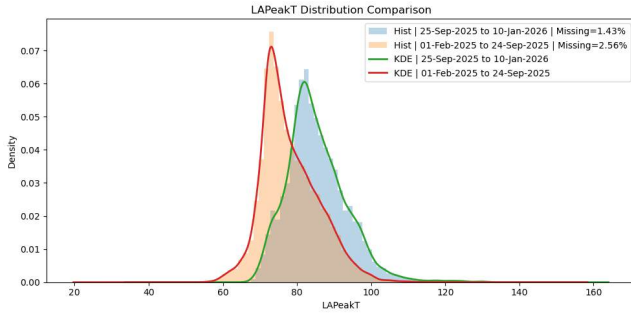


Figure 7: LAPeakT Festive-Nonfestive(Ballygunge Campus C.U.)

Similar plots have been provided in the appendix. The plots indicate towards festive periods having heavier tails for both impulsiveness and LAPeakT distributions.

Now, we consider the empirical survival probability plots(x against $1 - F_m(x)$) of the festive and non festive LAPeakT values(correspondingly, in another plot, the impulsiveness values), overlaid , to guess at the values where the survival probabilities start separating. We only put one here, similar other plots have been provided in the appendix and our guesses mostly align with, what is observed there as well.

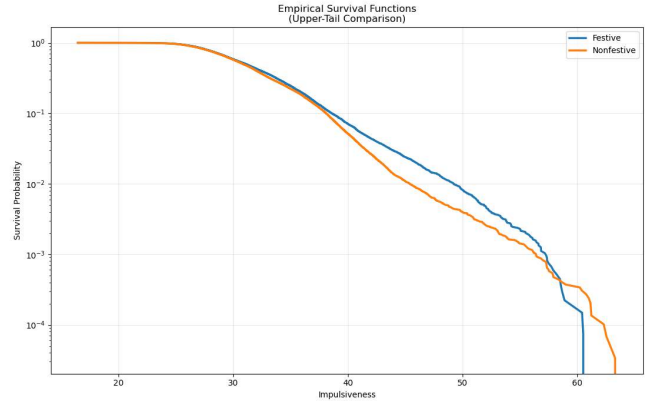


Figure 8: Empirical Survival probs: Impulsiveness, Festive-Nonfestive(Ballygunge Campus C.U.)

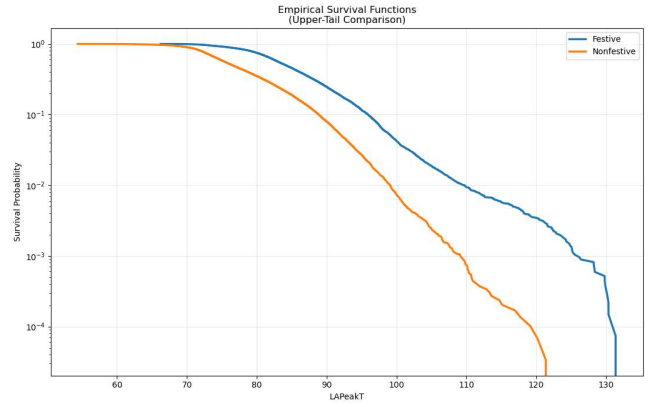


Figure 9: Empirical survival probs. : LAPeakT, Festive-Nonfestive(Ballygunge Campus C.U.)

From these, we may consider $LAPeakT > 100$ and $Impulsiveness > 30$ as reasonable indications of possible firecracker based noise.

2.3.2 Impulsiveness Distribution between the day and night period during festive periods

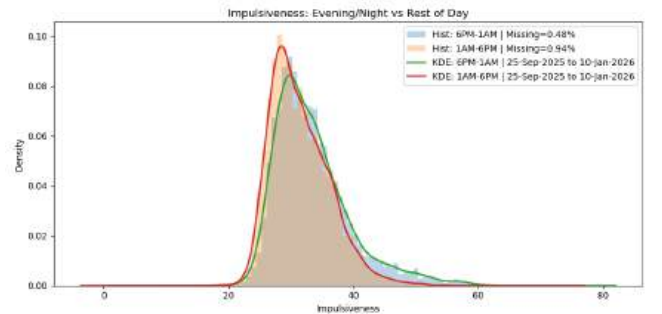


Figure 10: Impulsiveness Day-Night(Ballygunge Campus C.U.)

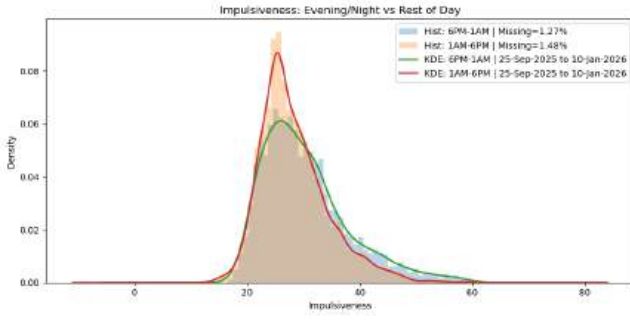


Figure 11: Impulsiveness Day-Night(Rajarhat Police Station)

The plots mostly show heavier tails for the night time. All these are supplemented by the one sided Anderson-Darling tests affirming the same.

2.3.3 Diurnal Profiles

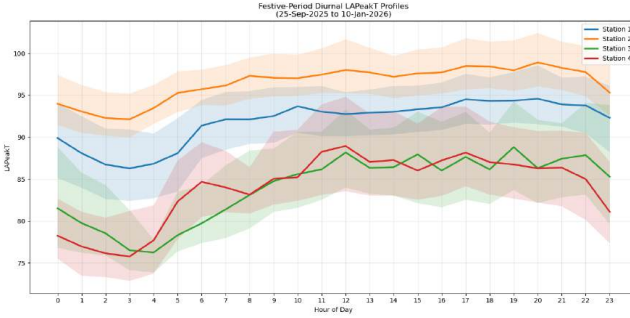


Figure 12: Diurnal pattern: median LApeakT hourly values, Festive period

2.4 Classification Rule

From these, we put forward the following classification rule:

We classify, based on the eq. noise levels in a given time window, a noise profile as coming from a,

- Firecracker based source if, $LA_{peakT} > 100$, $Impulsiveness > 30$ and it's nighttime(8PM- 4AM)
- Loud Music based if, $LC_{eqT} - LA_{eqT} > 10$ and, $Impulsiveness < 15$
- Traffic if, $LA_{eqT} > 60$, $LC_{eqT} - LA_{eqT} < 10$ and it's during rush hours(9AM-1PM, 5PM-8PM)
- Else, unclassified.

Results

We apply this classification rule on the noise level data from WBPCB, Saltlake Office, to obtain the following.

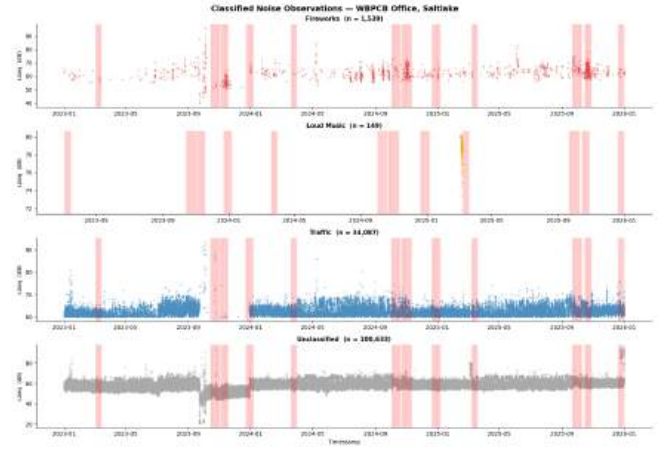


Figure 13: Classified Timeseries

From the points classified as being possible firecracker based noise, many of them exhibit spikes around October/November, which aligns with the months hosting festivities like Diwali/Durga Puja. Traffic noise classifications appear consistent throughout the year, which is reasonable. Possible music sources classified points seem to show a spike around March, 2025 with LAeq values occasionally exceeding 75dB(A), which may be attributed to Holi celebrations around that time(or possibly construction work persisting around that time). And, most points remain unclassified, which is somewhat reasonable considering we don't have access to the finer continuous noise level data, which may allow us to have better notions of the data.

3 Hotspot Identification

We present two different approaches, with two ways in which a hotspot(over a period of the year) has been defined.

3.1 Moran's I + Getis-Ord-Gi*

In this approach, we define a hotspot as a station with similar significantly higher values of a noise annoyance metric across the stations in it's neighbourhood than the outside. For a concerned period of the year(it may be a festive period, or a month of the year) and a time period(day period or night period), we take the metric as the proportion of days the integrated LAeq level exceeded the zone prescribed regulatory limits, for that time period. Then, we use Moran's I to test for the presence of global spatial autocorrelation, at a significance level of 10%. Rejecting this implies the presence of neighbourhoods with similar and significantly higher or lower values than the outside. Then, we proceed to test at the local level, to see if a certain neighbourhood around a station has similar significantly higher/lower values, using the Getis-Ord-Gi* statistic, and accordingly label it as a hotspot/coldspot.

3.1.1 Global Moran's I

The Global Moran's I involves a spatial weights matrix $W = ((w_{ij}))$, where the indices i, j vary over all the

stations present. Here, we use a binary distance matrix, which encodes as $w_{ij} = 1$, if $\text{distance}(i, j) \leq d$ and 0, otherwise; where, d is a preset distance threshold. Here, we shall work with $d = 4\text{km}$. By convention, $w_{ii} = 0 \forall i$.

The test statistic is the following:

$$I = \frac{n}{S_0} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (2)$$

where n is the number of stations; $x_i \in [0, 1]$ is the LAeq exceedance proportion at station i over the period of interest (which may be a calendar month or a festive period), $\bar{x}$ is the global mean exceedance proportion across all stations; and $S_0 = \sum_i \sum_j w_{ij}$. Under the null hypothesis of the absence of any global spatial autocorrelation, the expected value of I is $-1/(n-1)$, which approaches zero for large n .

The tests were performed via a permutation approach with $B = 999$ random permutations of the observed values across station locations, generating an empirical null distribution of I under H_0 . The p-value was computed as:

$$p = \frac{\#\{I^{(b)} \geq I^{\text{obs}}\} + 1}{B + 1} \quad (3)$$

A statistically significant positive Moran's I ($p < 0.10$) indicates that nearby stations tend to exhibit similar noise levels, confirming that spatial clustering is present and that local hotspot identification is statistically justified.

3.1.2 Getis-Ord G_i^*

For station i , the Getis-Ord-Gi* statistic is defined as:

$$G_i^* = \frac{\sum_j w_{ij} x_j - \bar{x} \sum_j w_{ij}}{s \sqrt{\frac{n \sum_j w_{ij}^2 - \left(\sum_j w_{ij}\right)^2}{n-1}}} \quad (4)$$

where s is the global standard deviation of the exceedance proportions. Under the null hypothesis, G_i^* follows an asymptotic standard normal distribution and is directly interpretable as a z -score. A large positive G_i^* indicates that station i and its neighbours jointly exhibit exceedance proportions significantly above the global mean, indicating a possible hot spot. A large negative G_i^* , on the other hand, indicates a cold spot.

As with Moran's I, inference was performed through permutation testing with $B = 999$ permutations, at 10% significance level.

Results

Festive Hotspots

We define the festive periods as the following, for 2023, 2024 & 2025.

For all the festive period $\times$ year $\times$ time period combinations, the Global Moran's I had the following results:

Based on those, the local clustering behaviours were looked into using Getis-Ord-Gi*. We present here only

Festival	2023	2024	2025
Holi	3 Mar–14 Mar	19 Mar–30 Mar	8 Mar–19 Mar
Durga Puja	14 Oct–31 Oct	2 Oct–18 Oct	21 Sep–9 Oct
Diwali	2 Nov–17 Nov	21 Oct–10 Nov	15 Oct–28 Oct
Christmas & New Year	22 Dec–31 Dec	20 Dec–31 Dec	20 Dec–31 Dec
New Year Carry-over	1 Jan–5 Jan	1 Jan–5 Jan	1 Jan–5 Jan

Table 1: Festive periods considered in the analysis for 2023–2025. The New Year carryover window (1–5 January) is associated with the preceding year's Christmas & New Year festival.

Festival	Year	Period	Moran's I	z	p (sim)	Decision
Diwali	2023	Day	0.5860	3.653	0.001	Rejected
	2023	Night	0.1748	1.371	0.092	Rejected
	2024	Day	0.4384	3.008	0.005	Rejected
	2024	Night	-0.0069	0.105	0.411	Not rejected
	2025	Day	0.0763	1.123	0.137	Not rejected
	2025	Night	0.0209	0.408	0.319	Not rejected

H_0 : No spatial autocorrelation in LAeq exceedance proportions. $\alpha = 0.10$, permutation-based p -value (999 permutations).

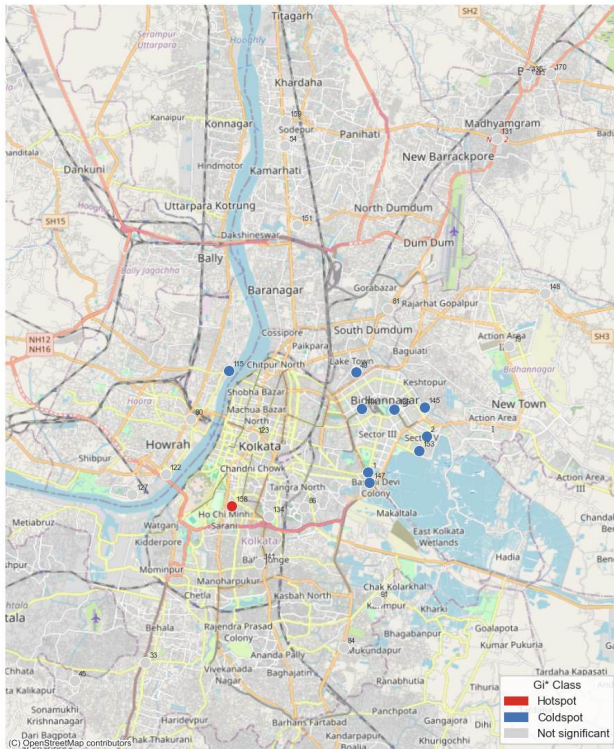
the results for Diwali and Durga Puja, over 2023-2024. The rest are given in the appendix.

The following were the results–

Festival	Year	Period	Moran's I	z	p (sim)	Decision
Durga Puja	2023	Day	0.6062	3.343	0.003	Rejected
	2023	Night	0.0333	0.513	0.216	Not rejected
	2024	Day	0.5356	3.648	0.001	Rejected
	2024	Night	0.0125	0.258	0.352	Not rejected
	2025	Day	0.0993	1.477	0.073	Rejected
	2025	Night	0.0758	1.202	0.120	Not rejected

H_0 : No spatial autocorrelation in LAeq exceedance proportions. $\alpha = 0.10$, permutation-based p -value (999 permutations).

Gi* Hotspots/Coldspots
Diwali 2023 æ" Day period



Gi* Hotspots/Coldspots
Diwali 2023 æ" Night period

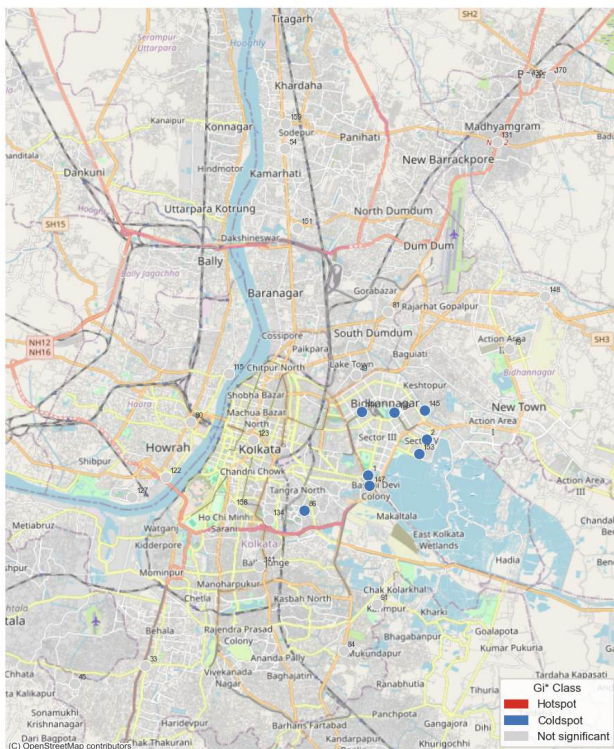
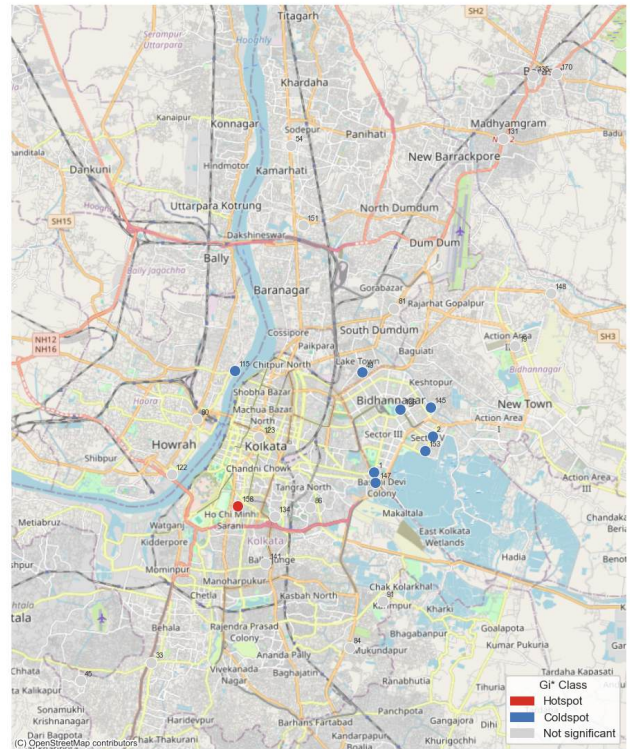


Figure 14: Getis-Ord-Gi, Diwali 2023 hotspots, day and night periods

Gi* Hotspots/Coldspots
Durga Puja 2023 æ" Day period



Gi* Hotspots/Coldspots
Durga Puja 2024 æ" Day period

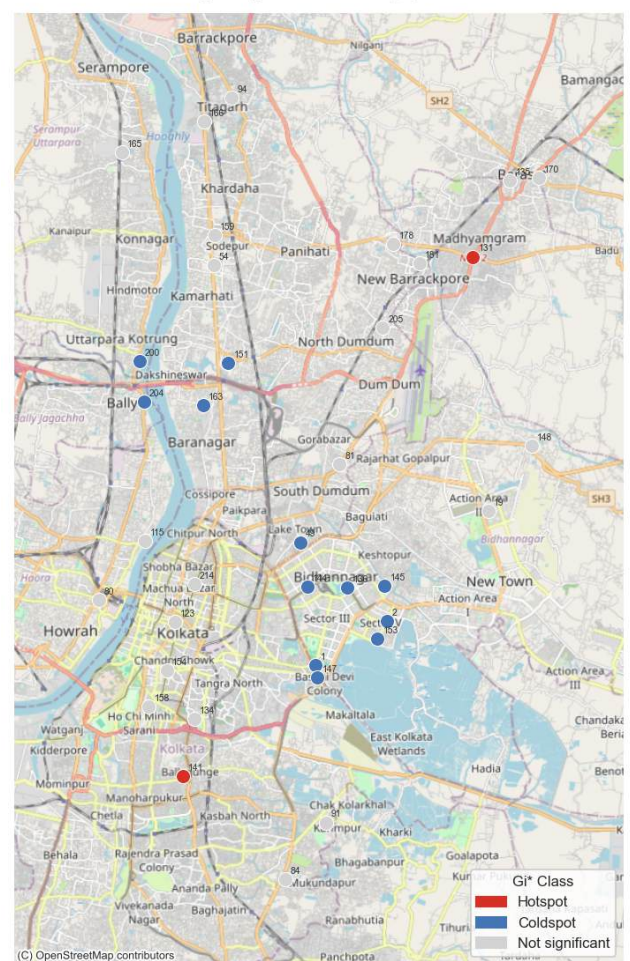


Figure 15: Getis-Ord-Gi, Durga Puja 2023 and 2024 hotspots, day period

The Global Moran's I + Getis-Ord-Gi* way of identifying hotspots is somewhat standard, but it may not illustrate full pictures if the environments are not quite suited. Moran's I + Getis-Ord-Gi* does a nice job, for the purpose of identifying similar significantly high/significantly low values than the outside, clustering nearby, and classifying them as hotspots/coldspots accordingly. Now, in a situation where the exceedance proportions are pretty high almost everywhere throughout the whole region(which is a very dire environmental condition and may happen during the peaks of festive periods), it classifies the ones that are in an even worse situation than the global pattern. And at the same time, spots that have high exceedance proportions(which is bad) but are not as high as the global values, are classified as coldspots. This might send wrong impressions about the prevailing patterns.

So, we present another approach to this. We define a hotspot differently.

3.2 Using Mean Exceedance Magnitudes over a period

3.2.1 Definition

For a station i and a period of interest $\mathcal{T}$ (which may be a calendar month or a festive period), let $\mathcal{T}_i \subseteq \mathcal{T}$ denote the set of days within $\mathcal{T}$ for which a valid (non-missing) LAeq observation is available at station i , and let $|\mathcal{T}_i|$ denote its cardinality. Let x_{it} denote the recorded LAeq value at station i on day t , and let $L_i^{(\text{per})}$ denote the zone-prescribed regulatory limit for station i under period $\text{per} \in \{\text{day}, \text{night}\}$, as specified in Table 2.

The *Mean Exceedance Magnitude* (MEM) for station i over period $\mathcal{T}$, separately for the day period (06:00–22:00) and the night period (22:00–06:00), is defined as:

$$\text{MEM}_i^{(\text{per})} = \frac{1}{|\mathcal{T}_i|} \sum_{t \in \mathcal{T}_i} \max(0, x_{it}^{(\text{per})} - L_i^{(\text{per})}) \quad (5)$$

On days where the recorded LAeq does not exceed the regulatory limit, the contribution to the sum is zero; on exceedance days, the contribution is the magnitude of the excess above the limit. A station that exceeds its limit rarely but by quite some margin, may be considered somewhat more severe than one that exceeds it frequently but marginally. Using exceedance proportions, however, would have reckoned the latter as more severe than the former.

Table 2: Zone-prescribed LAeq regulatory limits (CPCB norms).

Zone	Day Limit (dB)	Night Limit (dB)
Industrial	75	70
Commercial	65	55
Residential	55	45
Silence	50	40

3.2.2 Threshold Selection

To classify stations as hotspots on the basis of $\text{MEM}_i^{(\text{per})}$, a threshold $\tau^{(\text{per})}$ was selected separately for the day and night periods. For each period, the empirical distribution of $\text{MEM}_i^{(\text{per})}$ across all stations was examined via a histogram, and the threshold was chosen by visually inspecting the histograms.

3.2.3 Hotspot Classification

A station i is classified as a *hotspot* for period per during period of interest $\mathcal{T}$ if its mean exceedance magnitude exceeds the corresponding prescribed threshold:

$$\text{Hotspot}_i^{(\text{per})} \iff \text{MEM}_i^{(\text{per})} > \tau^{(\text{per})} \quad (6)$$

Classification is performed independently for the day and night periods.

3.2.4 Results

From inspecting the histograms(for the festive periods and months), the following thresholds were decided upon:

Table 3: Mean Exceedance Magnitude hotspot thresholds $\tau^{(\text{per})}$ (dB).

Period Type	Day Threshold (dB)	Night Threshold (dB)
Monthly	3.0	5.0
Festive	5.0	5.0

The histograms—

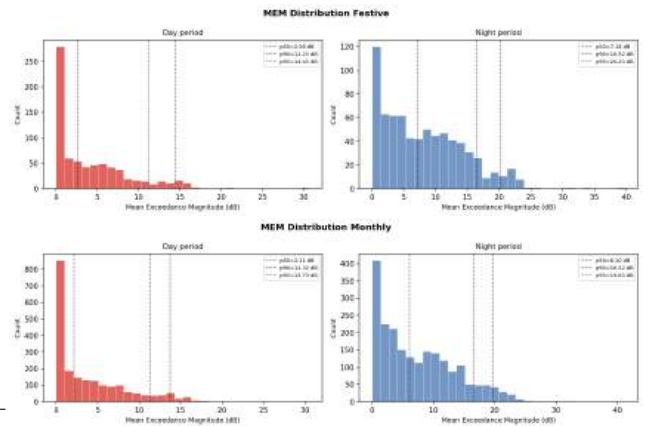


Figure 16: Histograms

Based on these, we present the results for Diwali and Durga Puja, in 2025, on both day and night period.

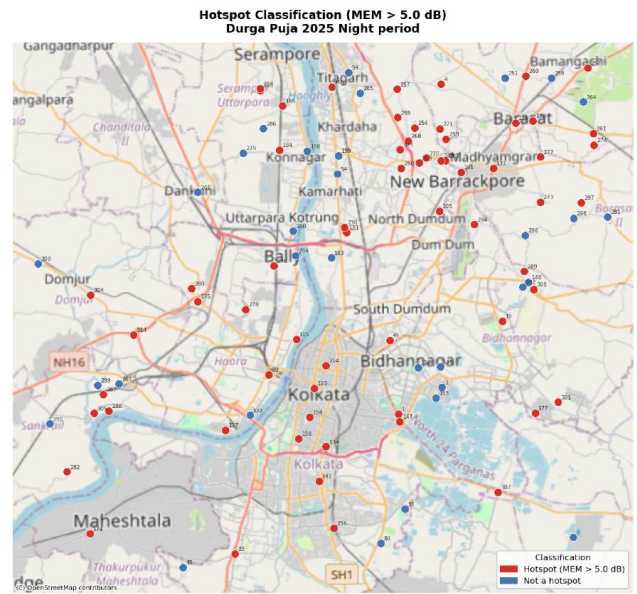
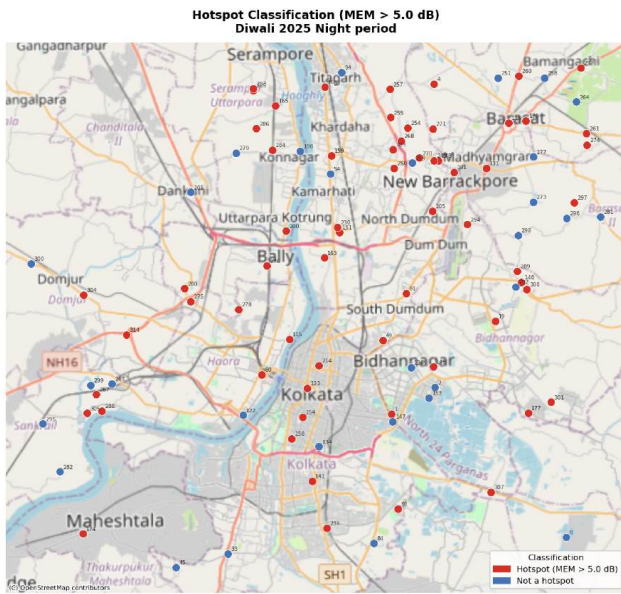
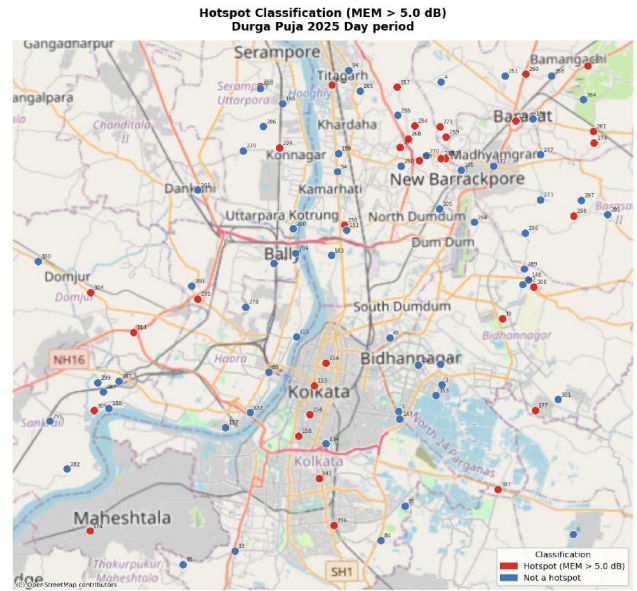
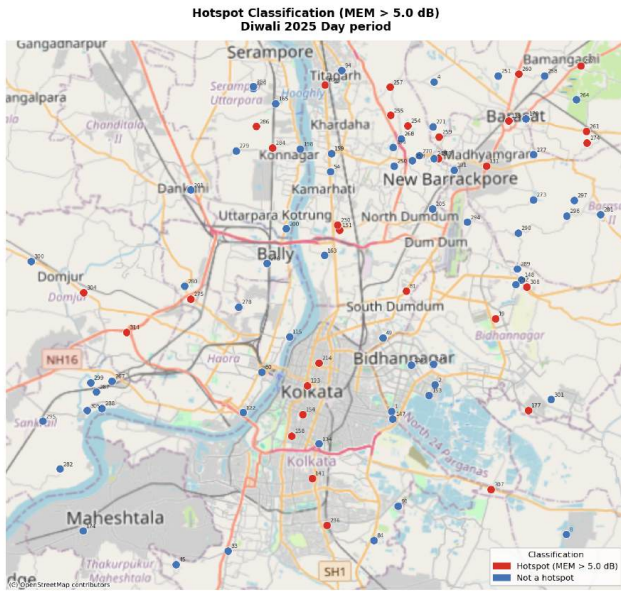


Figure 17: Hotspots for Diwali 2025, day and night periods

Figure 18: Hotspots 2025, Durga Puja, day and night period

The rest of the festive and monthly plots may be found in the appendix.

Characterizing Environmental and Impulsive Noise Dynamics in a Metropolitan Soundscape: A Longitudinal Study via Rule-Based Source Segregation

Abstract

Environmental noise exposure is a critical urban health concern in densely populated metropolitan regions like Kolkata, where noise levels escalate significantly due to traffic congestion, social gatherings, and firework-related festivals. This study investigates long-term environmental and impulsive noise exposure across multiple monitoring stations in Kolkata using high-resolution acoustic measurements collected at 10-minute intervals from January 2023 to January 2025. Analysis of acoustic indicators, including $L_{Aeq,T}$, $L_{Ceq,T}$, $L_{Zeq,T}$, and $L_{Apeak,T}$, reveals distinct temporal patterns, festival-related noise escalation, and frequent exceedances of Central Pollution Control Board (CPCB) standards. Furthermore, a rule-based acoustic segregation framework was employed to distinguish dominant urban noise sources, such as fireworks, traffic, and amplified music, based on their temporal and spectral characteristics. Preliminary findings indicate substantial violations of permissible day and night limits, particularly during major celebrations like Diwali, Kali Puja, and New Year. These results underscore the necessity of long-term urban acoustic monitoring for environmental health assessments and the formulation of evidence-based noise mitigation policies.

1 Introduction

Environmental noise is a major public health challenge in rapidly urbanising metropolitan areas. In densely populated cities like Kolkata, acoustic exposure is driven by vehicular traffic, commercial activities, construction, and frequent cultural celebrations. Among these sources, impulsive noise from fireworks during festivals such as Diwali, Kali Puja, and New Year represents one of the most intense forms of urban acoustic exposure.

While fireworks are integral to cultural festivities, research indicates that leisure-related noise exposure poses significant health risks. Children and young adults are particularly vulnerable to noise-induced hearing impairment[4]. High-intensity recreational sound sources can lead to temporary or permanent hearing loss, sleep disruption, and elevated stress levels [1,4].

1.1 Impulsive noise in urban leisure activities

Unlike continuous noise from traffic, impulsive noise consists of short-duration, high-intensity acoustic events. In urban settings, these occur during firework displays, public celebrations, and sudden industrial or transport-related sounds. Impulsive noise often carries a higher auditory risk than continuous sound because:

- Extreme peak sound levels can cause immediate mechanical damage to the inner ear.
- The short duration may lead to an underestimation of the actual acoustic risk based on perceived loudness.
- Festive contexts often lead to a disregard for noise hazards, as these sounds are culturally associated with celebration.

In Kolkata, festivals like Diwali and Kali Puja coincide with intensive firework activity and significantly elevated nighttime noise levels. The high population density and narrow street geometry can further amplify and sustain these high-intensity sound events.

1.2 Environmental noise limits and regulations

International organisations and national governments have established various frameworks to mitigate noise exposure. In India, the Central Pollution Control Board (CPCB) regulates ambient noise limits across industrial, commercial, residential, and silence zones [2]. Specific restrictions also govern the permissible sound pressure levels for fireworks during celebrations [2].

Globally, guidelines such as the WHO Environmental Noise Guidelines and the European Union Directive 2013/29/EU provide standards for managing both continuous and impulsive noise sources [1]. These regulations reflect a growing recognition of the health implications of recreational and environmental acoustic exposure.

In addition to quantifying overall environmental noise exposure, identifying the dominant contributors to urban acoustic escalation is also important for effective mitigation planning. Urban sound environments are typically composed of overlapping sources such as vehicular traffic, public celebrations, and amplified music sys-

tems. Distinguishing these acoustic sources using high-resolution monitoring data may provide deeper insight into the temporal behaviour and impulsive characteristics of metropolitan noise exposure.

Despite these regulatory frameworks, long-term monitoring studies in Indian metropolitan regions remain scarce. This study addresses this empirical gap by characterizing environmental and impulsive noise dynamics in Kolkata using high-resolution acoustic data spanning from January 2023 to January 2025. The investigation evaluates temporal trends, festival-induced noise escalations, and acoustic source segregation, while assessing adherence to CPCB-mandated standards. Beyond conventional environmental noise assessment, a rule-based classification framework is implemented to identify and differentiate dominant urban acoustic contributors, such as pyrotechnics, vehicular traffic, and amplified music systems.

2 Methodology

2.1 Study Area and Data Acquisition

This study utilizes longitudinal acoustic monitoring data collected from a high-resolution station located within the Avidipta Housing Complex in Kolkata. The monitoring site represents a typical high-density urban residential environment, characterized by a complex interplay of vehicular traffic from adjacent arterial roads, community-driven social activities, and periodic cultural celebrations.

Acoustic measurements were recorded continuously at 10-minute intervals from January 1, 2023, to December 31, 2024. The sampling process yielded a robust dataset comprising 78,214 valid observations. Each record includes a precise timestamp alongside multiple integrated acoustic indicators designed to capture both steady-state and transient environmental noise characteristics.

2.2 Acoustic Indicators

To provide a comprehensive assessment of the urban soundscape, the following acoustic indicators were analyzed:

- **Equivalent Continuous A-weighted Sound Level ($L_{Aeq,T}$):** Represents the energy-equivalent sound level over the measurement interval, emphasizing frequencies to which the human ear is most sensitive.
- **Equivalent Continuous C- and Z-weighted Levels ($L_{Ceq,T}, L_{Zeq,T}$):** Used to evaluate low-frequency components and unfiltered acoustic energy, respectively.
- **Peak A-weighted Sound Level ($L_{Apeak,T}$):** Captures the maximum instantaneous sound pressure level, essential for identifying impulsive events.

The raw dataset underwent rigorous quality control to ensure chronological consistency and to identify potential outliers or instrumental errors. Temporal features,

including hour of the day, day of the week, and seasonal month, were extracted to facilitate multi-scale temporal analysis.

2.3 Characterization of Impulsive Noise

To quantify the impulsive nature of the acoustic environment, an *Impulsive Index* (I_{idx}) was calculated for each observation:

$$I_{idx} = L_{Apeak,T} - L_{Aeq,T} \quad (1)$$

This index serves as a proxy for the crest factor of the ambient sound, where higher values signify the presence of high-energy, short-duration acoustic transients, such as firework detonations or abrupt traffic-related bursts.

Furthermore, statistical noise levels (L_{10} , L_{50} , and L_{90}) were derived to describe the noise distribution. The L_{10} level (sound level exceeded for 10% of the time) represents peak activity, while L_{90} serves as a measure of the background noise floor. The difference ($L_{10} - L_{90}$) was utilized to assess the acoustic climate variability.

2.4 Rule-Based Acoustic Segregation

To identify the dominant contributors to the monitored urban soundscape, a rule-based acoustic segregation framework was developed using acoustic indicators together with temporal activity patterns. Urban environmental noise typically arises from multiple overlapping sources, including vehicular traffic, firecrackers, amplified music systems and community activities. The proposed framework aims to distinguish these major acoustic categories using high-resolution monitoring data.

The classification procedure utilized impulsiveness measures, equivalent sound levels and low-frequency dominance indicators along with contextual temporal information such as nighttime activity periods and rush-hour intervals.

The impulsiveness index was defined as

$$I_{idx} = L_{Apeak,T} - L_{Aeq,T}.$$

Firecracker-related acoustic events were identified using extreme impulsive characteristics during nighttime periods. An observation was classified as a firecracker-based source if

$$L_{Apeak,T} > 100,$$

$$I_{idx} > 30,$$

and the observation occurred during nighttime hours (8 PM – 4 AM).

Amplified music activity was characterized by strong low-frequency dominance together with relatively low impulsiveness. An observation was classified as loud music if

$$L_{Ceq,T} - L_{Aeq,T} > 10,$$

and

$$I_{idx} < 15.$$

Traffic-related acoustic signatures were identified during rush-hour intervals using elevated equivalent sound levels and limited low-frequency dominance. An observation was classified as traffic-related if

$$L_{Aeq,T} > 60,$$

$$L_{Ceq,T} - L_{Aeq,T} < 10,$$

and the observation occurred during rush-hour periods (9 AM – 1 PM and 5 PM – 8 PM).

Observations failing to satisfy any predefined criteria were assigned to an unclassified category, representing mixed or ambiguous acoustic environments.

2.5 Temporal Classification and Regulatory Framework

In addition to ambient environmental noise standards, the study also considered the Indian regulatory framework governing impulsive firecracker noise. According to the Environment (Protection) Rules and subsequent amendments issued by the Central Pollution Control Board (CPCB), the manufacture, sale and use of firecrackers generating noise levels exceeding 125 dB(AI) or 145 dB(C) peak at a distance of 4 meters from the point of bursting are prohibited.

The regulations further prescribe standardized procedures for acoustic measurements, including free-field measurement conditions and the use of approved sound level meters. These guidelines were introduced to reduce the harmful effects associated with impulsive acoustic exposure during festivals and public celebrations.

The present study utilizes these regulatory thresholds as reference values for interpreting peak environmental noise exposure observed during festival periods and other high-intensity impulsive acoustic events.

2.6 Statistical Analysis

The analytical framework included descriptive statistics, exceedance probability analysis, and non-parametric hypothesis testing (e.g., Mann-Whitney U test) to determine the statistical significance of noise level variations between festival and non-festival periods. Correlation analysis (Spearman's rank correlation) was performed to explore the interdependence between continuous energy-based metrics and impulsive peak levels.

3 Results and Discussion

The long-term monitoring data revealed a persistently elevated acoustic environment within the studied residential region of Kolkata. A total of 78,214 valid observations were analysed between January 2023 and December 2024 using 10-minute interval measurements. The monitored soundscape exhibited substantial temporal variability associated with traffic activity, local human movement, commercial operations and seasonal festive celebrations.

The equivalent A-weighted sound level ($L_{Aeq,T}$) varied between 46.16 dB and 93.47 dB, with an overall mean value of approximately 57.90 dB. The corresponding peak sound levels ($L_{Apeak,T}$) ranged from 67.29 dB to 145.56 dB, indicating the occurrence of several extremely high-intensity impulsive acoustic events during the monitoring period. The mean values of the additional acoustic indicators were 65.86 dB for $L_{Ceq,T}$ and 70.58 dB for $L_{Zeq,T}$.

The statistical acoustic descriptors calculated from the $L_{Aeq,T}$ distribution were:

$$L_{10} = 63.44 \text{ dB}, \quad L_{50} = 57.69 \text{ dB}, \quad L_{90} = 52.38 \text{ dB}.$$

The relatively large difference between L_{10} and L_{90} suggests the presence of strong temporal fluctuations in environmental noise levels. This behaviour is characteristic of densely populated metropolitan environments where intermittent high-energy acoustic events are superimposed on an already elevated urban background sound level.

Figure 1 presents the temporal evolution of the daily average equivalent sound level throughout the study period. Although the overall acoustic environment remained relatively stable during ordinary days, several noticeable elevations were observed during festival periods and high-activity urban intervals.

The hourly acoustic profile demonstrated a clear diurnal structure associated with human activity patterns. Figure 2 shows that environmental noise levels increased progressively during the morning hours, remained elevated throughout the daytime period and gradually declined during late-night hours.

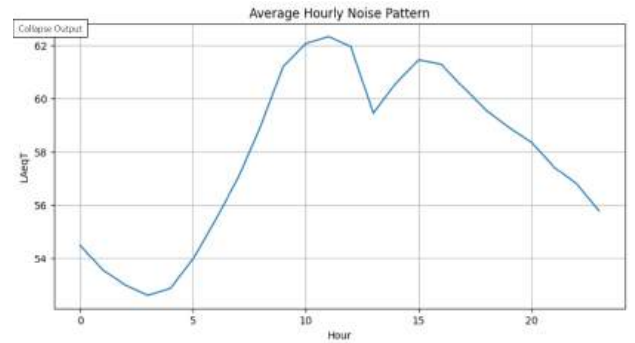


Figure 2: Average hourly variation of environmental noise levels.

Daytime noise levels were consistently higher than nighttime levels. The average daytime $L_{Aeq,T}$ was approximately 59.77 dB, compared with 54.13 dB during nighttime periods. Similarly, daytime peak sound levels and impulsive indices were also greater, indicating intensified urban activity during working and commercial hours.

To characterize abrupt high-energy acoustic events, an impulsive index was evaluated using the difference between peak and equivalent sound levels:

$$I_{idx} = L_{Apeak,T} - L_{Aeq,T}.$$

The impulsive index demonstrated considerable variability throughout the monitoring period, reflecting the

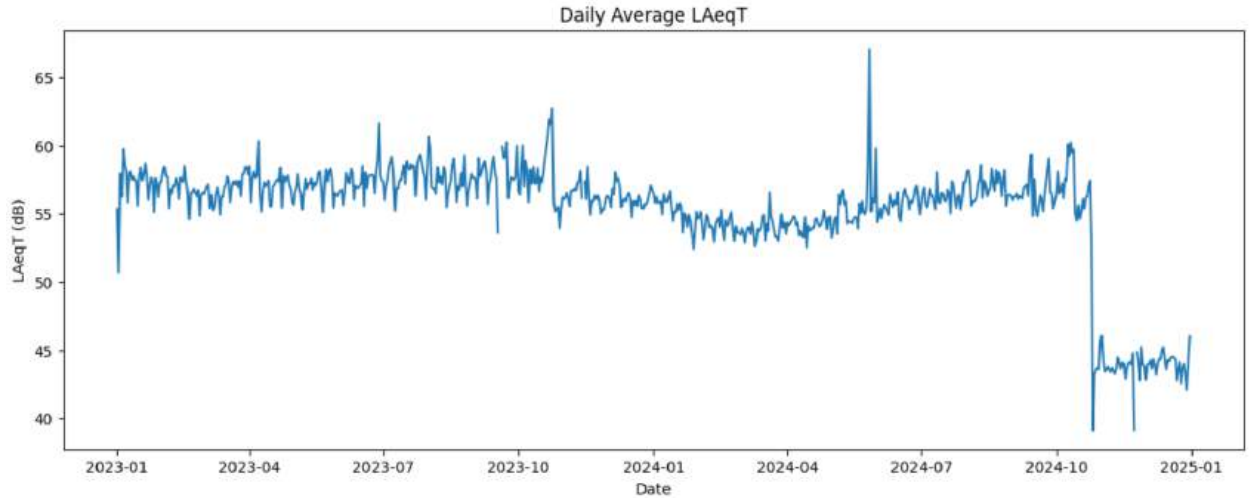


Figure 1: Daily average equivalent sound level ($L_{Aeq,T}$) observed between January 2022 and December 2023.

irregular occurrence of short-duration acoustic bursts. Figure 3 illustrates the temporal evolution of impulsive acoustic activity across the study duration.

Beyond the analysis of overall sound levels and impulsive acoustic behaviour, an additional objective of the study was to identify the dominant urban activities contributing to elevated environmental noise exposure. Since metropolitan acoustic environments are generated by multiple overlapping sources, a rule-based noise segregation framework was developed to distinguish between different classes of urban acoustic events using both acoustic indicators and temporal activity patterns.

To further investigate the dominant contributors to the observed urban acoustic environment, the rule-based noise segregation framework described in the methodology section was applied to the complete monitoring dataset. The classification results revealed distinct temporal patterns associated with different urban acoustic sources.

Figure 4 illustrates the temporal distribution of classified acoustic events throughout the monitoring period. Firework-related observations appeared predominantly during festive nighttime intervals and were characterized by strong impulsive behaviour. These events were concentrated mainly around major celebration periods such as Diwali, Kali Puja and New Year.

Traffic-related noise remained continuously distributed across ordinary daytime urban activity, particularly during morning and evening rush-hour periods. The traffic component formed the dominant continuous urban background acoustic environment throughout much of the study duration.

Loud music events were comparatively less frequent and occurred primarily during late-evening and nighttime periods. These observations exhibited enhanced low-frequency acoustic characteristics consistent with amplified entertainment-related sound sources.

A substantial proportion of observations also remained within the unclassified category, indicating the presence of overlapping or mixed urban acoustic conditions that could not be uniquely associated with a single dominant source type using the predefined rule-based

criteria.

These events are likely associated with high-intensity impulsive urban activities such as fireworks, amplified public gatherings, construction bursts or abrupt vehicular disturbances. The observed peak levels approach or exceed several internationally recognized environmental noise thresholds for impulsive sound exposure.

Figure 5 displays the temporal distribution of these extreme peak acoustic events.

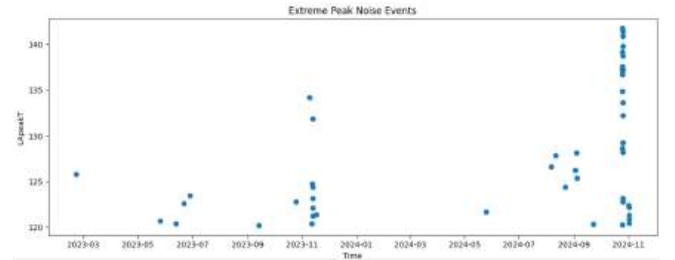


Figure 5: Extreme peak acoustic events exceeding 120 dB peak sound level.

Festival-related analysis revealed measurable increases in both environmental and impulsive noise exposure. Major celebration periods including Holi, Diwali, Kali Puja, Durga Puja and New Year were isolated and compared against ordinary non-festival periods. In 2023, Diwali was observed on November 12, while Durga Puja celebrations occurred between October 20 and October 24. In 2024, Diwali fell on October 31 and Durga Puja was celebrated from October 9 to October 13.

Festival-related analysis revealed measurable increases in both environmental and impulsive noise exposure. Major celebration periods including Holi, Diwali, Kali Puja, Durga Puja and New Year were isolated and compared against ordinary non-festival periods. Note that in 2023, Diwali fell on November 12 and Durga Puja occurred from October 20 to 24, while in 2024, Diwali was observed on October 31 and Durga Puja took place from October 9 to 13.

Figure 6 presents the distribution of peak acoustic levels during festival and ordinary periods. The festival distribution exhibited a noticeably heavier upper tail

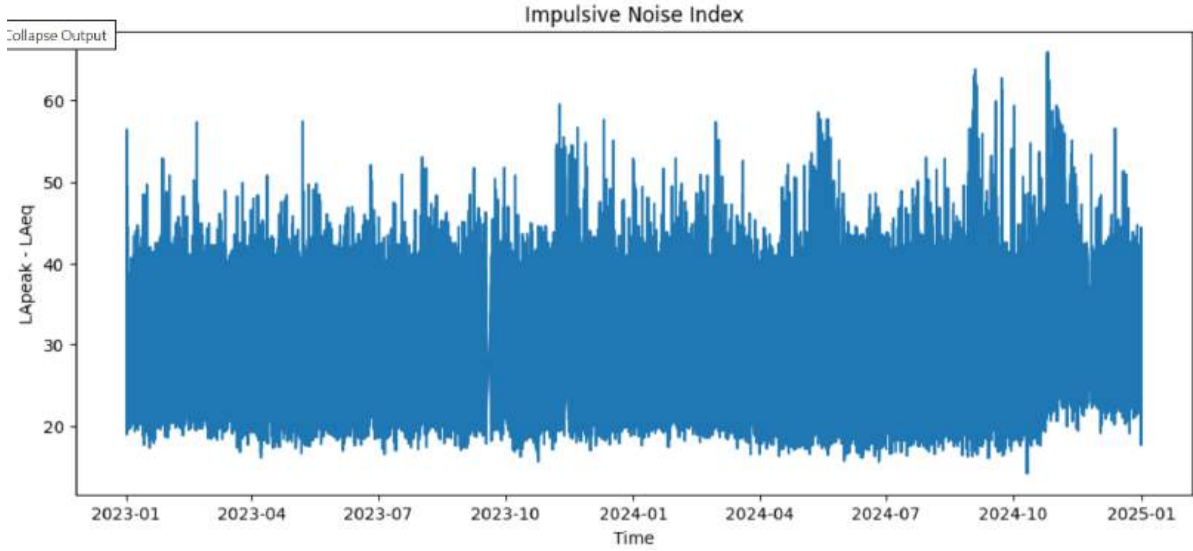


Figure 3: Temporal variation of the impulsive noise index during the monitoring period.

together with stronger variability in high-intensity peak events. This behaviour suggests deviations from normality and motivates the use of non-parametric statistical procedures for comparing the acoustic distributions.

For the following analyses, observations were assumed to be independent and identically distributed (i.i.d.) within the festival and ordinary-period groups.

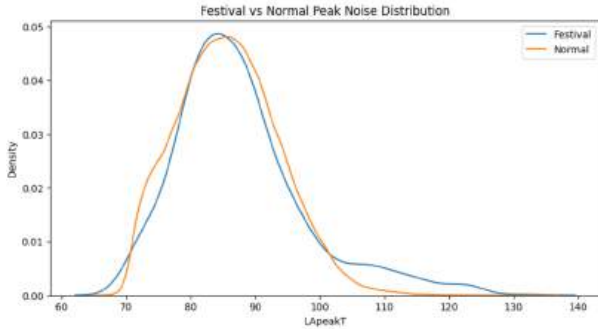


Figure 6: Kernel density distribution of peak sound levels during festival and ordinary periods.

To compare the overall distributions of peak acoustic exposure between festival and ordinary periods, a two-sample Kolmogorov–Smirnov (KS) test was performed.

The hypotheses were defined as:

$$H_0 : F_{\text{festival}}(x) = F_{\text{ordinary}}(x)$$

$$H_1 : F_{\text{festival}}(x) \neq F_{\text{ordinary}}(x)$$

The analysis yielded:

$$D = 0.0608, \quad p = 0.0022.$$

The KS test indicated a statistically significant difference between the two acoustic distributions. The observed changes were associated primarily with increased variability, stronger impulsive acoustic behaviour and heavier upper-tail characteristics during festival periods.

Additional non-parametric analysis was performed on the impulsive index:

$$I_{idx} = L_{Apeak,T} - L_{Aeq,T}.$$

The average impulsive index increased from approximately 27.90 dB during ordinary periods to 29.41 dB during festival periods, indicating stronger short-duration impulsive acoustic activity during celebrations.

A Mann–Whitney U test was conducted to compare impulsive-index distributions.

The hypotheses were defined as:

H_0 : Festival and ordinary periods have similar impulsive-index distributions.

H_1 : Festival periods exhibit larger impulsive-index values.

The resulting test statistics were:

$$U = 38,827,721.5, \quad p = 5.47 \times 10^{-6}.$$

The Mann–Whitney analysis demonstrated that festival periods are associated with significantly stronger impulsive acoustic activity compared with ordinary urban conditions. The analysis of regulatory exceedance further demonstrated the severity of the urban acoustic environment. According to CPCB residential standards, permissible limits are 55 dB during daytime and 45 dB during nighttime periods. Exceedance analysis revealed that approximately 96.11% of all observations exceeded the prescribed limits.

Daytime exceedance reached approximately 94.18%, while nighttime exceedance approached complete non-compliance. The persistence of elevated nighttime acoustic exposure is particularly important because nighttime noise has been strongly associated with sleep disturbance, stress-related physiological effects and reduced urban environmental quality.

Overall, the results indicate that the monitored urban residential environment experiences chronic environmental noise exposure combined with episodic high-intensity

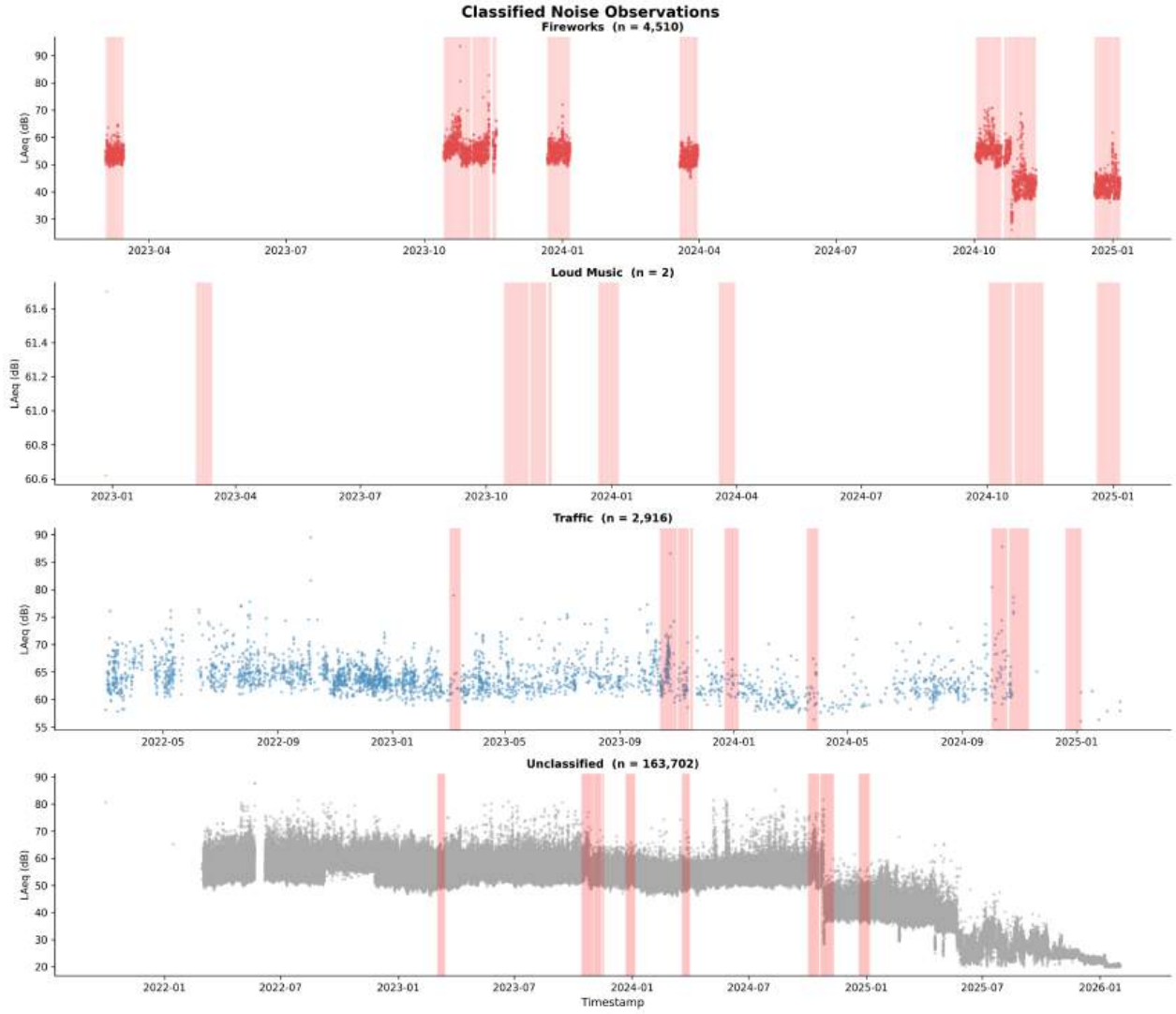


Figure 4: Temporal distribution of classified urban acoustic events based on rule-based noise segregation.

impulsive acoustic events. While continuous traffic-related sound forms the dominant background acoustic climate, festival periods and occasional impulsive disturbances substantially intensify the urban soundscape.

These findings emphasize the importance of long-term environmental acoustic monitoring in rapidly urbanizing metropolitan regions such as Kolkata. The observed exceedance of regulatory standards and the occurrence of extreme impulsive events suggest the need for improved enforcement of environmental noise regulations, particularly during festive periods involving firecrackers and amplified public activities.

4 Conclusion

This study investigated long-term environmental and impulsive noise exposure in Kolkata using high-resolution acoustic monitoring data collected between January 2022 and December 2023. The analysis revealed that the monitored urban residential environment experiences persistently elevated sound levels with frequent exceedances of CPCB residential noise standards.

The acoustic environment was characterized by sub-

stantial temporal variability driven by daily urban activity patterns, vehicular movement, and festival-related celebrations. Statistical acoustic indicators demonstrated the presence of both elevated continuous background noise and intermittent high-intensity impulsive events. Several extreme peak events exceeded 120 dB, with the maximum observed peak level reaching 145.56 dB.

Festival periods including Diwali, Kali Puja, Holi and New Year exhibited significantly higher impulsive acoustic behaviour compared with ordinary periods. Statistical testing confirmed that these differences were significant, indicating the important contribution of celebratory activities to urban acoustic escalation.

The exceedance analysis demonstrated that environmental noise levels remained above CPCB permissible limits for the majority of the monitoring period, particularly during nighttime hours. These findings suggest the persistence of chronic environmental acoustic stress within densely populated urban residential regions.

Overall, the results highlight the growing importance of continuous environmental acoustic monitoring in rapidly urbanizing metropolitan areas such as Kolkata. Improved enforcement of environmental noise

regulations, increased public awareness regarding impulsive acoustic exposure, and stricter management of festival-related firecracker activity may help reduce long-term public health risks associated with excessive urban noise exposure.

Analysis of Daily and Annual Noise Pollution Cycles Using Clustering and FPCA

1 Data Preparation and Preliminary Diurnal Analysis

1.1 Data Integration and Cleaning

Noise-monitoring data were obtained from multiple monitoring stations distributed across several districts of West Bengal. The raw data files were available separately for individual stations in .csv format. A total of 311 station-level files were imported and merged into a unified master dataset using R.

The final merged dataset contained approximately 36.6 million observations and the following variables:

- **timestamp**: date-time of observation,
- **laeqt**: equivalent continuous A-weighted sound level,
- **lceqt**: equivalent continuous C-weighted sound level,
- **lzeqt**: equivalent continuous Z-weighted sound level,
- **lapeakt**: peak A-weighted sound level,
- **District**: district identifier,
- **Station**: station identifier.

Temporal variables including year, month, weekday, hour, minute and day-of-year were subsequently extracted from the timestamp variable for further temporal analysis.

During preliminary inspection, a small number of observations with anomalous timestamps from the year 1970 were detected, likely arising from device initialization or timestamp corruption. Since these observations were clearly outside the study period, all observations prior to the year 2020 were excluded from subsequent analyses.

No missing values were observed in the principal acoustic variables after cleaning.

1.2 Hourly Aggregation

Since the original observations were recorded at very high temporal resolution (approximately minute-level or second-level observations), direct analysis of the raw data was computationally inefficient and scientifically unnecessary for studying diurnal variation.

Therefore, hourly aggregation was performed. For each district, station, calendar date and hour, the following quantities were computed:

- mean $LA_{eq,T}$,
- mean $LC_{eq,T}$,
- mean $LZ_{eq,T}$,
- mean $LA_{peak,T}$,
- maximum $LA_{peak,T}$,
- number of raw observations contributing to the hourly average.

The resulting hourly dataset formed the basis of all subsequent temporal analyses.

1.3 Overall Diurnal Pattern Estimation

To estimate the average daily noise cycle, hourly averages were computed across all stations and dates. Figure 1 displays the estimated mean diurnal profiles for $LA_{eq,T}$, $LC_{eq,T}$, and $LZ_{eq,T}$.

The estimated profiles exhibited a clear and interpretable diurnal structure. Noise levels remained relatively low during late-night and early-morning hours (approximately 00:00–04:00 hrs), followed by a gradual increase beginning in the early morning. A pronounced rise was observed during the morning activity period, after which the sound levels remained elevated during daytime and evening hours before declining again at night.

The ordering

$$LZ_{eq,T} > LC_{eq,T} > LA_{eq,T}$$

was consistently observed throughout the day. This pattern is acoustically meaningful because Z-weighting preserves the full frequency spectrum, C-weighting retains substantial low-frequency components, whereas A-weighting attenuates low-frequency sound in accordance with human auditory sensitivity.

The observed diurnal behaviour strongly suggests the influence of anthropogenic activities such as road traffic, commercial movement and urban crowding on environmental noise dynamics.

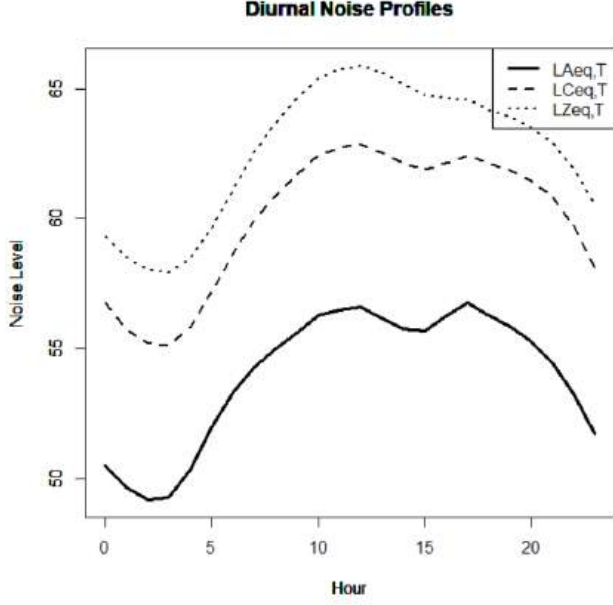


Figure 1: overall diurnal

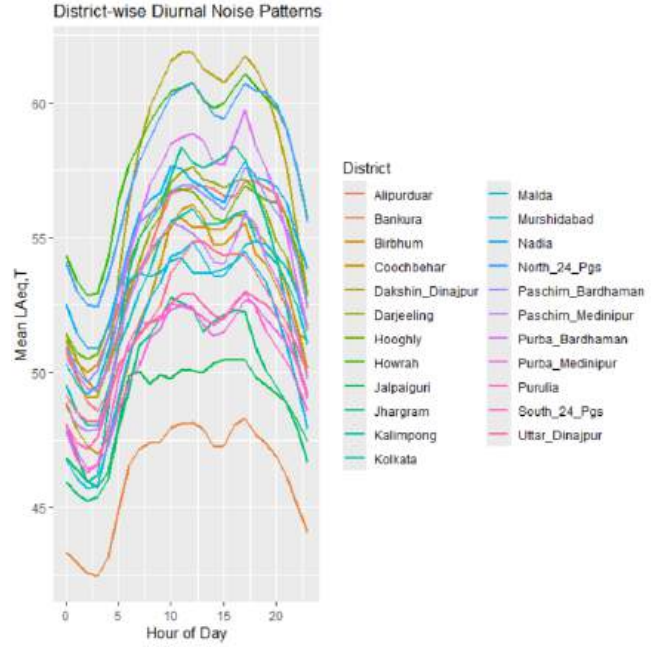


Figure 2: District Wise

1.4 District-wise Diurnal Variation

To investigate spatial heterogeneity in temporal noise behaviour, district-wise mean diurnal profiles were estimated using hourly aggregated data. Figure 2 presents the district-level diurnal curves for $LA_{eq,T}$.

Several important features emerged from the analysis:

1. All districts exhibited a broadly similar diurnal structure characterised by lower nighttime levels and elevated daytime activity.
2. Considerable spatial variation was nevertheless observed in both the magnitude and shape of the diurnal cycles.
3. Certain districts consistently exhibited higher average noise levels throughout the day, indicating greater urbanisation, traffic density or commercial activity.
4. Some districts displayed sharper morning and evening peaks, suggestive of commuting-related traffic effects.
5. Other districts exhibited relatively flatter profiles, indicating more stable or less traffic-dominated acoustic environments.

The large degree of overlap and heterogeneity among district-level curves further motivated the need for dimension-reduction and clustering approaches, which are explored in subsequent sections.

2 Reduction and Representation of Large Numbers of Daily Cycles

2.1 Motivation

One of the major analytical challenges in the present study arises from the extremely large number of temporal noise trajectories generated across stations and dates. Since each station-day combination produces a complete 24-hour noise profile, direct reporting of all daily cycles becomes impractical and visually uninformative.

To address this issue, a curve-reduction strategy based on clustering of daily diurnal profiles was adopted. The objective was to identify a small number of representative temporal noise patterns capable of summarising the dominant forms of diurnal variation observed across the dataset.

2.2 Construction of Daily Functional Curves

Using the hourly aggregated dataset, each station-day observation was converted into a 24-dimensional vector representing the hourly evolution of the equivalent continuous sound level ($LA_{eq,T}$) over a complete day.

Mathematically, each daily observation may be represented as

$$X_i(t), \quad t = 0, 1, \dots, 23,$$

where $X_i(t)$ denotes the average noise level during hour t for the i -th station-day.

The data were transformed into a wide functional format in which:

- each row corresponded to one complete daily cycle,

- each column corresponded to one hour of the day.

After removal of incomplete daily records containing missing hourly observations, a total of 226,367 complete daily curves remained available for analysis.

2.3 Normalization of Daily Curves

Direct clustering of raw noise curves primarily captures differences in overall loudness levels between stations rather than differences in temporal structure. Since the primary objective was to identify characteristic *shapes* of diurnal variation, each daily curve was individually standardized prior to clustering.

Specifically, for each daily curve:

$$Z_i(t) = \frac{X_i(t) - \bar{X}_i}{s_i},$$

where:

- $\bar{X}_i$ denotes the daily mean noise level,
- s_i denotes the daily standard deviation.

This normalization removes absolute level effects and preserves only the relative temporal behaviour of the daily noise trajectory.

Consequently, the clustering procedure focused on identifying similarities in:

- morning rise patterns,
- evening peaks,
- nighttime decline,
- bimodal or unimodal activity structures.

2.4 Selection of Number of Clusters

A representative random sample of 10,000 normalized daily curves was selected for computational efficiency during exploratory clustering.

The k -means clustering algorithm was applied for values of $k = 1, \dots, 10$, and the within-cluster sum of squares (WCSS) criterion was examined using the elbow method.

The elbow plot (Figure 3) demonstrated a substantial reduction in WCSS up to approximately $k = 4$, after which the rate of improvement diminished considerably. Therefore, a four-cluster solution was adopted as a balance between model complexity and interpretability.

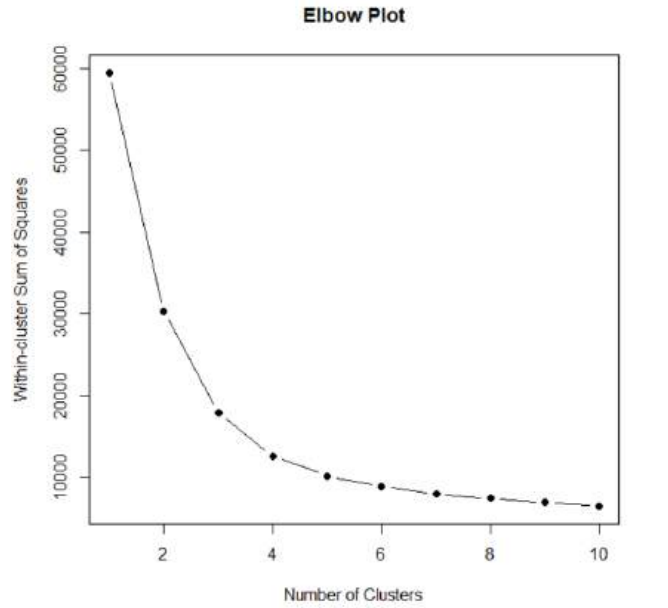


Figure 3: Elbow Plot

2.5 Representative Daily Noise Patterns

The final clustering procedure identified four dominant classes of diurnal noise behaviour. Cluster centres were extracted and interpreted as representative daily noise cycles. The curves are represented in figure 4.

The resulting representative profiles revealed several distinct forms of temporal acoustic behaviour:

1. A profile characterised by pronounced daytime elevation and gradual nighttime decline.
2. A morning-dominant pattern suggestive of commuter or traffic-driven activity.
3. A late-evening or nighttime-elevated profile potentially associated with commercial or urban social activity.
4. A comparatively flatter but temporally shifted profile exhibiting delayed peak behaviour.

These representative cluster-centre curves provide a compact yet informative summary of the large collection of observed daily cycles, thereby addressing the principal objective of Challenge .

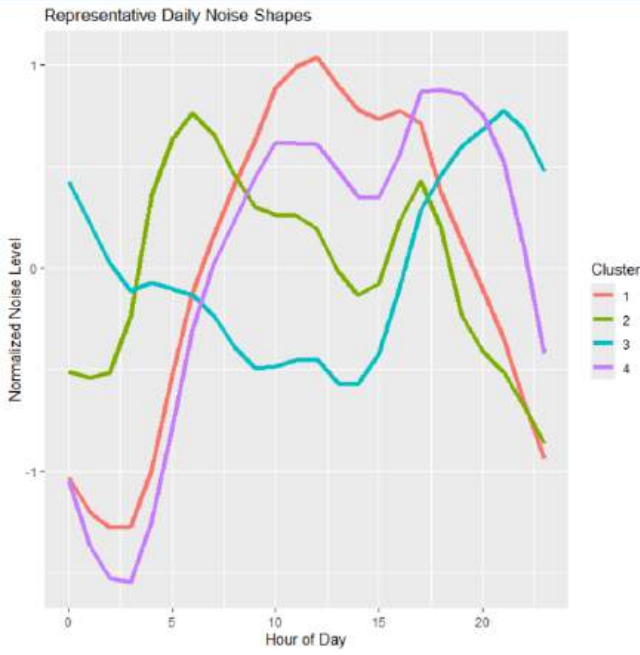


Figure 4: Representative daily noise-cycle shapes obtained from clustering of normalized daily curves.

2.6 Functional Principal Component Analysis of Daily Noise Curves

Although clustering provides representative archetypal daily profiles, an additional and statistically more flexible approach to summarising large collections of temporal curves is Functional Principal Component Analysis (FPCA). In the present study, since the daily noise trajectories were observed on a discrete hourly grid (24 hourly observations per day), a standard Principal Component Analysis (PCA) applied to the discretized curves serves as an effective approximation to FPCA.

The analysis was performed on the normalized daily curves in order to study dominant modes of temporal variation independent of absolute noise intensity levels.

Let

$$X_i(t), \quad t = 0, 1, \dots, 23,$$

denote the normalized noise trajectory corresponding to the i -th station-day. The functional PCA representation may be expressed as

$$X_i(t) = \mu(t) + \sum_{k=1}^{\infty} \xi_{ik} \phi_k(t),$$

where:

- $\mu(t)$ denotes the mean daily curve,
- $\phi_k(t)$ denotes the k -th functional principal component,
- ξ_{ik} denotes the corresponding principal component score.

For computational feasibility, a representative random sample of 10,000 normalized daily curves was used for PCA estimation.

2.6.1 Variance Explained by Principal Components

The first principal component (PC1) alone explained approximately 52.36% of the total variability in daily noise trajectories. The first two principal components together explained approximately 62.39% of total variability, while the first five components accounted for approximately 76.5% of the observed variation.

These results indicate that the highly complex collection of more than 226,000 daily curves can be effectively summarized using only a few dominant latent temporal structures.

The dominance of the first few principal components strongly suggests the existence of common underlying behavioural mechanisms governing environmental noise evolution across districts and stations.

```
> summary(pca_result)
Importance of components:
          PC1    PC2    PC3    PC4    PC5    PC6    PC7    PC8    PC9    PC10
Standard deviation  3.4704 1.5152 1.2312 0.97313 0.88973 0.84145 0.72256 0.6853 0.62034 0.60268
Proportion of Variance 0.5236 0.1003 0.0659 0.04117 0.03441 0.03078 0.02272 0.0206 0.01673 0.01579
Cumulative Proportion 0.5236 0.6239 0.6898 0.73099 0.76540 0.79619 0.81891 0.8395 0.85624 0.87203

          PC11   PC12   PC13   PC14   PC15   PC16   PC17   PC18   PC19   PC20
Standard deviation 0.57127 0.54799 0.52451 0.50024 0.48575 0.47435 0.46550 0.45190 0.44795 0.43615
Proportion of Variance 0.01419 0.01305 0.01196 0.01088 0.01026 0.00978 0.00942 0.00888 0.00872 0.00827
Cumulative Proportion 0.88621 0.89927 0.91123 0.92211 0.93237 0.94219 0.95157 0.96045 0.96917 0.97744

          PC21   PC22   PC23   PC24
Standard deviation 0.43279 0.42154 0.39231 7.087e-15
Proportion of Variance 0.00814 0.00773 0.00669 0.000e+00
Cumulative Proportion 0.98558 0.99331 1.00000 1.000e+00
```

Figure 5: PCA Summary

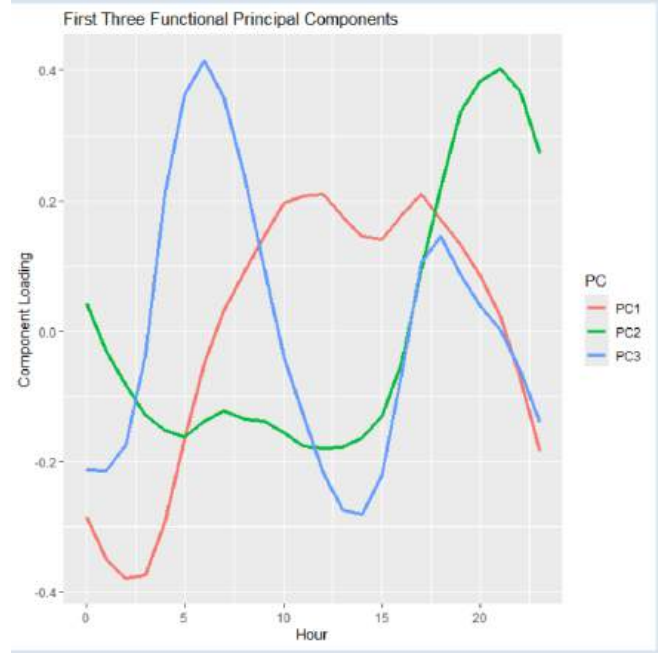


Figure 6: First three functional principal components estimated from normalized daily noise curves.

2.6.2 Interpretation of Principal Components

The estimated principal component functions revealed several important modes of temporal variability:

1. The first principal component primarily represented the contrast between elevated daytime activity and quieter nighttime behaviour, effectively capturing the overall day-night acoustic intensity structure.

2. The second principal component captured variations between morning-dominant and evening-dominant activity patterns, potentially reflecting differences in commuting and commercial activity schedules.
3. The third principal component reflected secondary fluctuations and nighttime irregularities, indicating the presence of localized episodic acoustic behaviour.

Together, the functional principal components provide a parsimonious low-dimensional representation of the extremely large collection of observed daily noise trajectories.

2.7 Annual Pattern Estimation

In addition to short-term diurnal variation, long-term annual behaviour of environmental noise was also investigated. For each station and calendar date, the hourly average noise levels were aggregated to obtain a daily average $LA_{eq,T}$ value. (Figure 7)

Subsequently, average daily noise levels were computed across all stations for each day of the year in order to estimate the annual noise trajectory.

The estimated annual pattern demonstrated clear seasonal variability, indicating that environmental noise levels are not temporally homogeneous across the calendar year. Several broad phases of elevated and reduced acoustic activity were observed, together with occasional sharp local fluctuations.

A smoothed annual curve was estimated using locally weighted regression (LOESS) smoothing in order to reveal the underlying seasonal structure while reducing short-term day-to-day fluctuations. (Figure 8)

The resulting annual trajectory suggested the existence of systematic long-term temporal variation potentially associated with:

- seasonal traffic intensity,
- climatic conditions,
- tourism and commercial activity,
- public events and festive periods.

Several sharp spikes observed in the raw annual curve may correspond to episodic events or festival-related acoustic disturbances, motivating future event-specific analyses.

3 Conclusion

The present study conducted a large-scale exploratory and functional analysis of environmental noise dynamics using high-frequency acoustic measurements collected across multiple districts and monitoring stations of West Bengal.

The analysis revealed pronounced and interpretable diurnal variation in environmental noise levels, characterised by elevated daytime activity and reduced nighttime sound intensity. Considerable spatial heterogeneity

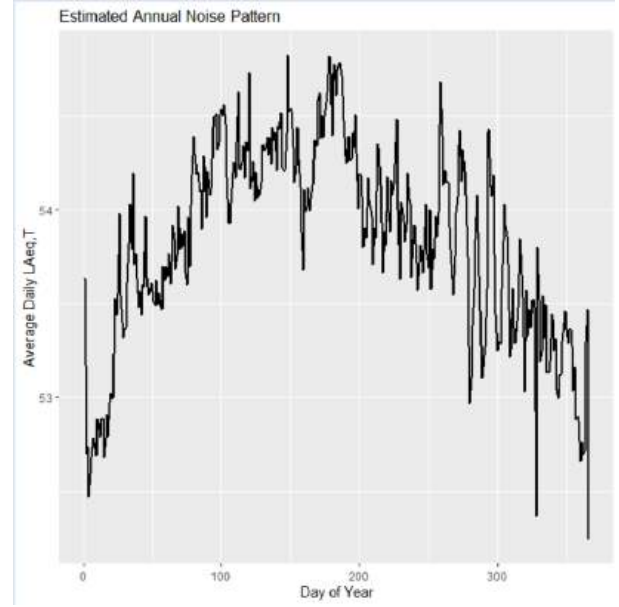


Figure 7: Estimated annual noise pattern based on daily average $LA_{eq,T}$ values.

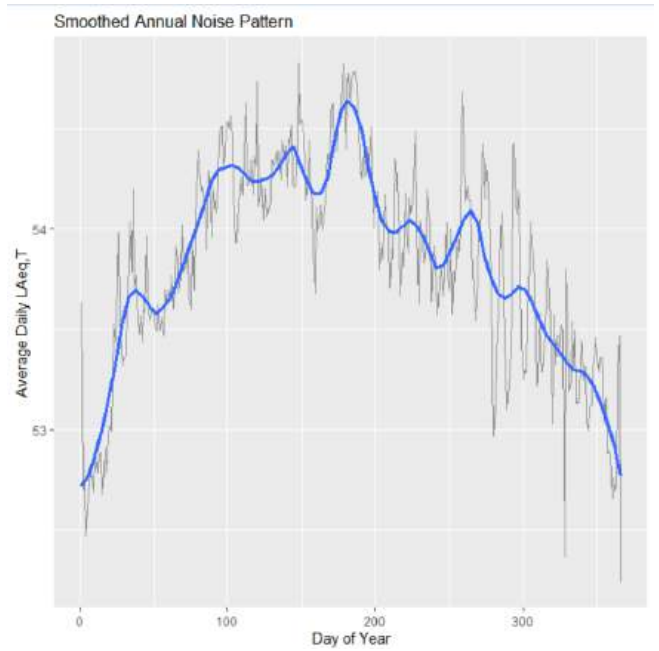


Figure 8: Smoothed annual noise trajectory obtained using LOESS smoothing.

was also observed across districts, suggesting the influence of differing levels of urbanisation, traffic density and commercial activity. To address the analytical challenge posed by the extremely large number of daily temporal trajectories, clustering and functional principal component approaches were employed. Representative cluster-centre curves successfully summarised dominant forms of daily acoustic behaviour, while FPCA demonstrated that a relatively small number of latent temporal components explained a substantial proportion of total variability.

The annual pattern analysis further revealed the presence of long-term seasonal variation and potential event-related fluctuations in environmental noise.

Overall, the study demonstrates the usefulness of functional and unsupervised statistical techniques for analysing large-scale environmental acoustic datasets and provides a compact yet informative representation of complex temporal noise dynamics.

4 Acknowledgment

We gratefully acknowledge **West Bengal Pollution Control Board (WBPCB)** for the availability of long-term environmental acoustic monitoring data used in this study. We would like to express my deep gratitude to **Dr. Debasish Sengupta** and **Dr. Suman Majumdar** and **Dr. Kalyan Rudra** for their invaluable guidance, patience, and feedback throughout this project. Their insights helped shape our research and deepened our understanding of dynamics of noise pollution. We also acknowledge the use of AI-assisted tools for language refinement, manuscript organization, and improvement of scientific presentation. All analyses, interpretations, statistical evaluations, and conclusions presented in this work were independently verified by the each group member.

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Appendix

Appendix A: Seasonal Decomposition Plots

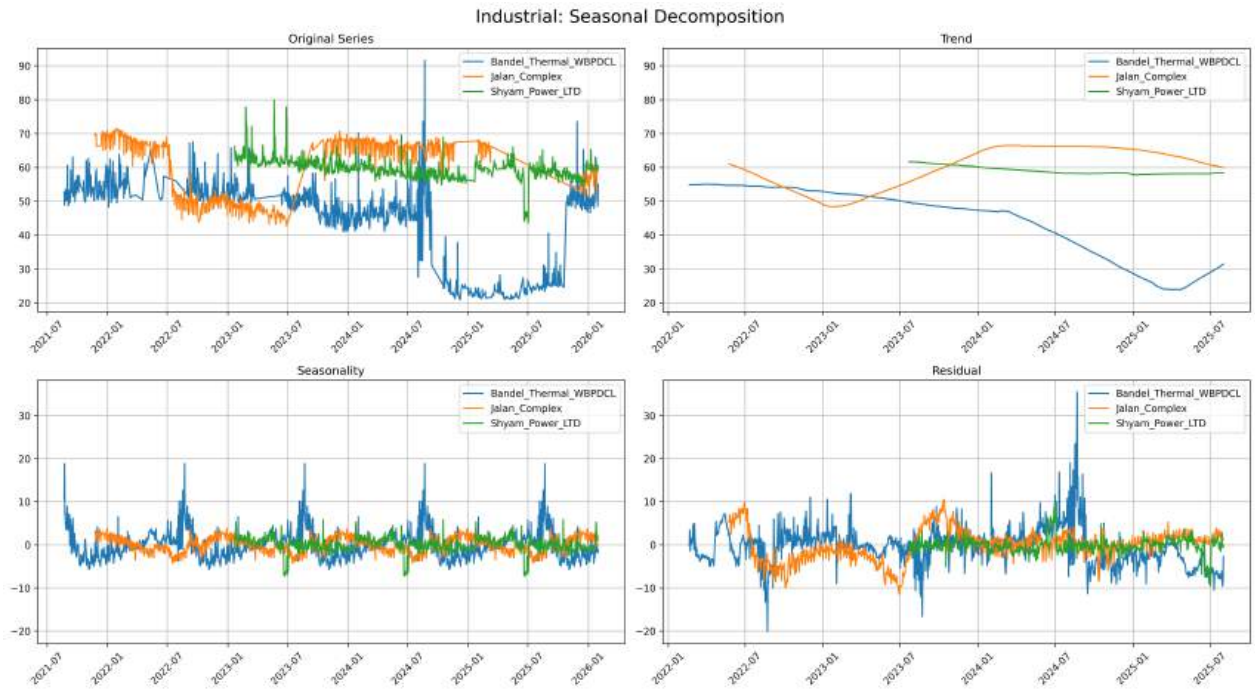


Figure 1: Industrial Zone: Daytime Seasonal Decomposition

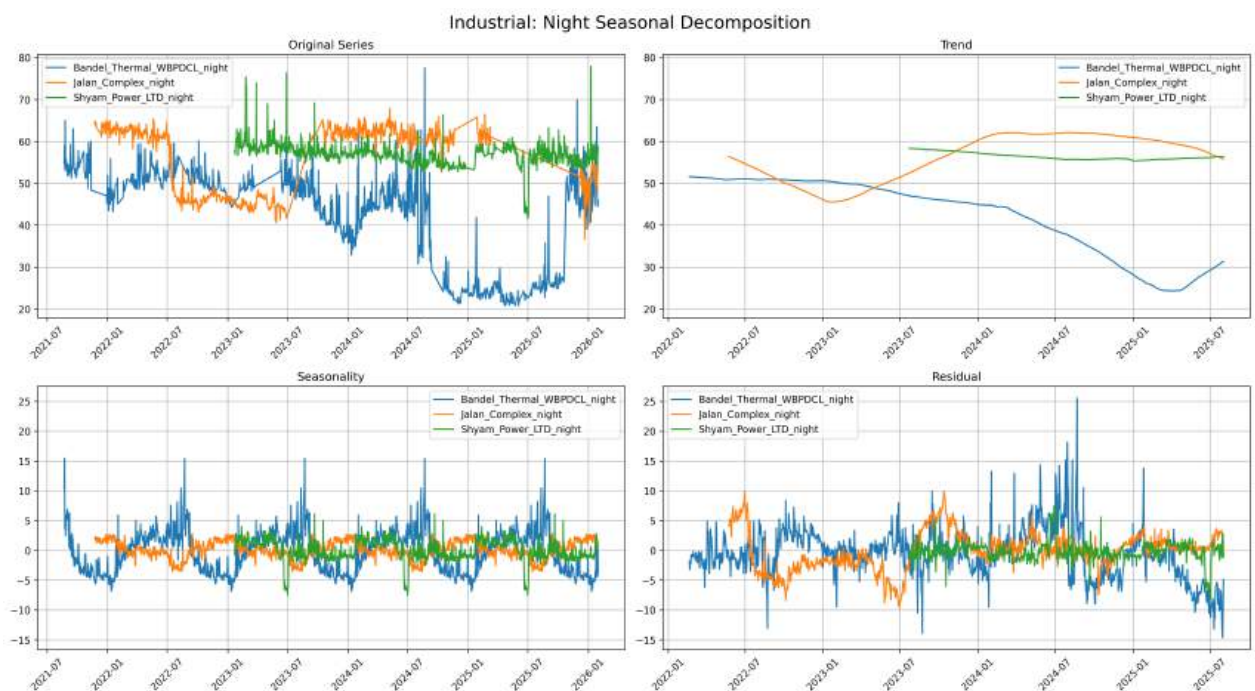


Figure 2: Industrial Zone: Nighttime Seasonal Decomposition

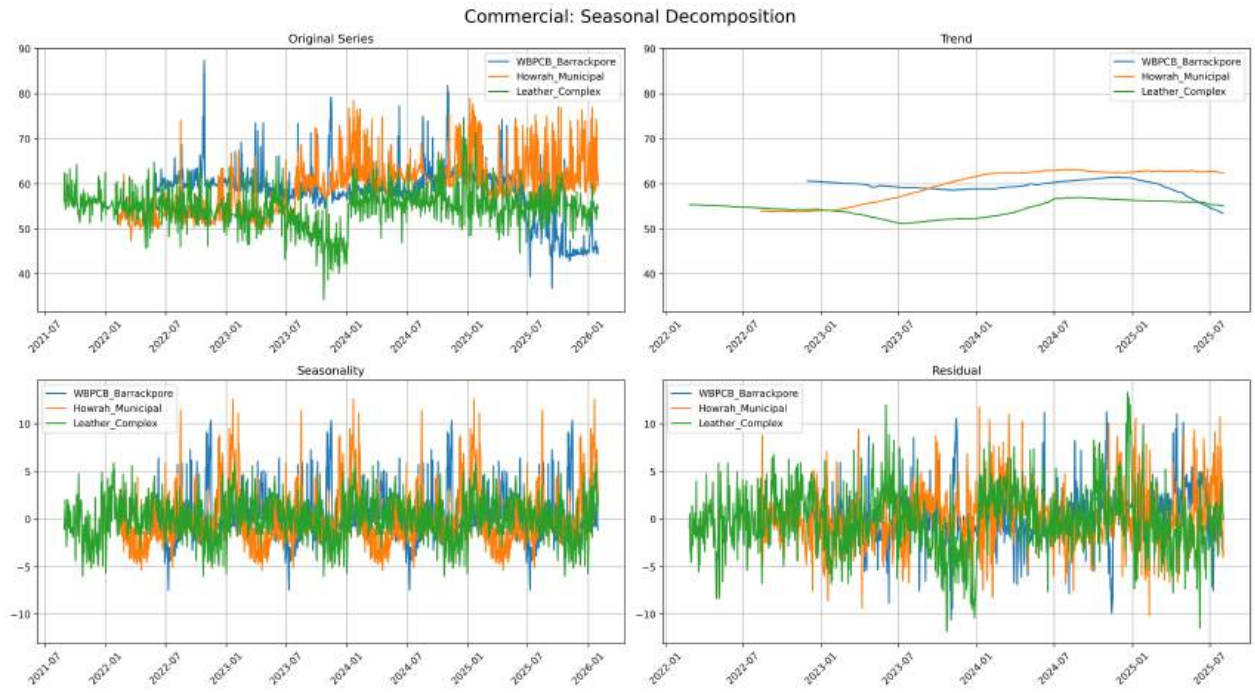


Figure 3: Commercial Zone: Daytime Seasonal Decomposition

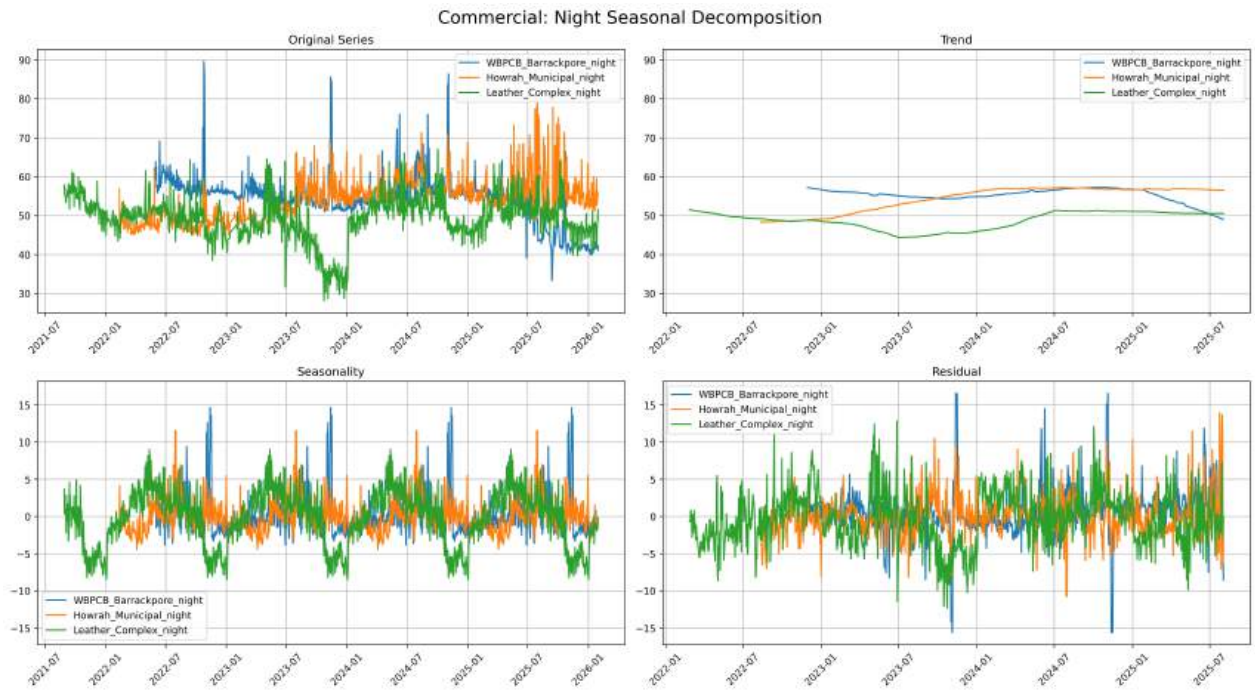


Figure 4: Commercial Zone: Nighttime Seasonal Decomposition

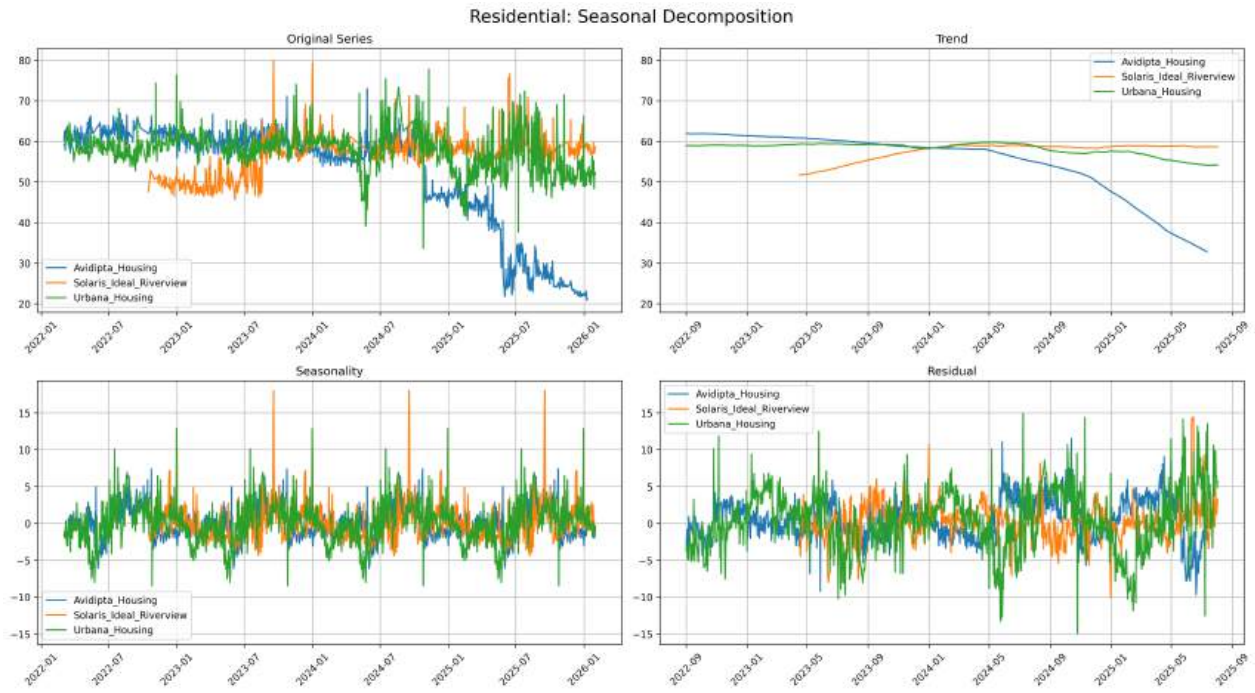


Figure 5: Residential Zone: Daytime Seasonal Decomposition

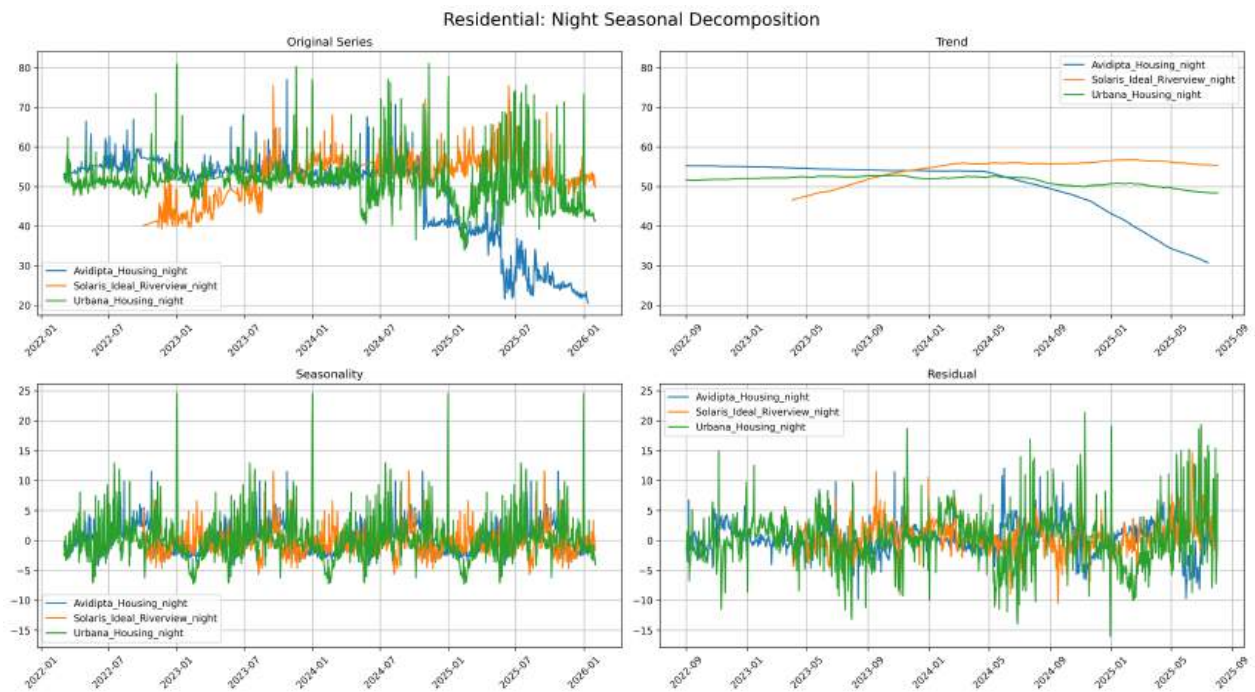


Figure 6: Residential Zone: Nighttime Seasonal Decomposition

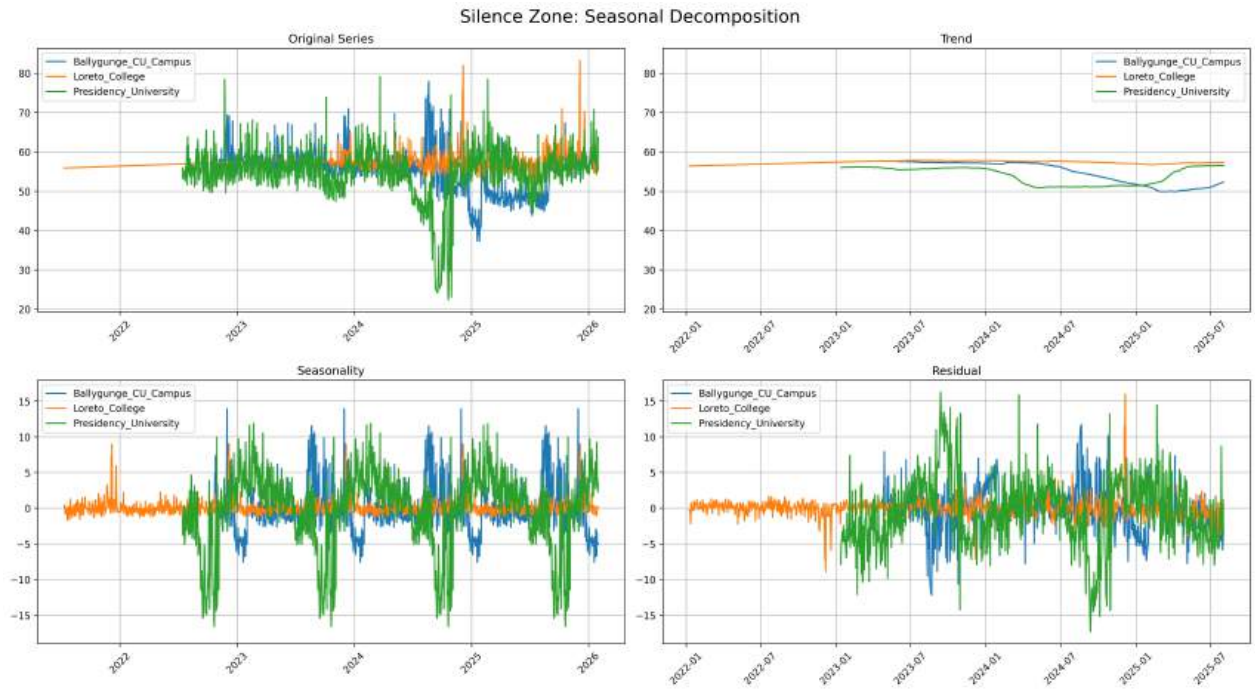


Figure 7: Silence Zone: Daytime Seasonal Decomposition

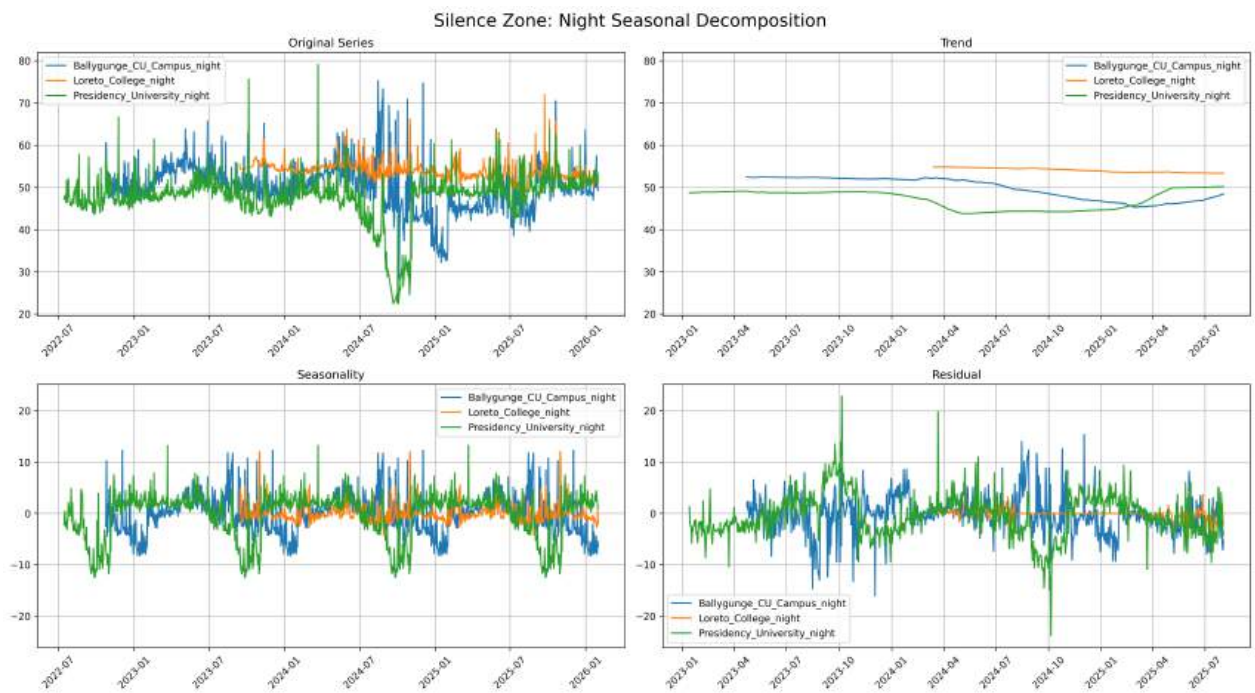


Figure 8: Silence Zone: Nighttime Seasonal Decomposition

Appendix B: Distribution Comparison Plots

LAeqT Comparisons

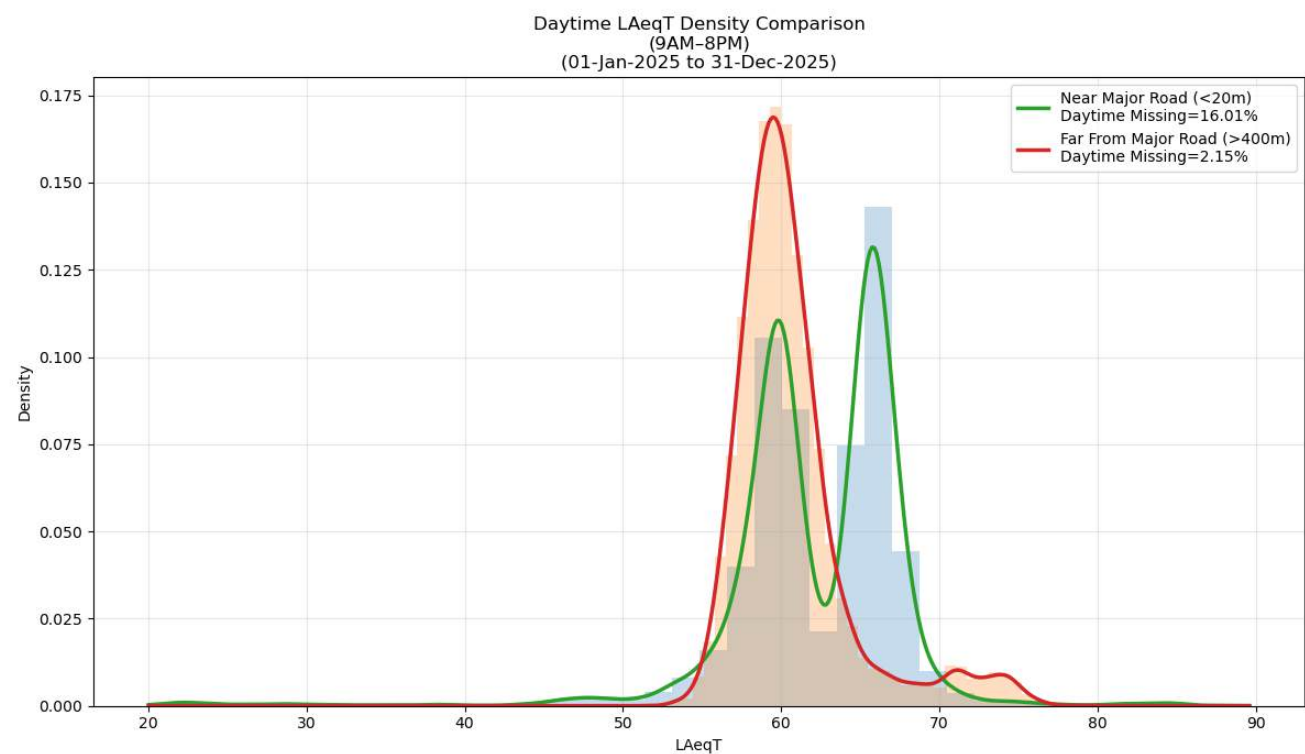


Figure 1: Belghoria-Polerhat Police Station

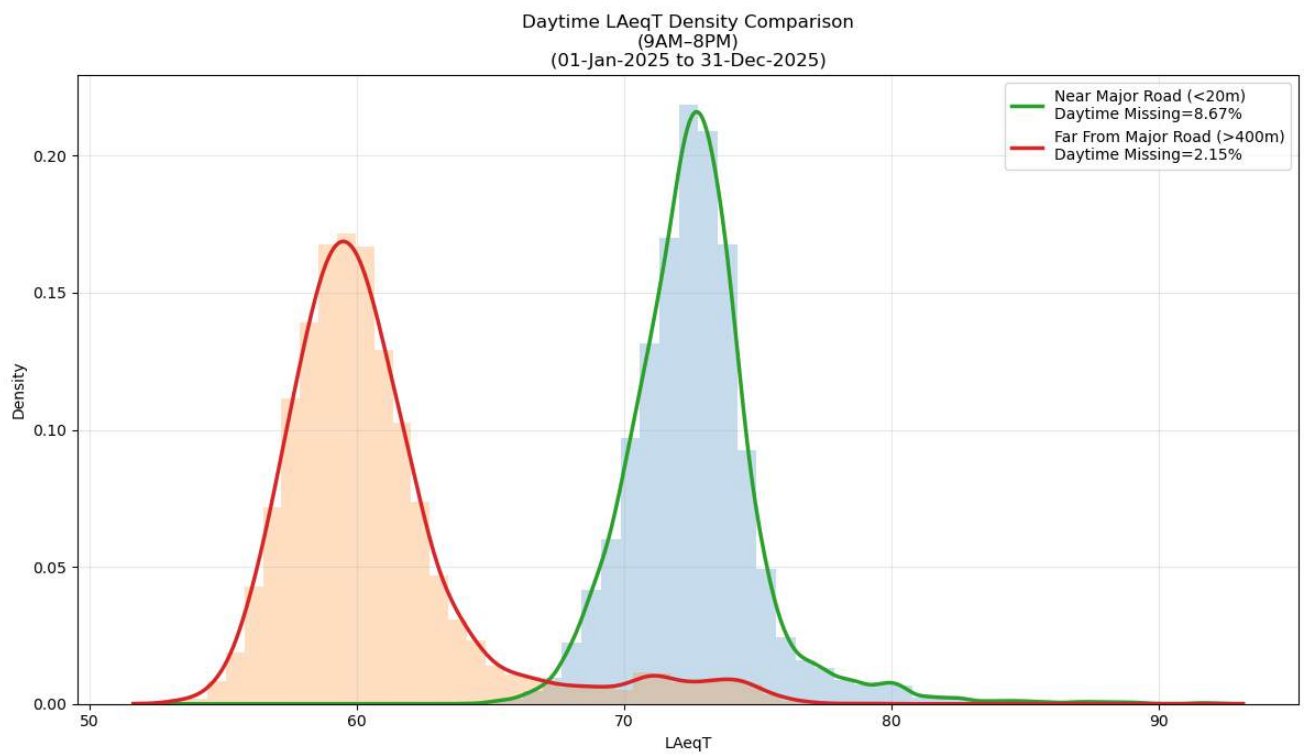


Figure 2: Kamarhati Municipality-Polerhat Police Station

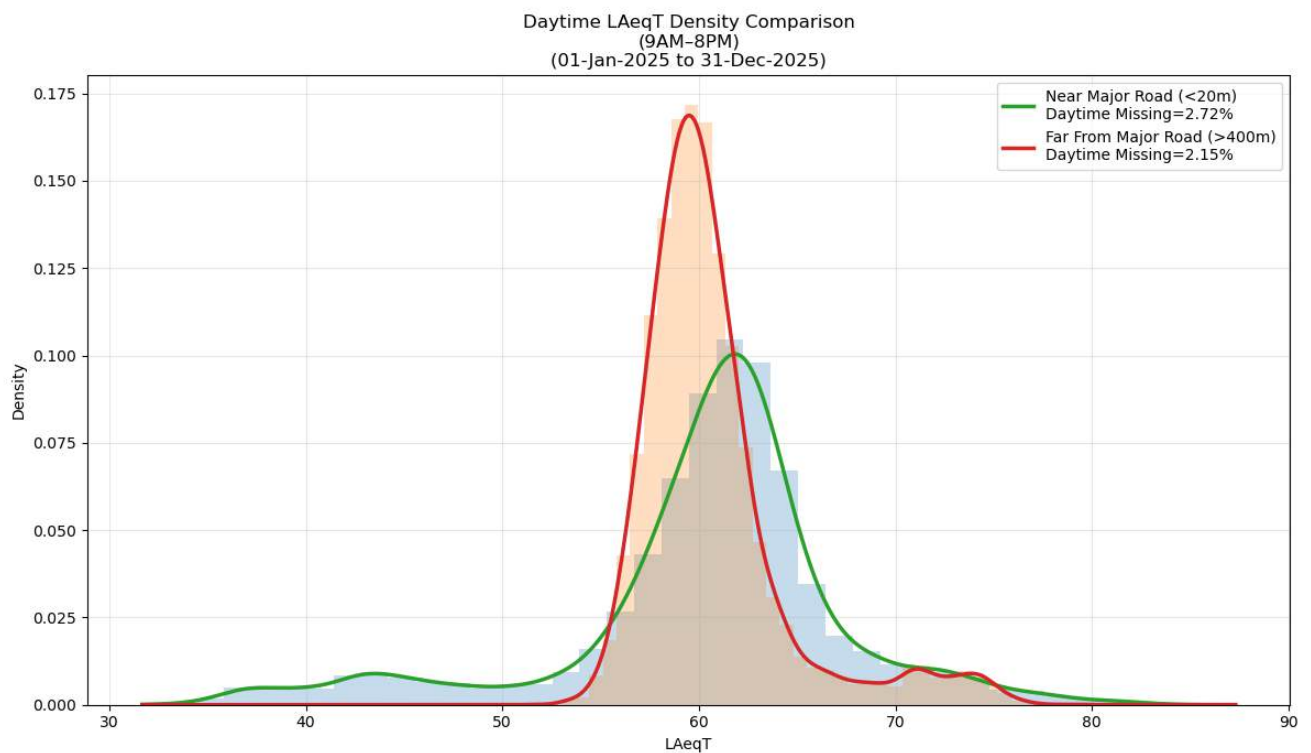


Figure 3: Karunamoye Bus Stand-Polerhat Police Station

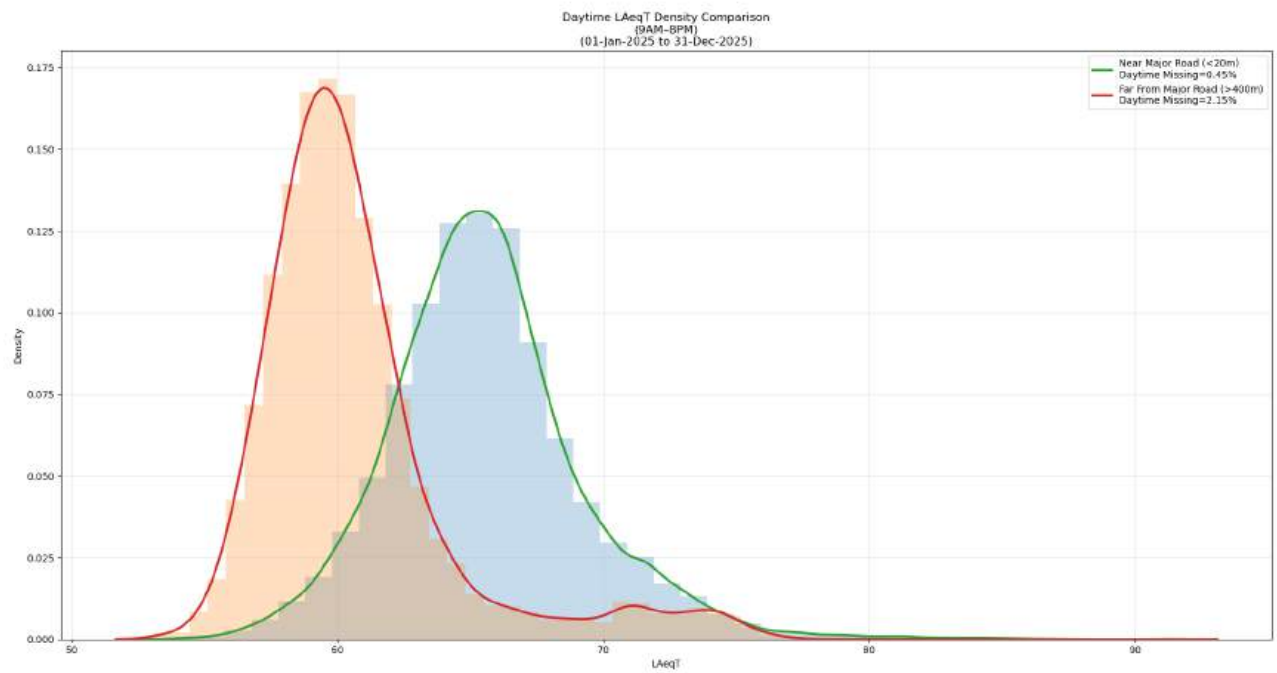


Figure 4: Panihati Municipality- Polerhat Police Station

Diurnal Traffic Profile Differences between stations near roads and stations further away from ones

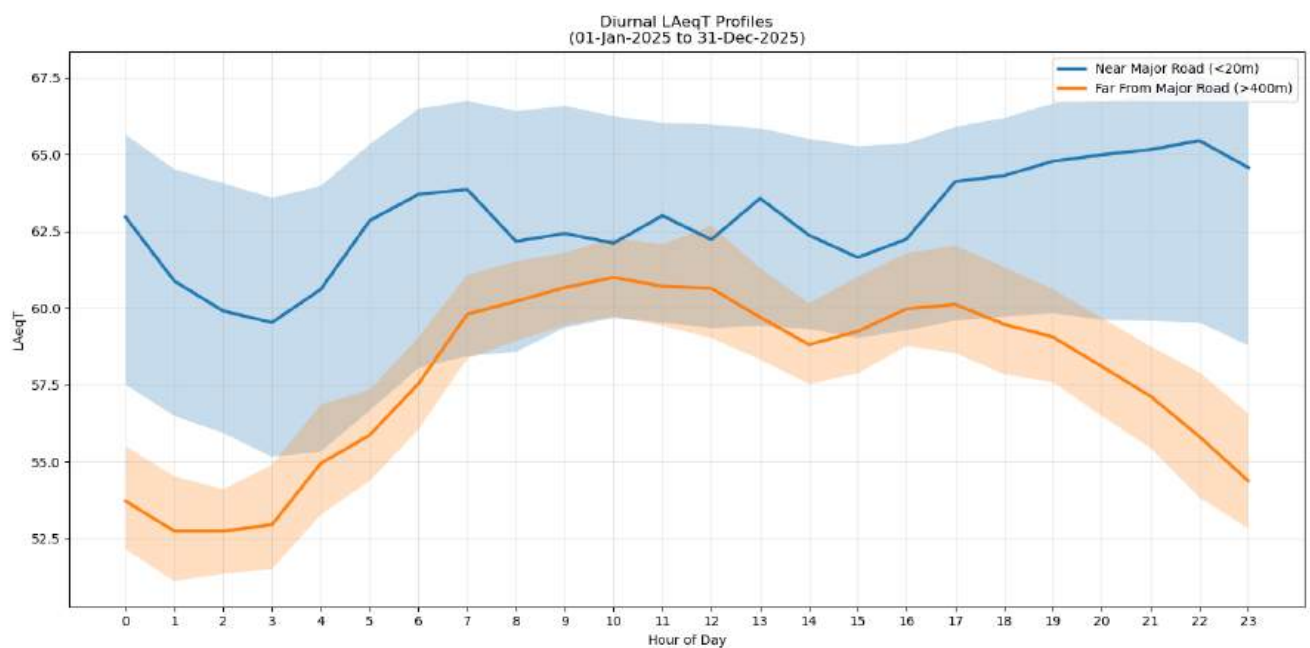


Figure 5: Belghoria- Polerhat Police Station

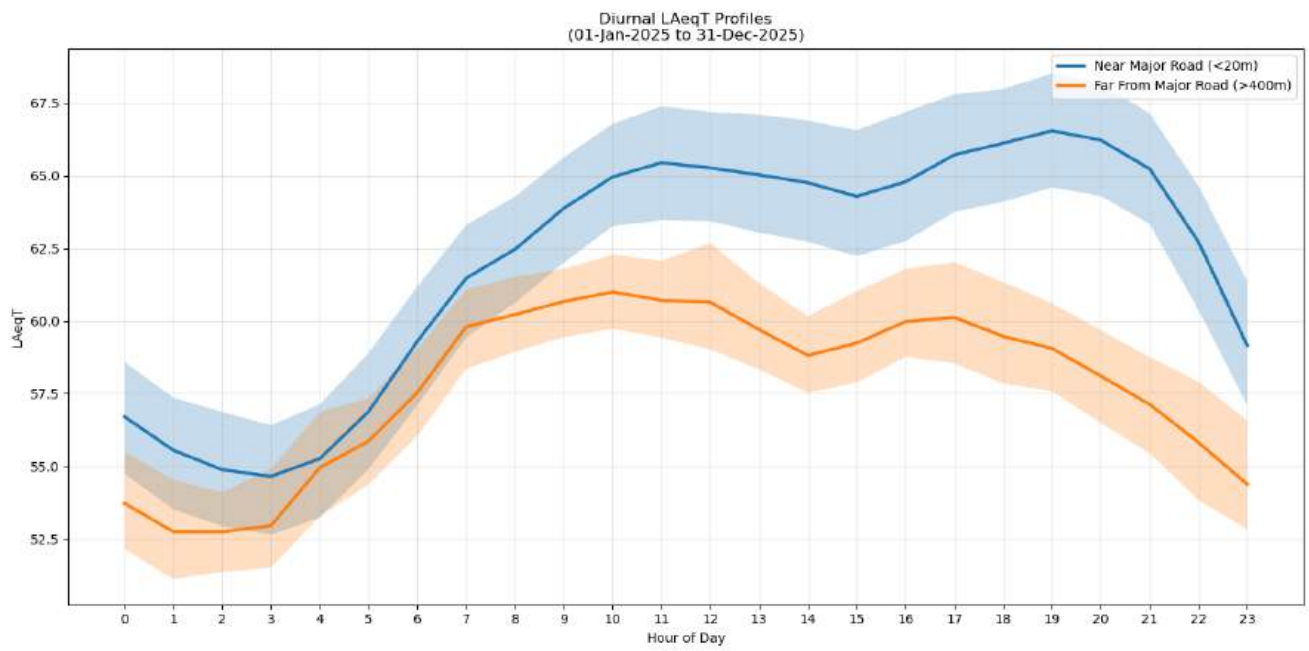


Figure 6: Panihati Municipality- Polerhat Police Station

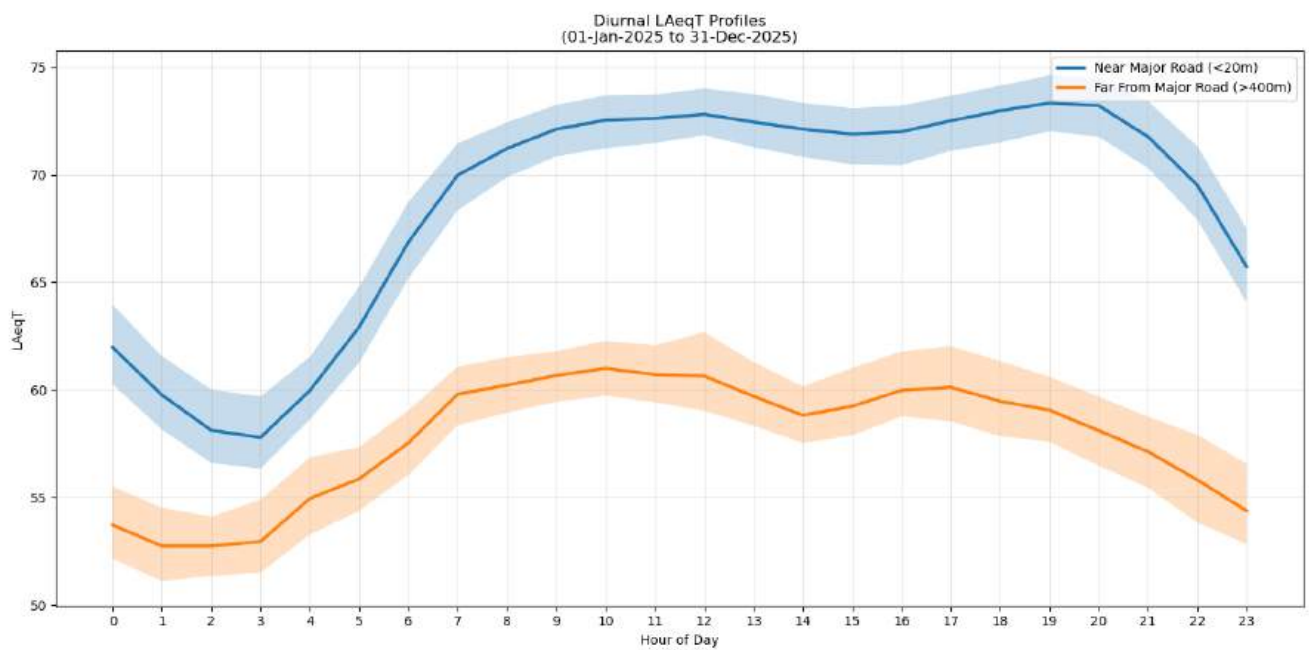


Figure 7: Kamarhati Municipality- Polerhat Police Station

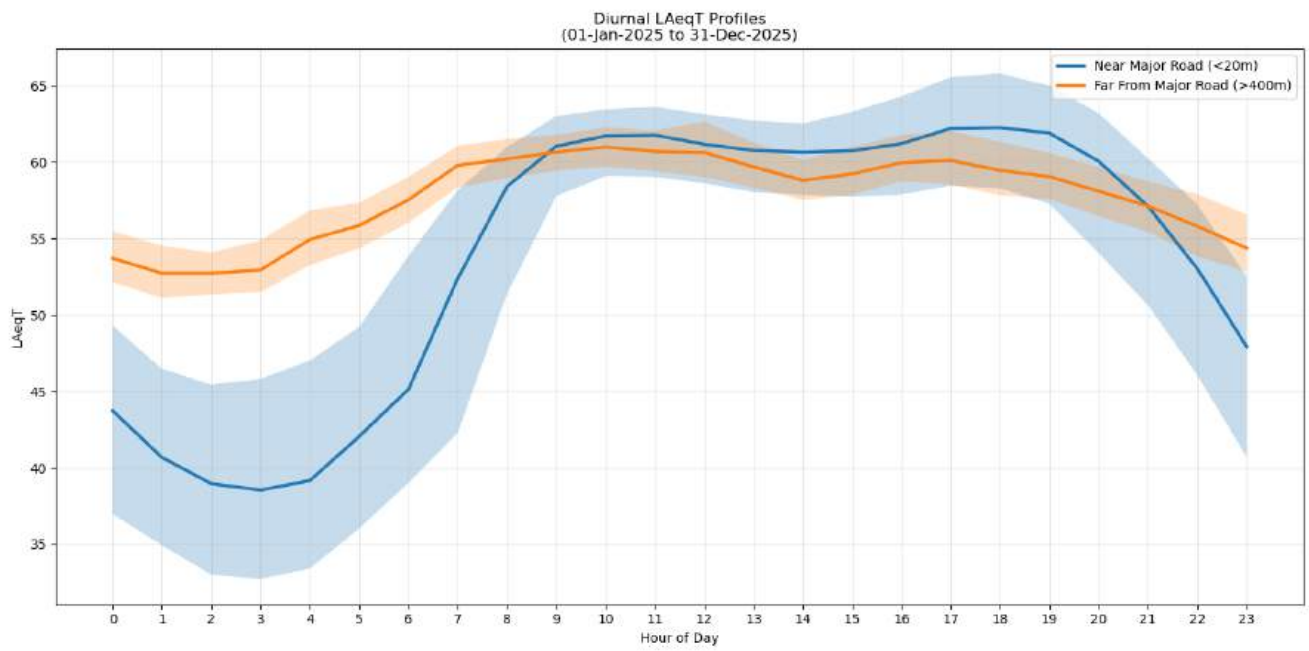


Figure 8: Karunamoye Bus Stand- Polerhat Police Station

Impulsiveness Festive-Nonfestive and the empirical survival plots

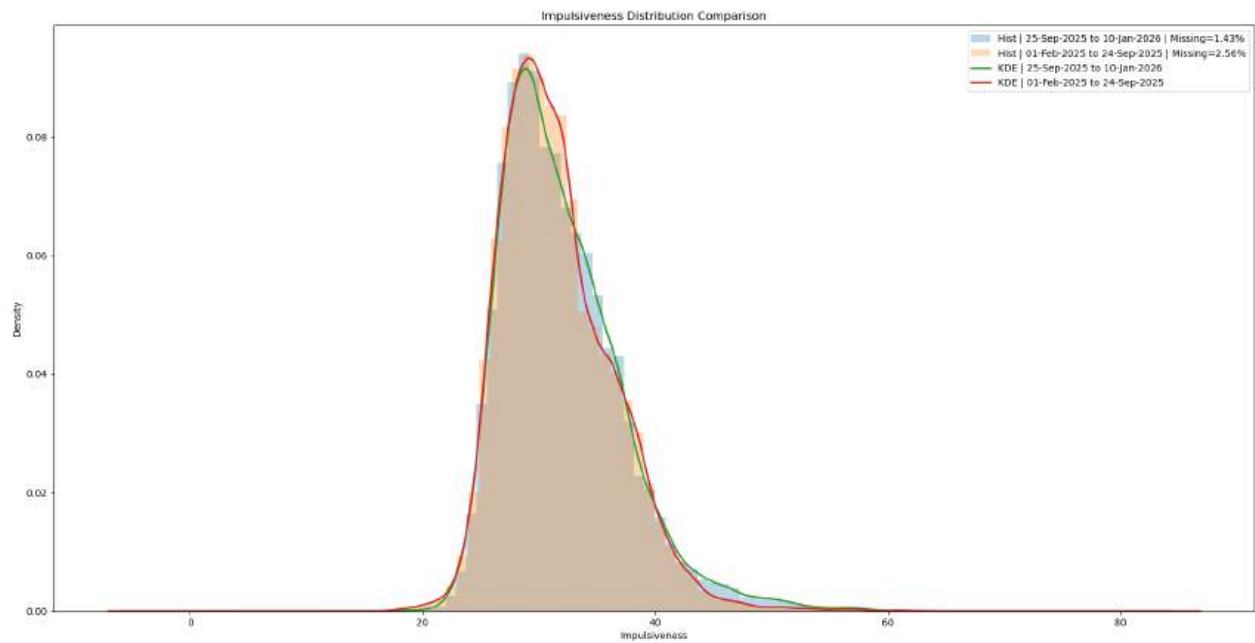


Figure 9: Ballygunge C.U.

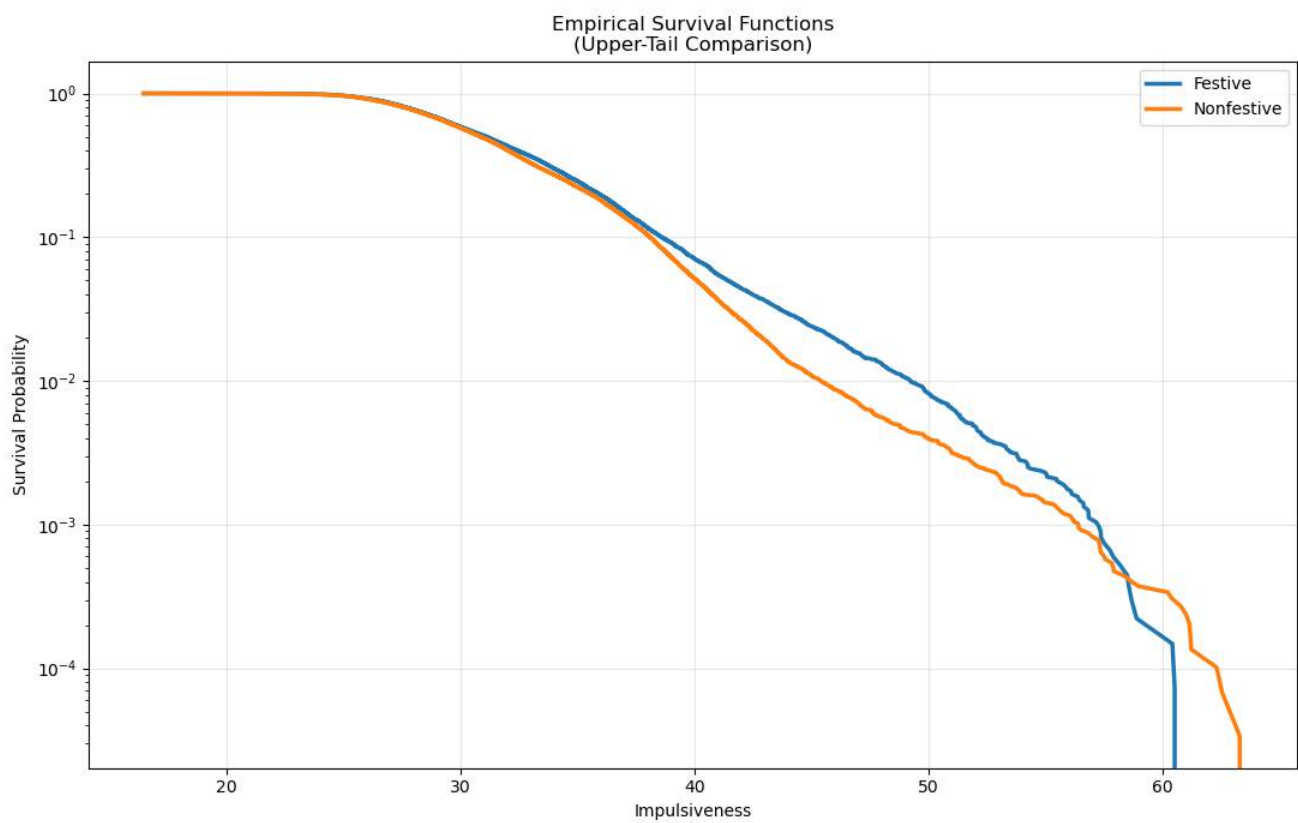


Figure 10: Ballygunge C.U.

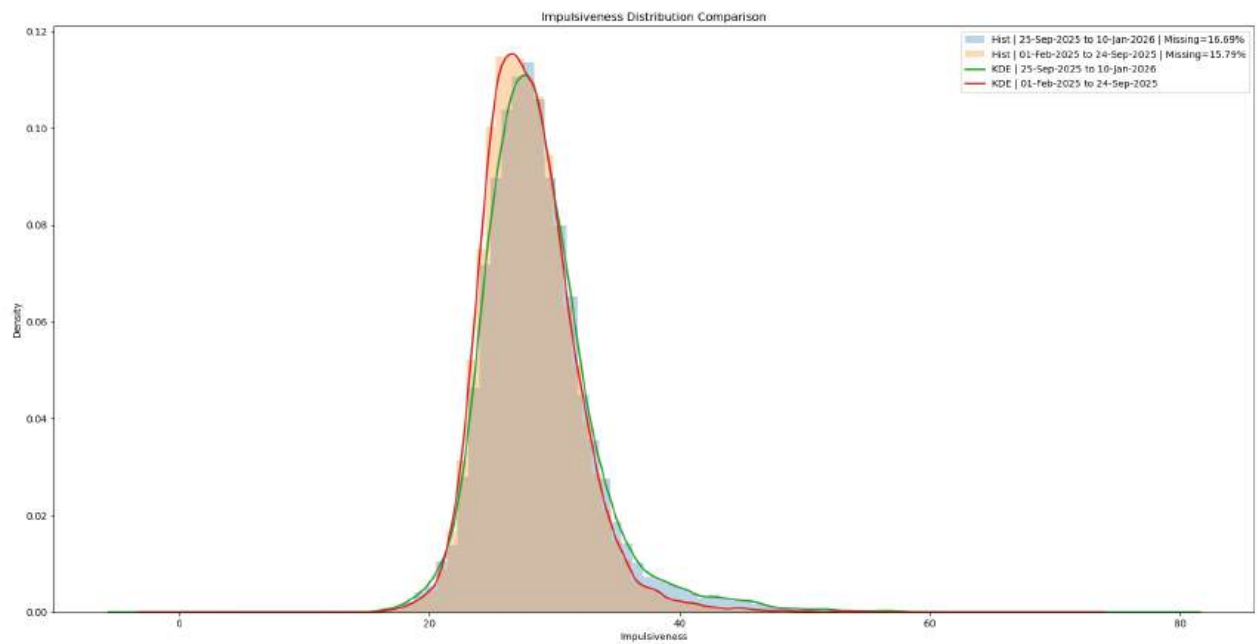


Figure 11: Madhyamgram Municipality

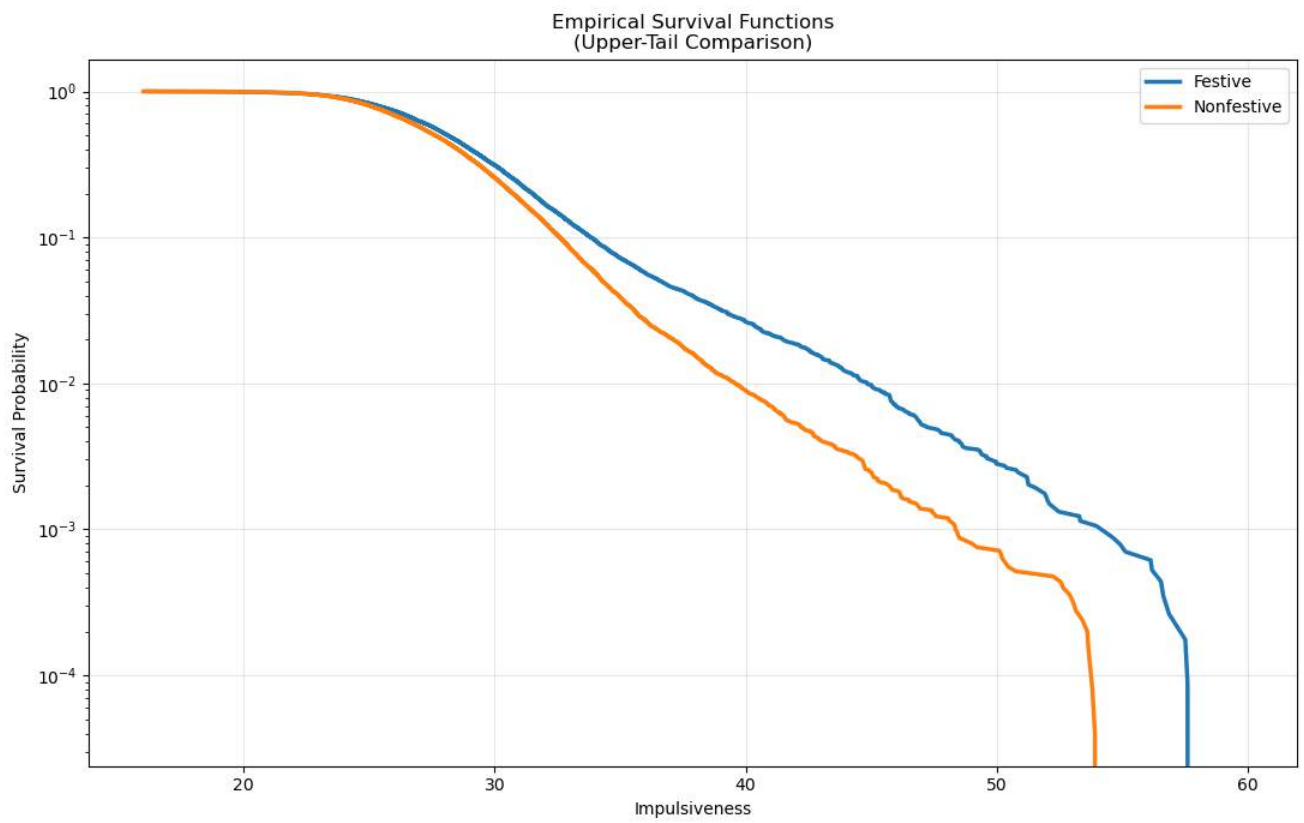


Figure 12: Madhyamgram Municipality

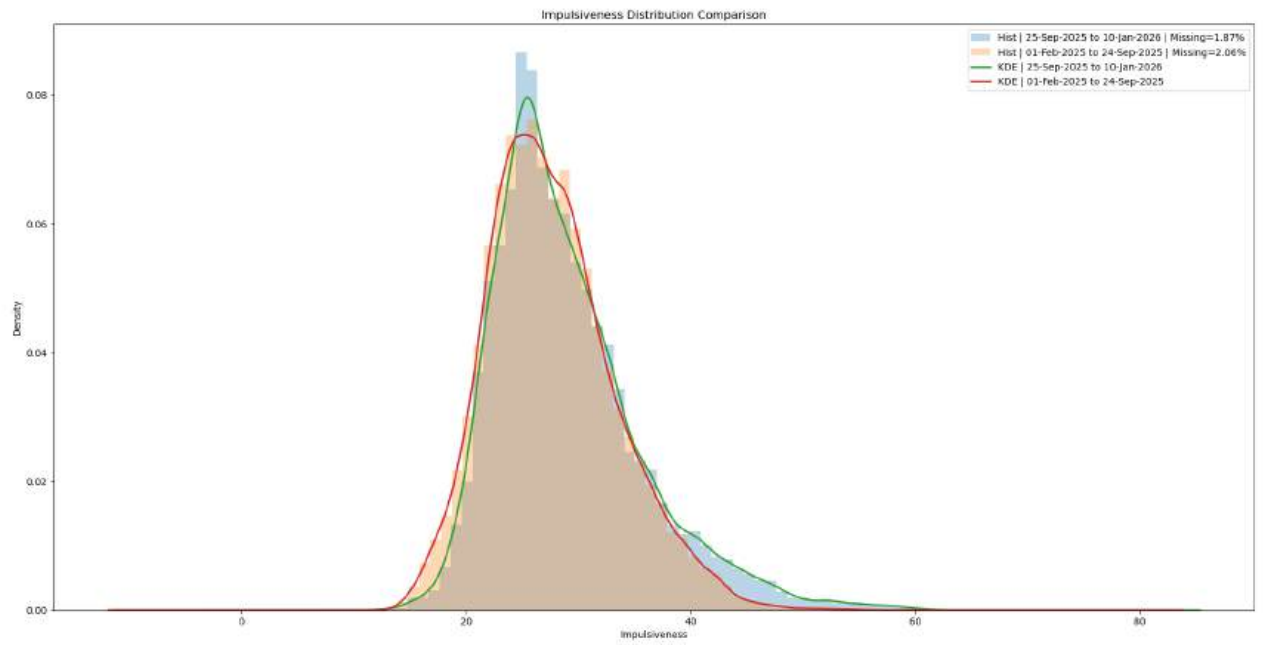


Figure 13: Rajarhat Police Station

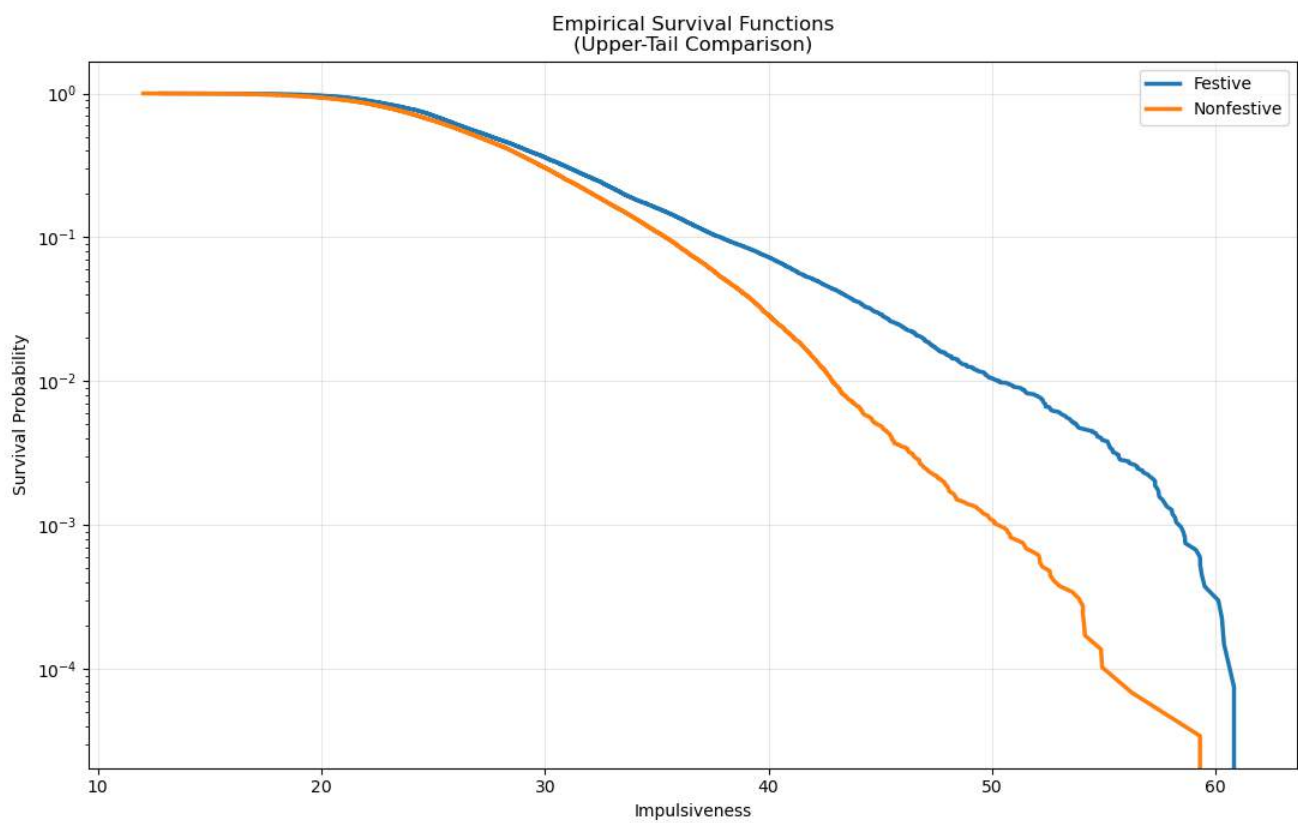


Figure 14: Rajarhat Police Station

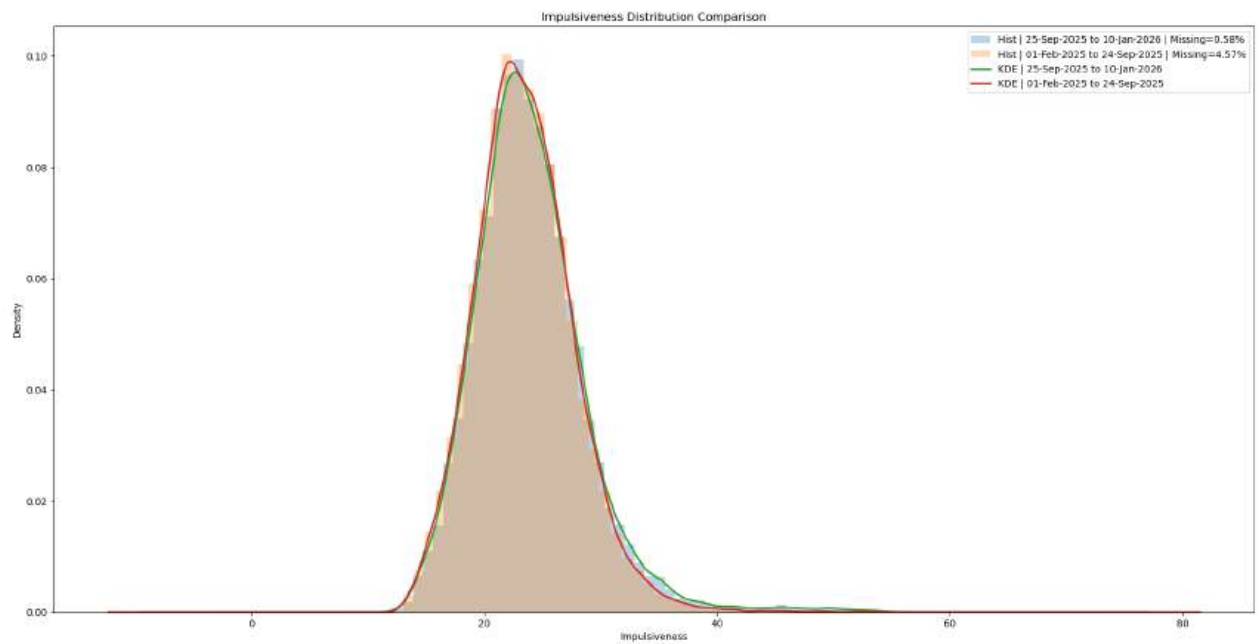


Figure 15: Titagarh Municipality

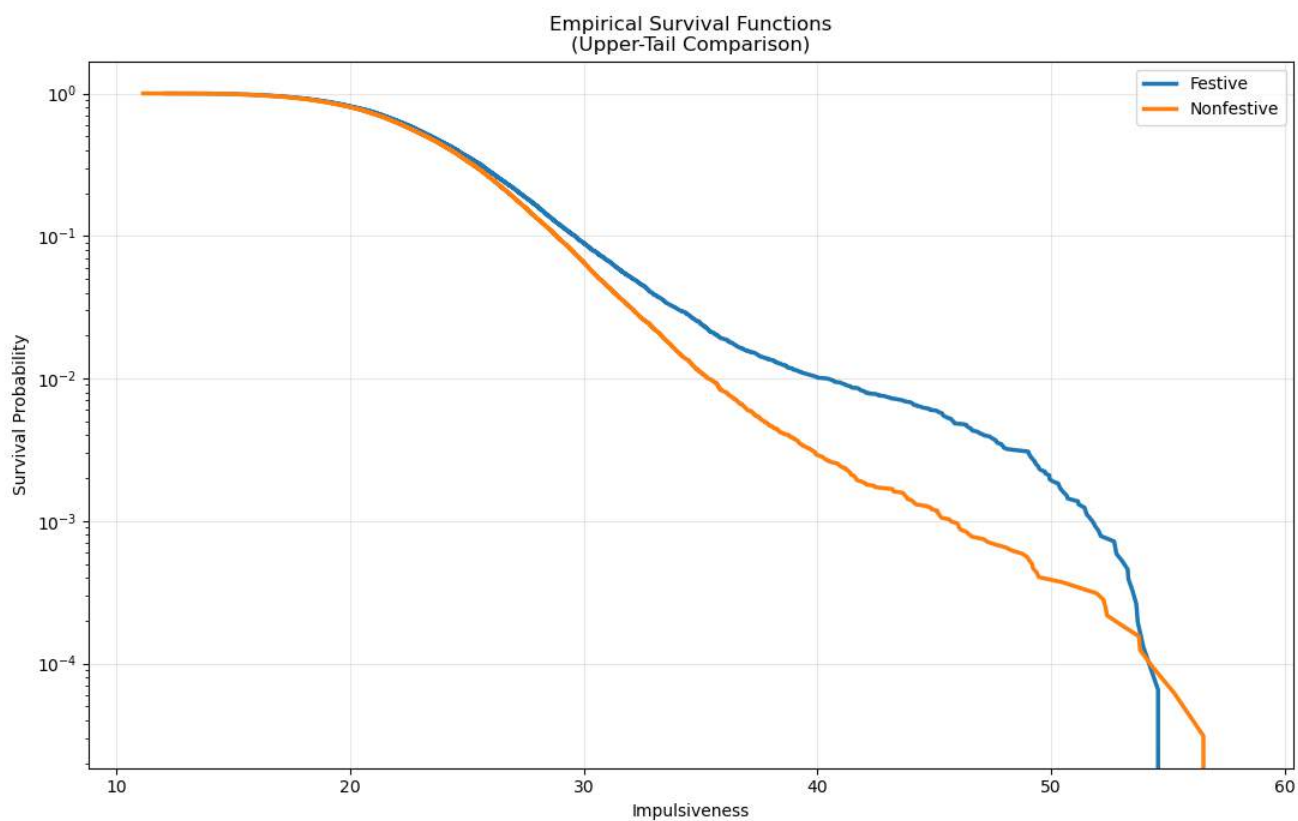


Figure 16: Titagarh Municipality

LAPeakT distributions Festive-Nonfestive

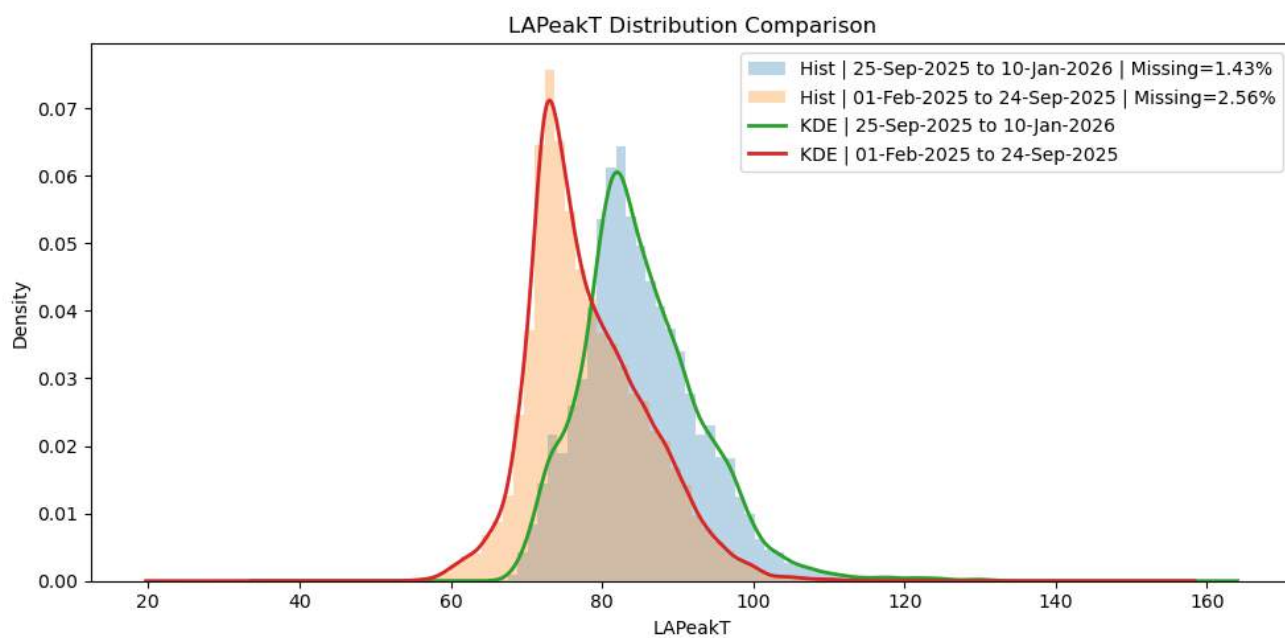


Figure 17: Ballygunge Campus C.U.

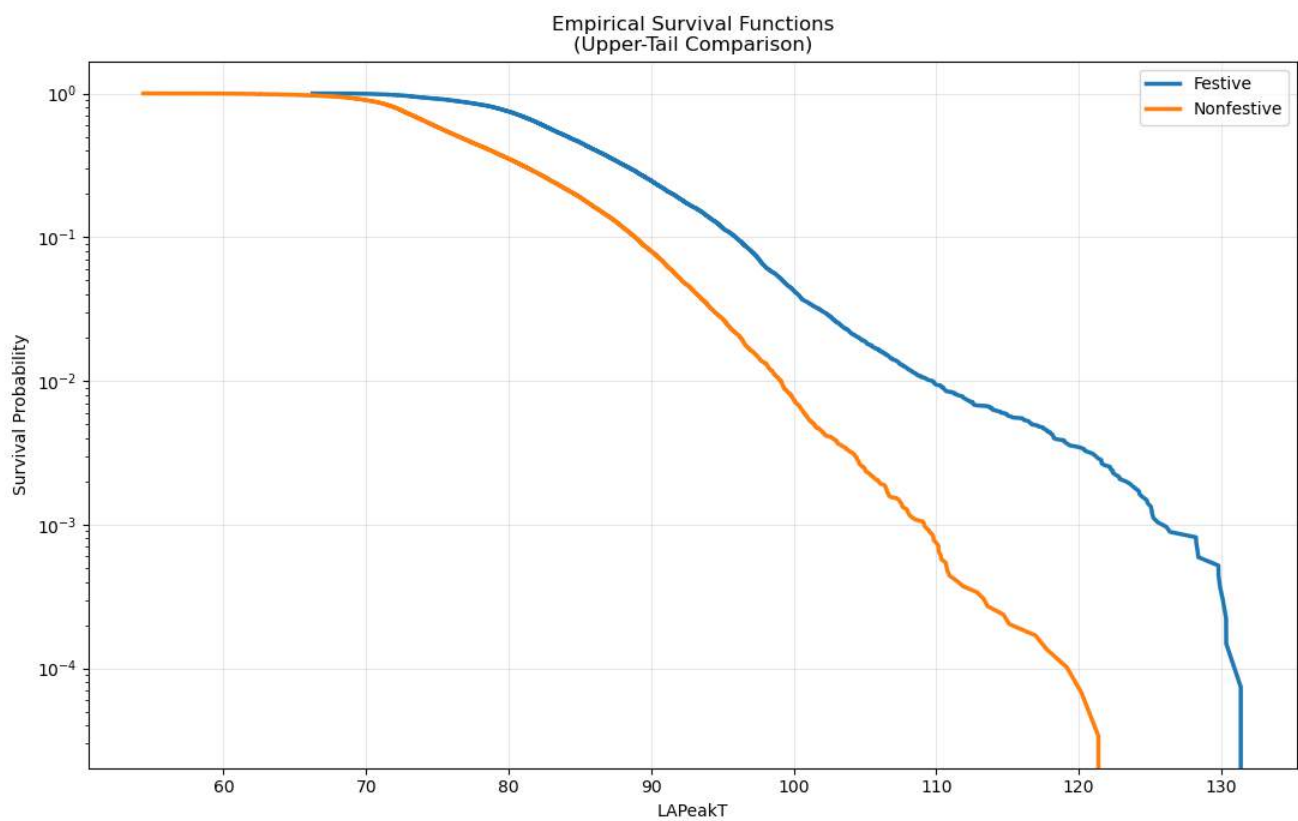


Figure 18: Ballygunge Campus C.U.

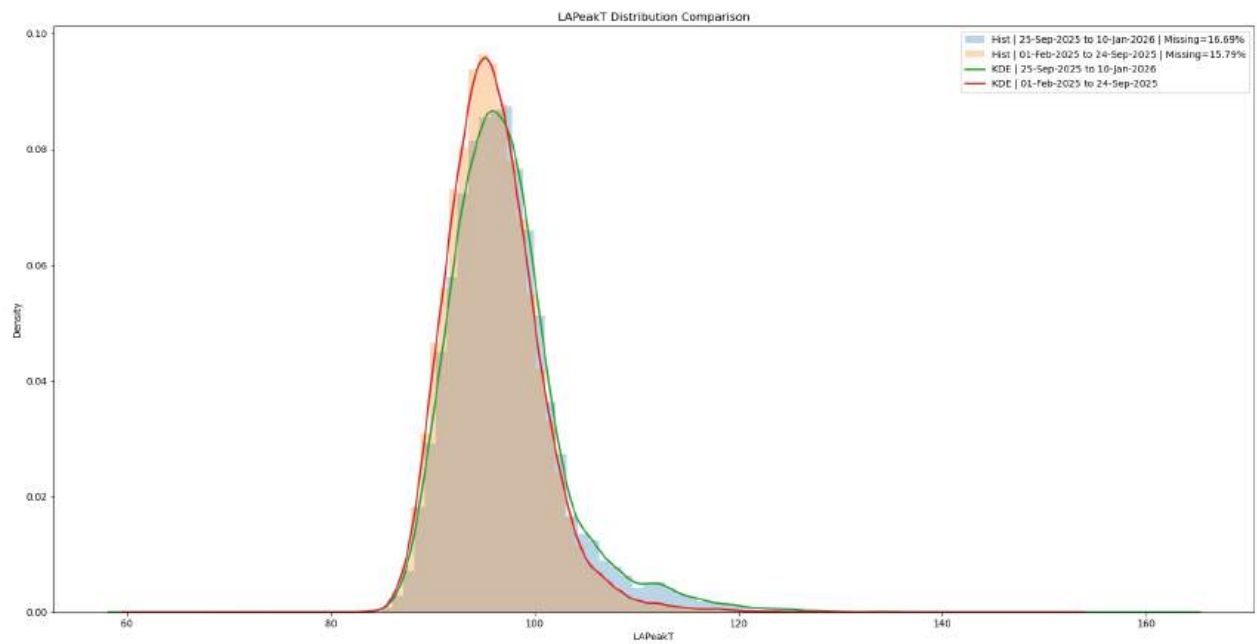


Figure 19: Madhyamgram Municipality

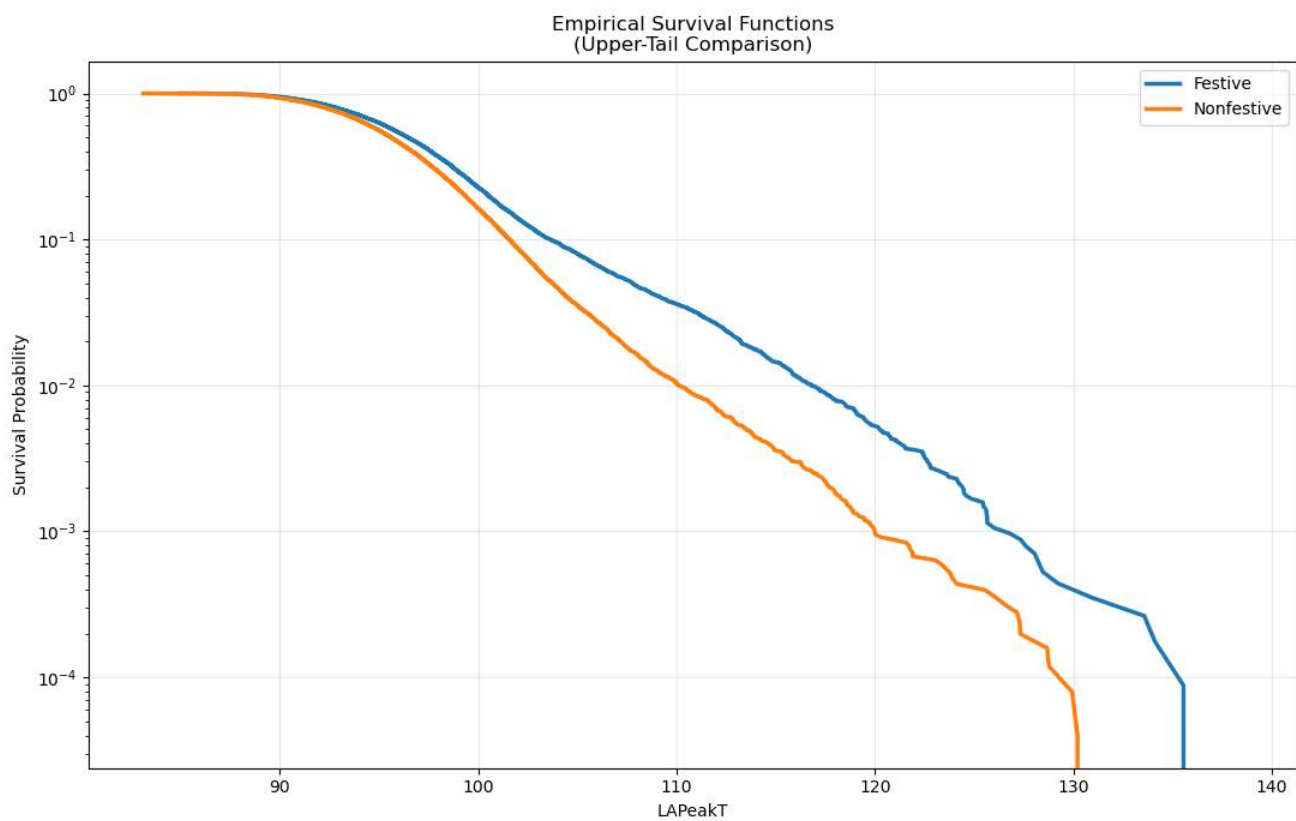


Figure 20: Madhyamgram Municipality

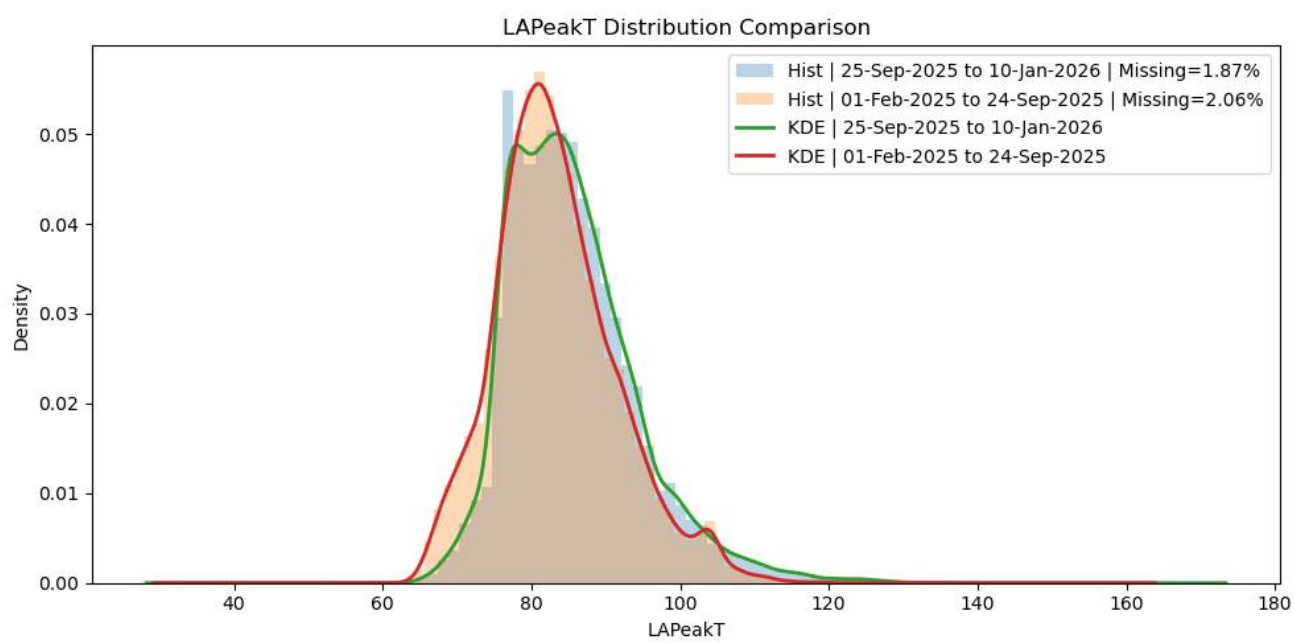


Figure 21: Rajarhat Police Station

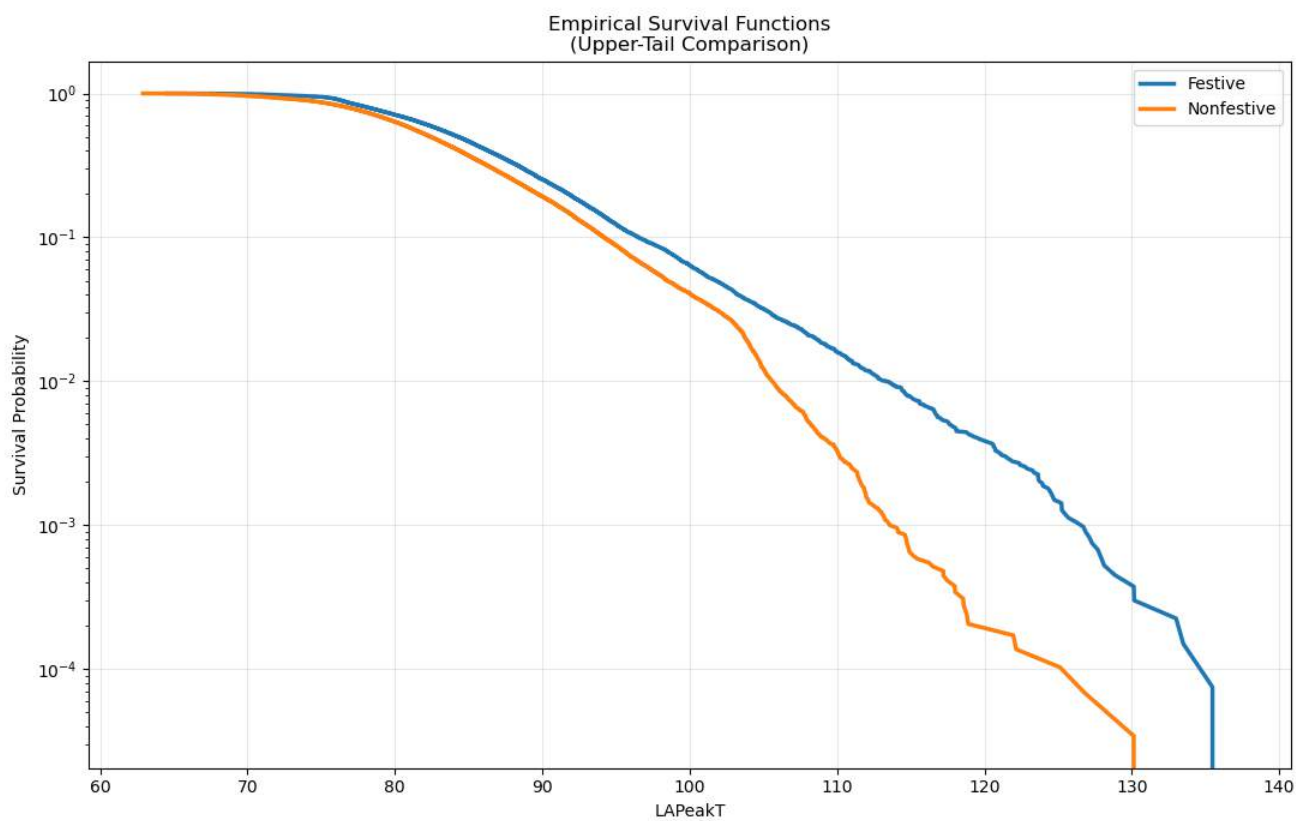


Figure 22: Rajarhat Police Station

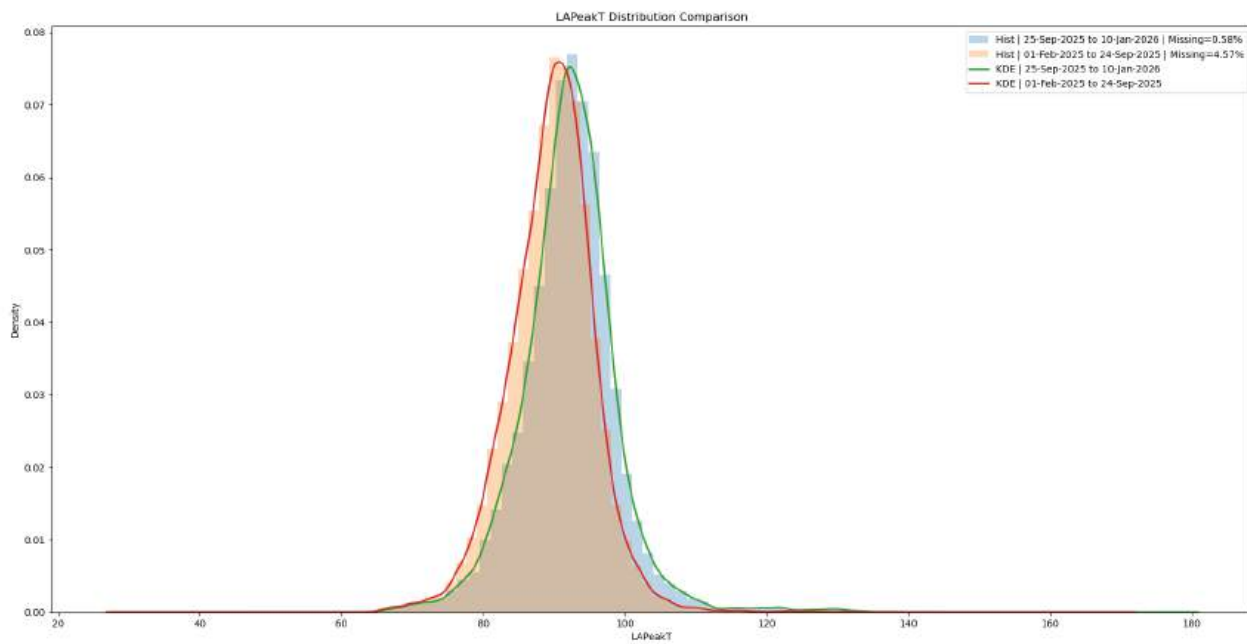


Figure 23: Titagarh Municipality

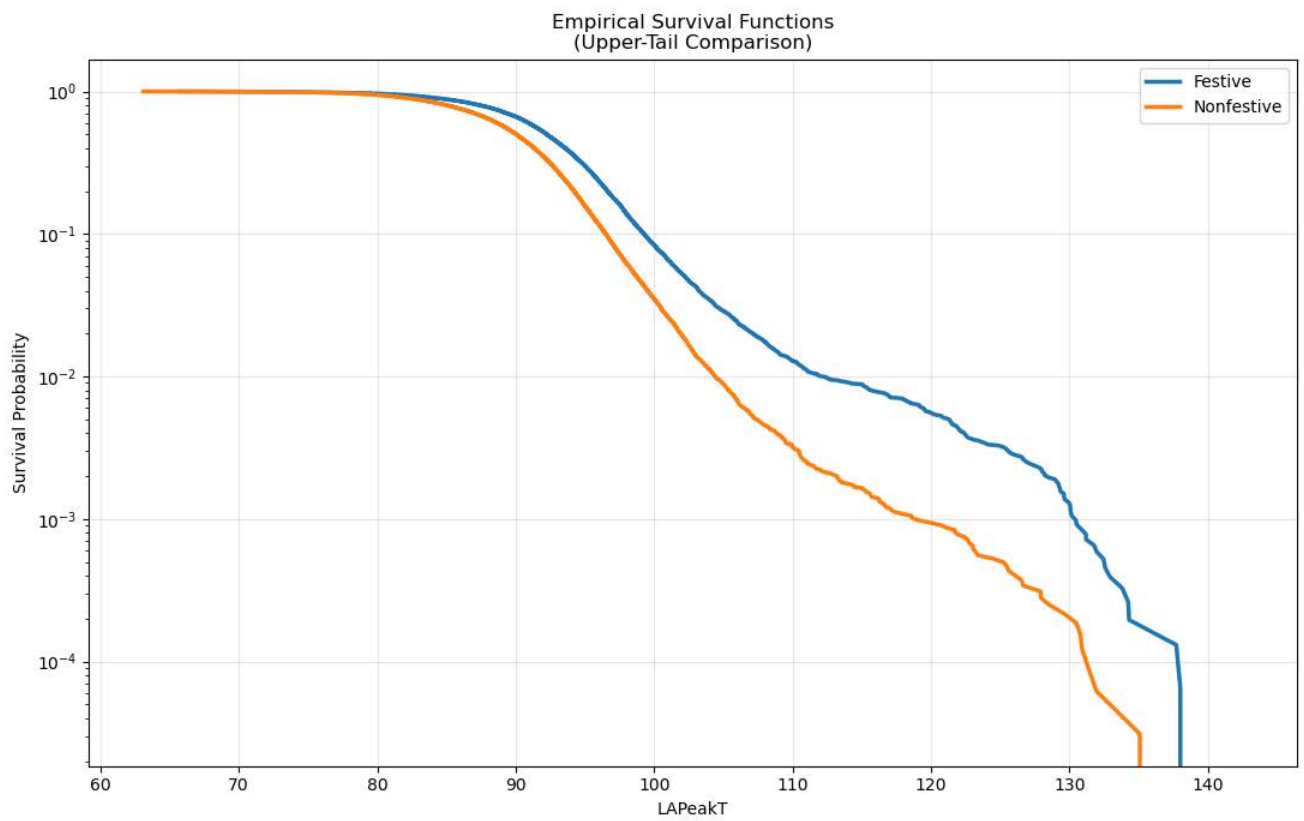


Figure 24: Titagarh Municipality

0.1 Diurnal LAPeakT patterns

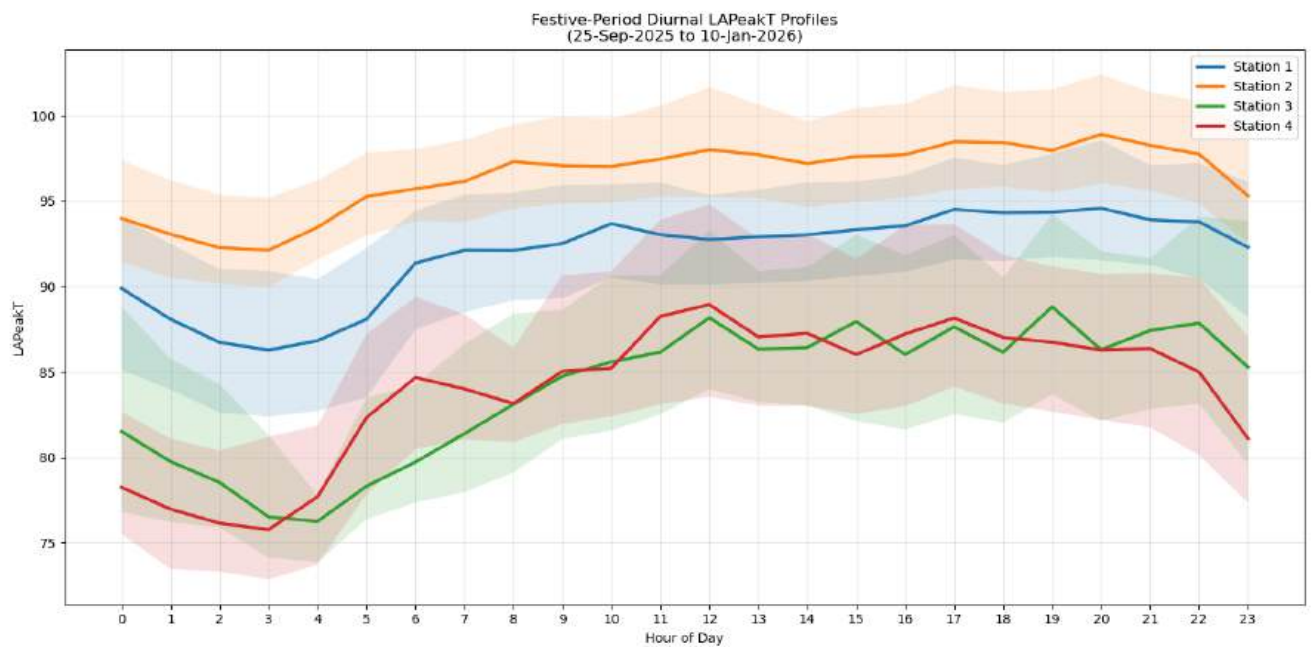


Figure 25: Diurnal Pattern, median LAPeakT hourly values, festive period

Getis-Ord-Gi* hotspots for festive periods and months

Gi* Hotspots/Coldspots Diwali 2023 – Day period

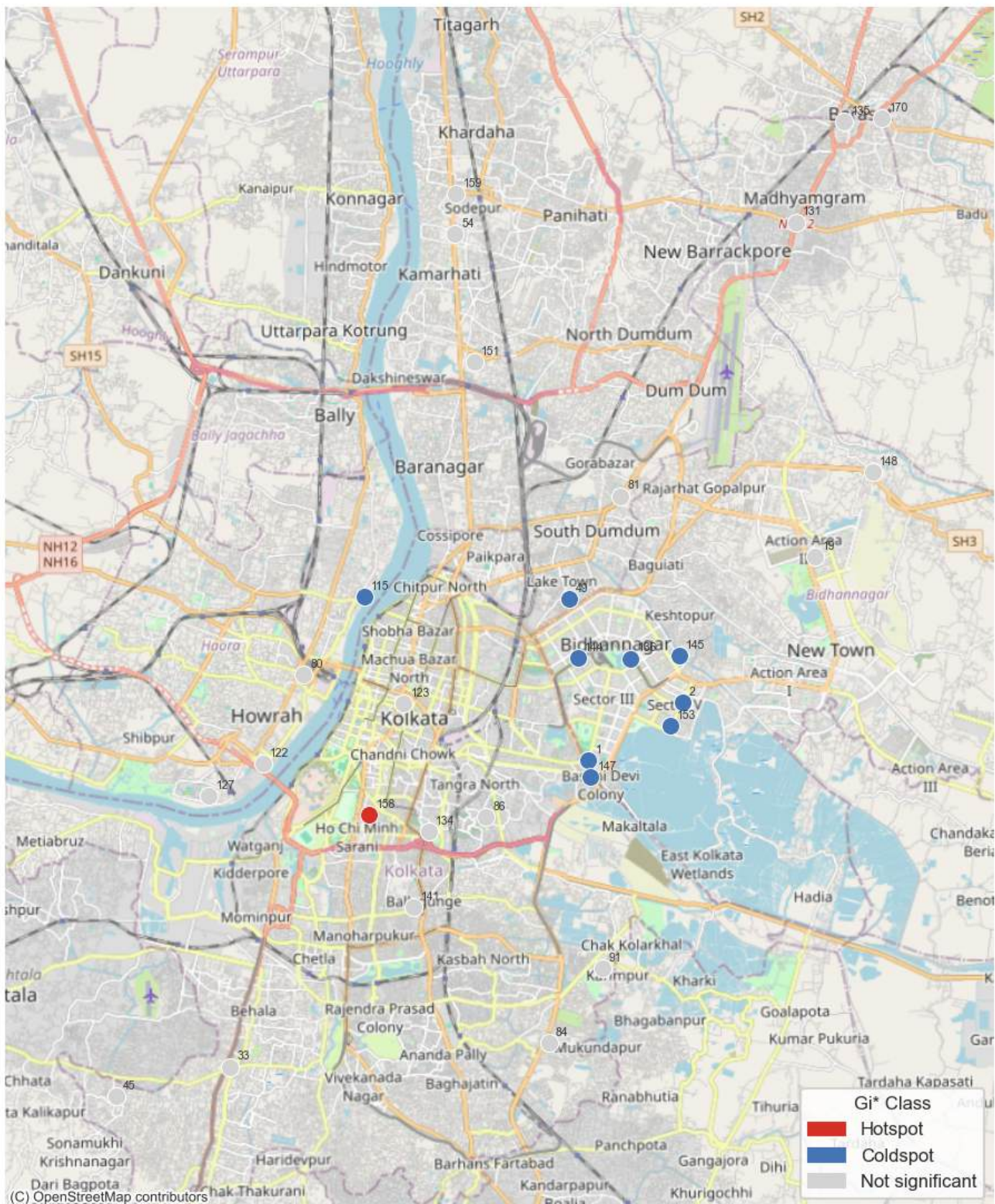


Figure 26

Gi* Hotspots/Coldspots **Diwali 2023 – Night period**

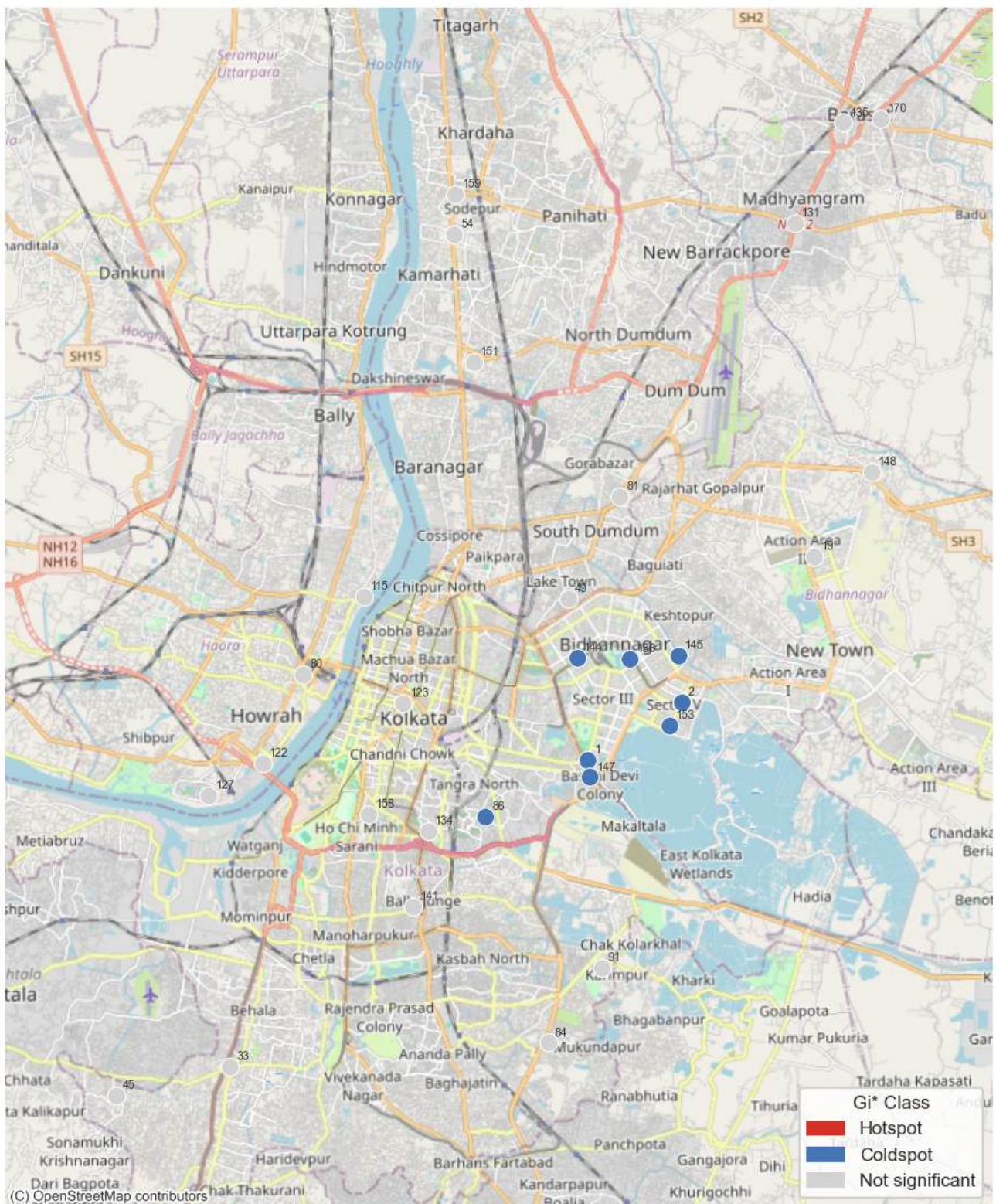


Figure 27

**Gi* Hotspots/Coldspots
Diwali 2024 – Day period**

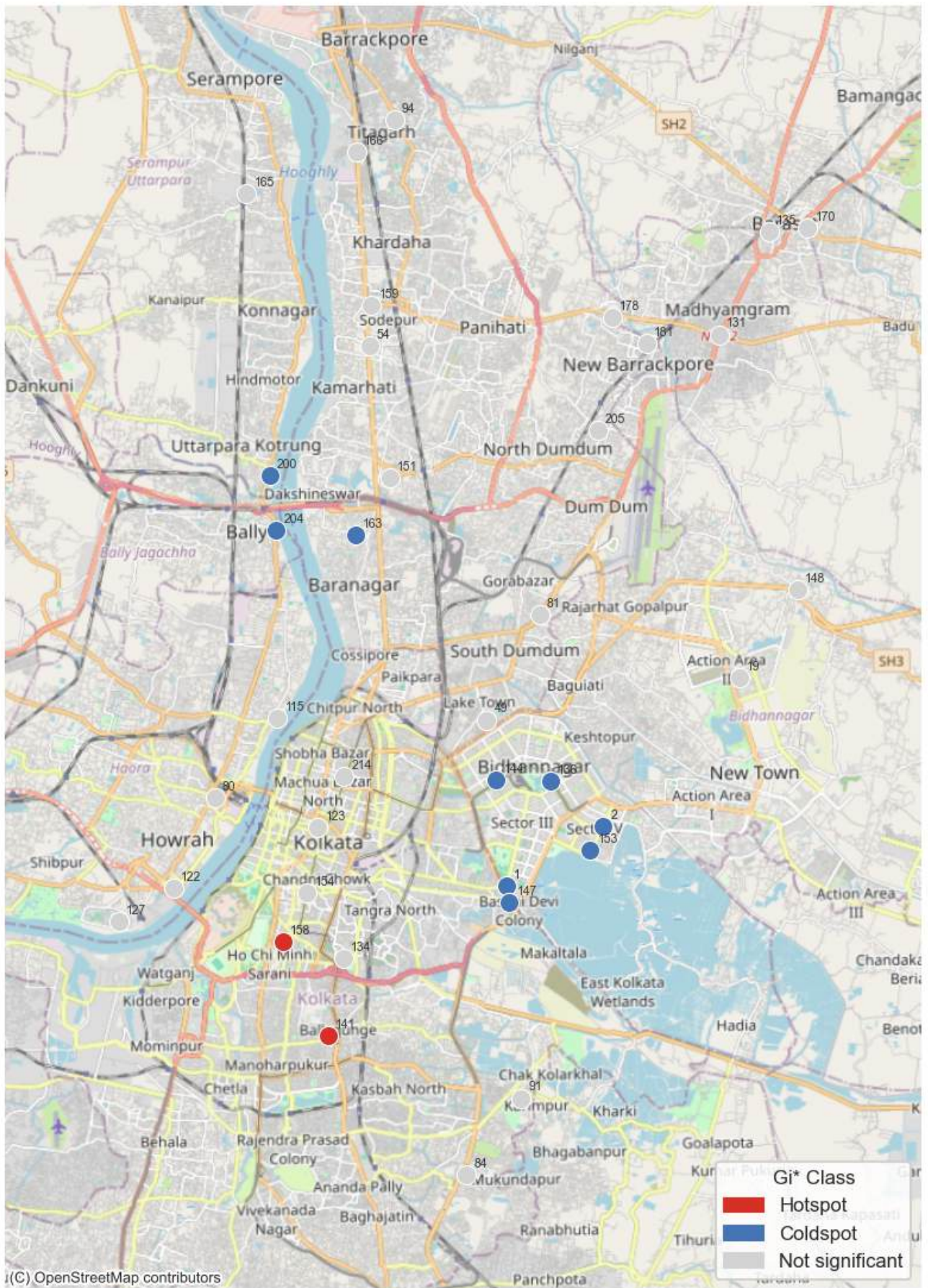


Figure 28

Gi* Hotspots/Coldspots
Diwali 2024 – Night period

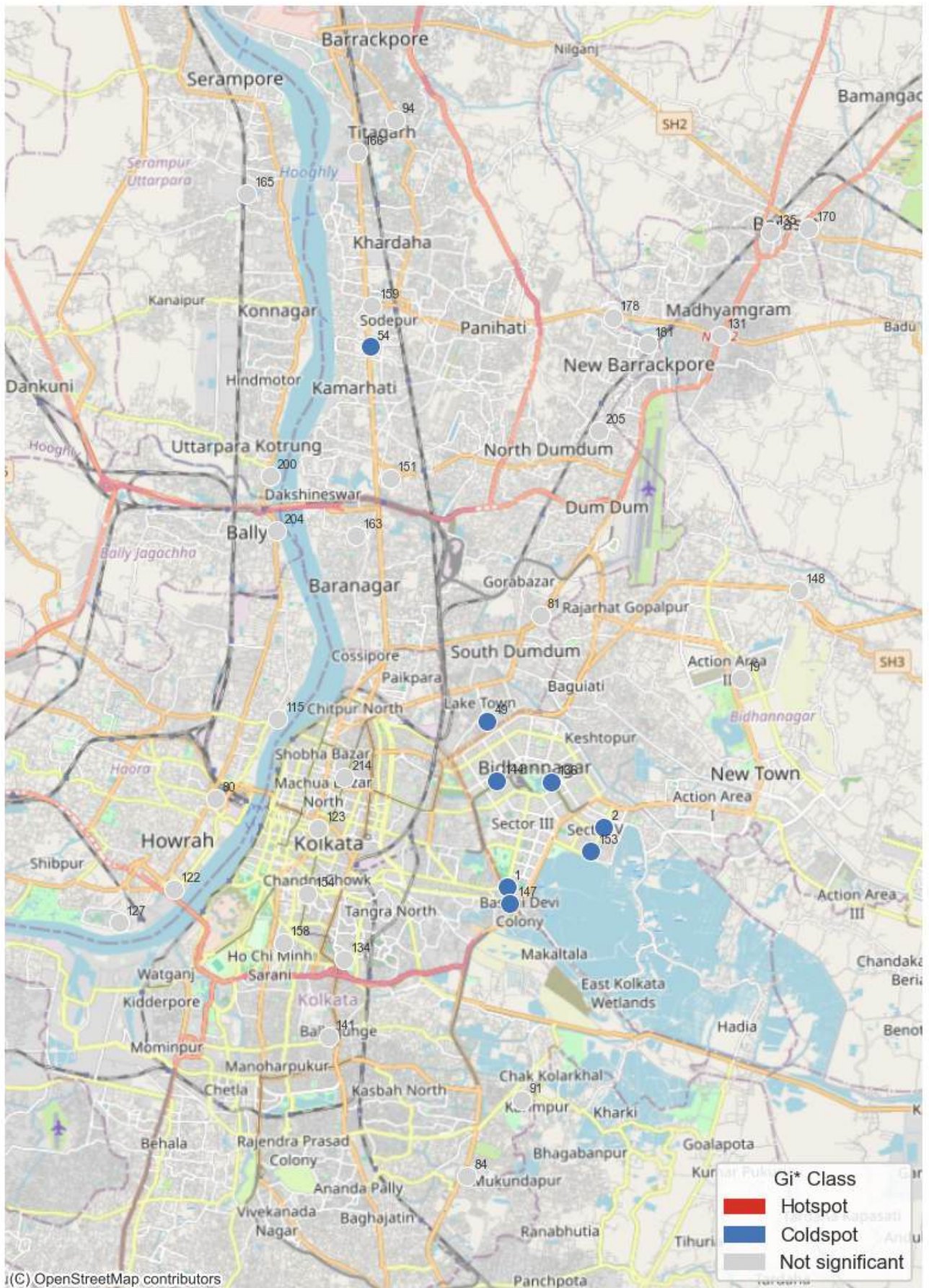


Figure 29

This map displays the Kolkata region, including areas like Serampore, Titagarh, Khardaha, Konnagar, Kamrhati, New Barrackpore, North Dum Dum, Dum Dum, South Dum Dum, Bidhannagar, and the city of Kolkata. The Hooghly River is shown flowing through the area. The map is overlaid with a grid of points categorized by Gi* Class: Hotspot (red), Coldspot (blue), and Not significant (grey). Major roads like NH16 and SH1 are also visible.

(C) OpenStreetMap contributors

**Gi* Hotspots/Coldspots
Diwali 2025 – Night period**

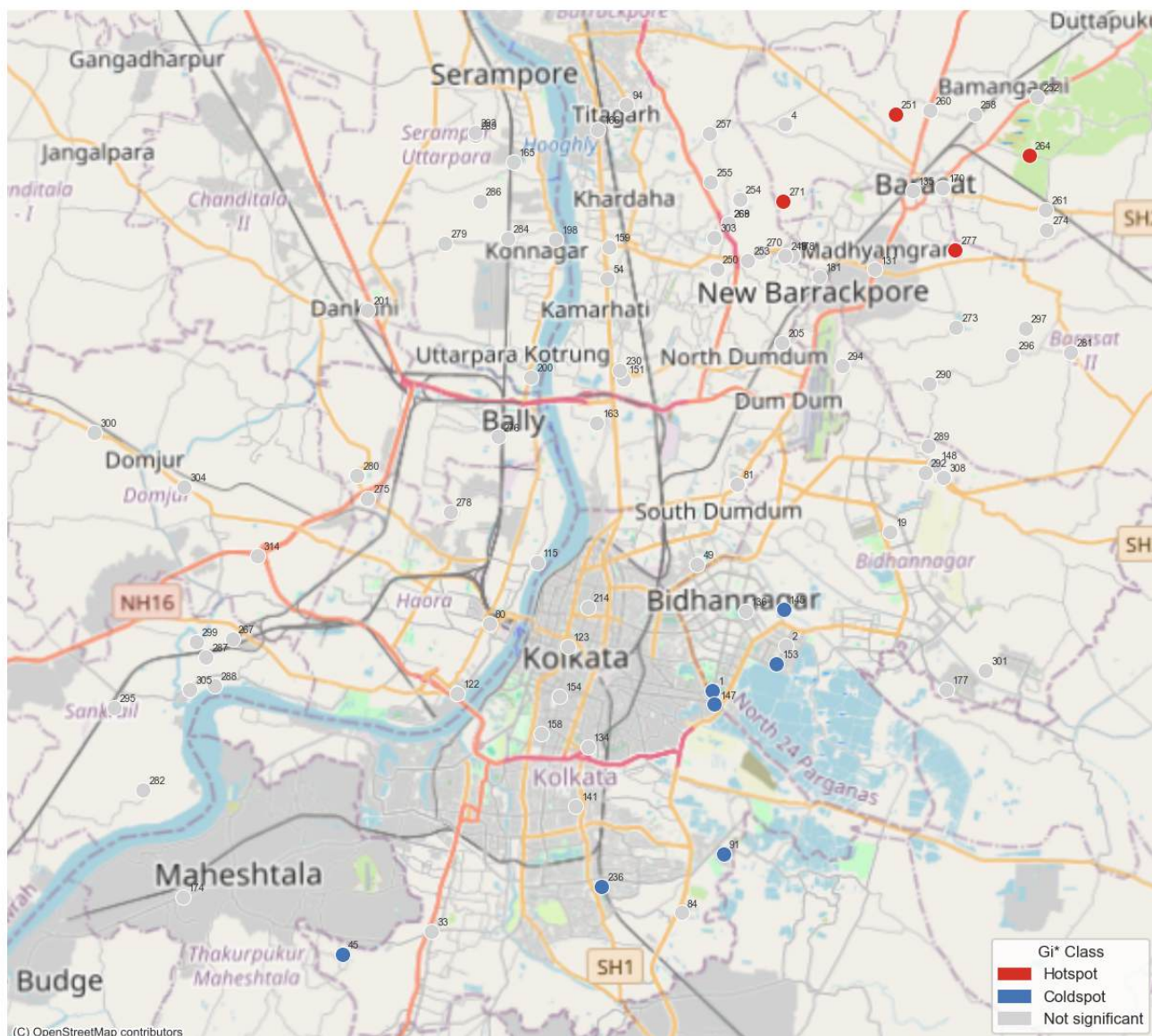


Figure 31

**Gi* Hotspots/Coldspots
Durga Puja 2023 – Day period**

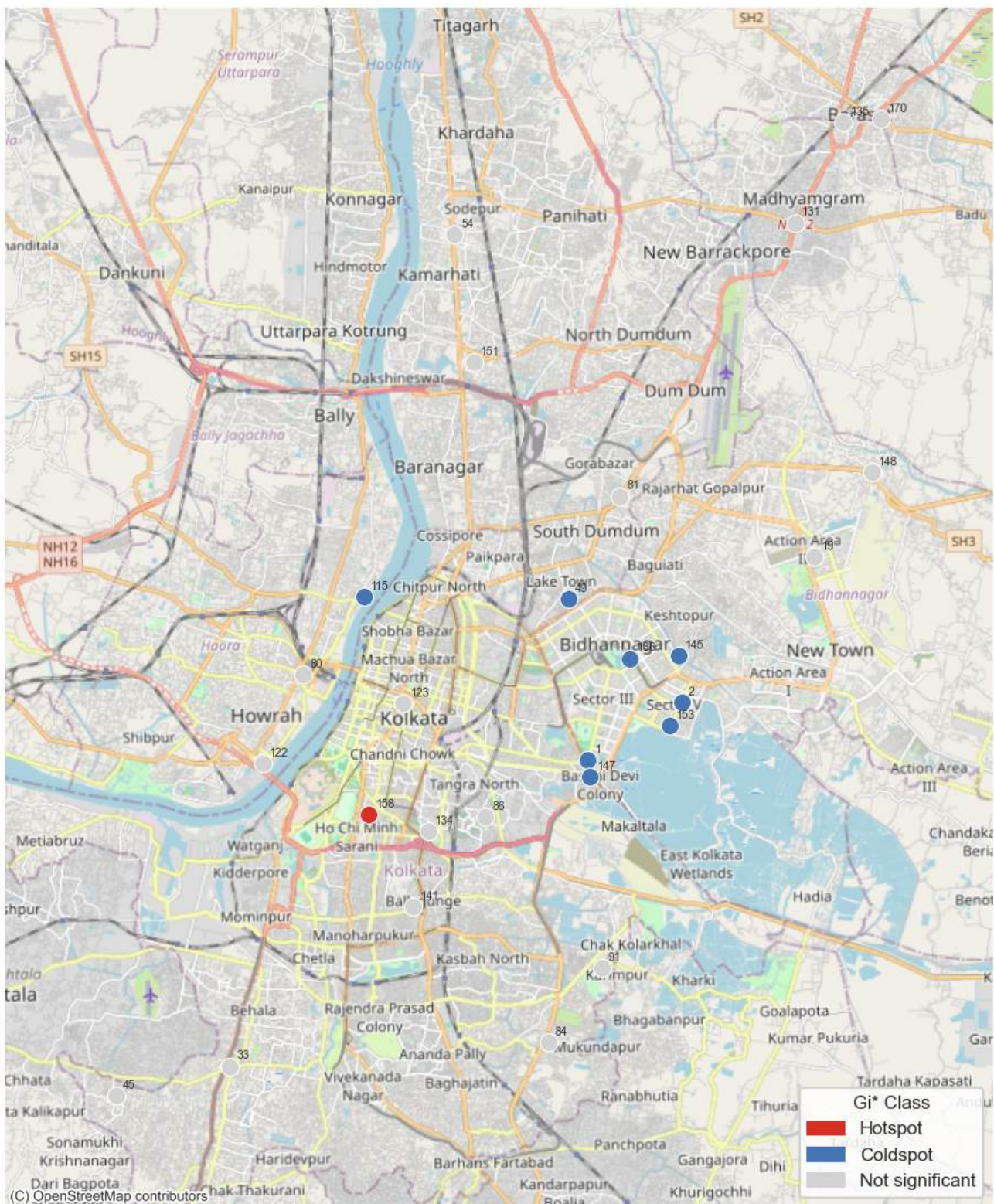
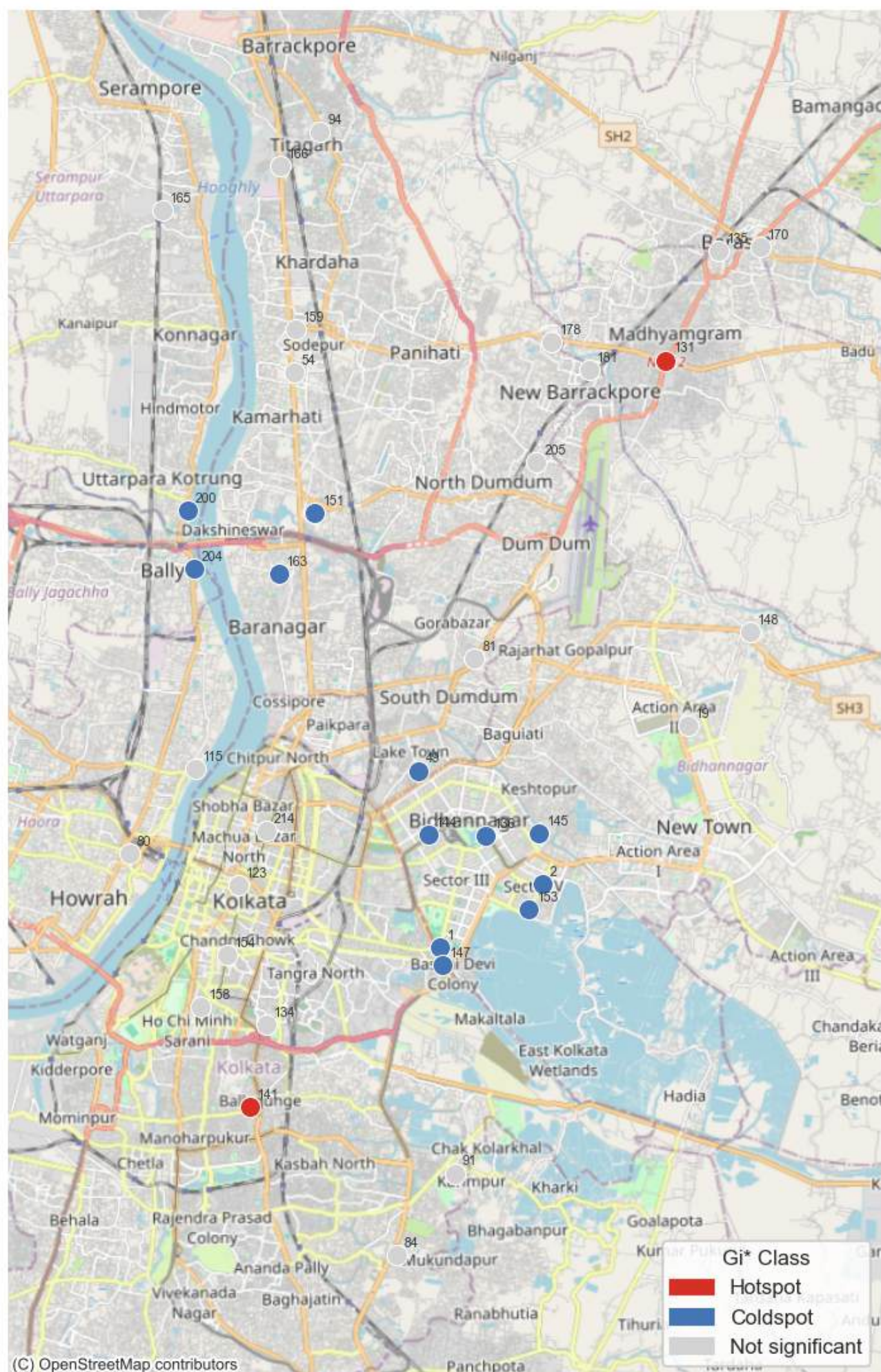


Figure 32

Gi* Hotspots/Coldspots
Durga Puja 2024 – Day period



**Gi* Hotspots/Coldspots
Durga Puja 2025 à€” Day period**

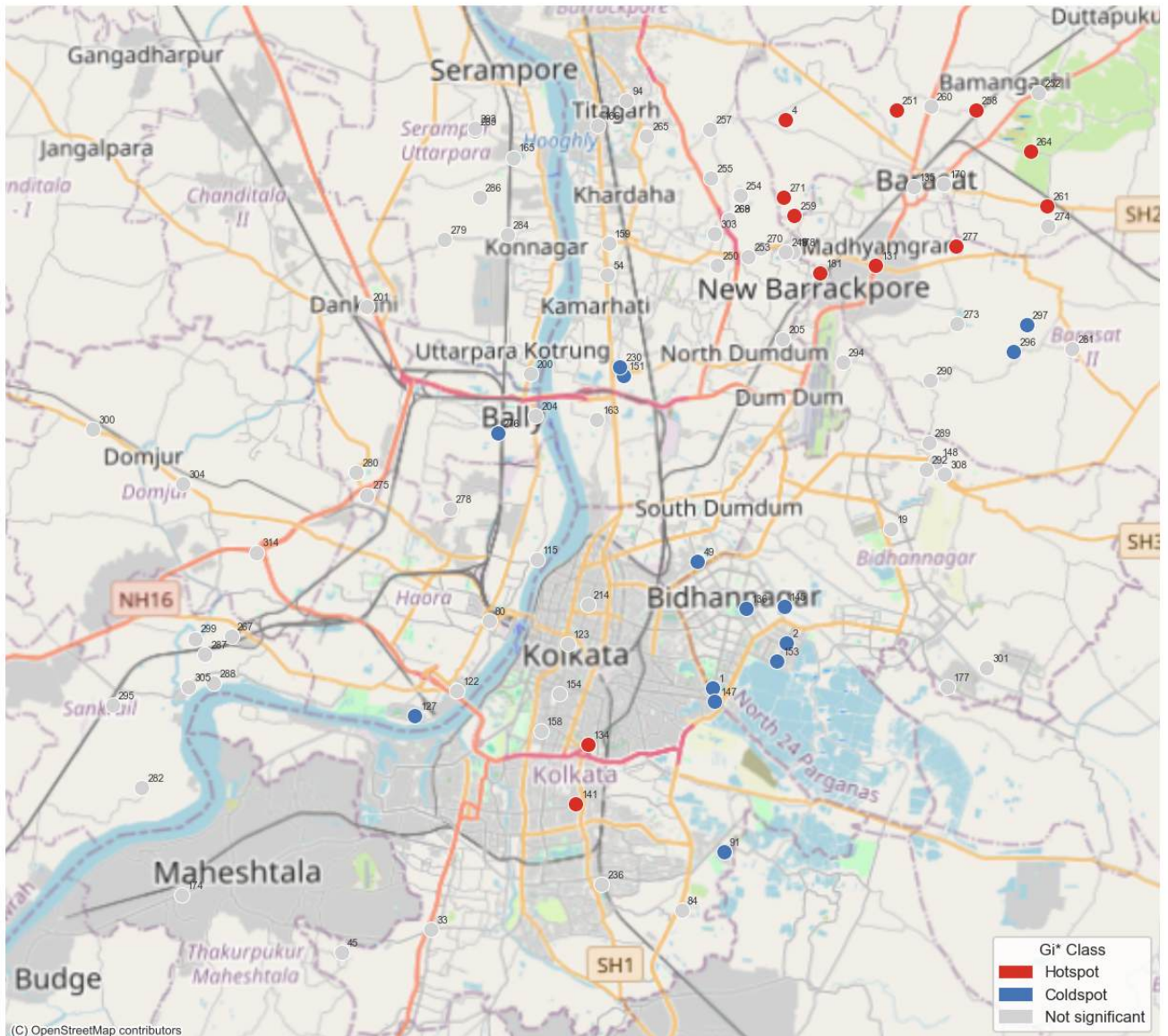


Figure 34

Gi* Hotspots/Coldspots
Durga Puja 2025 “ Night period

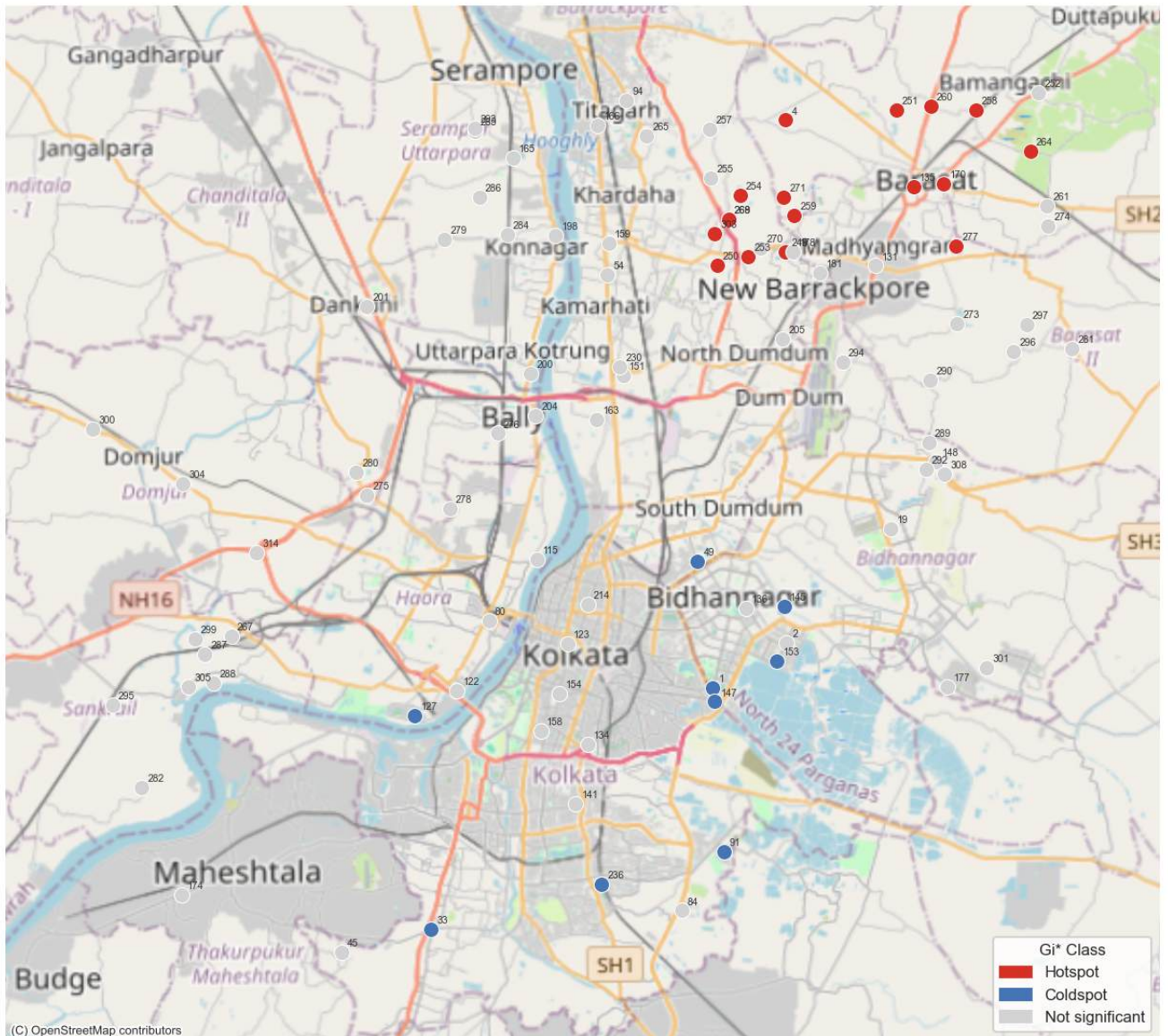


Figure 35

**Gi* Hotspots/Coldspots
Christmas NewYear 2025 æ” Day period**



Figure 36

Gi* Hotspots/Coldspots
Christmas NewYear 2025 – Night period



Figure 37

Hotspot maps from considering Mean Exceedance Magnitudes

Hotspot Classification (MEM > 5.0 dB)
Diwali 2024 Day period

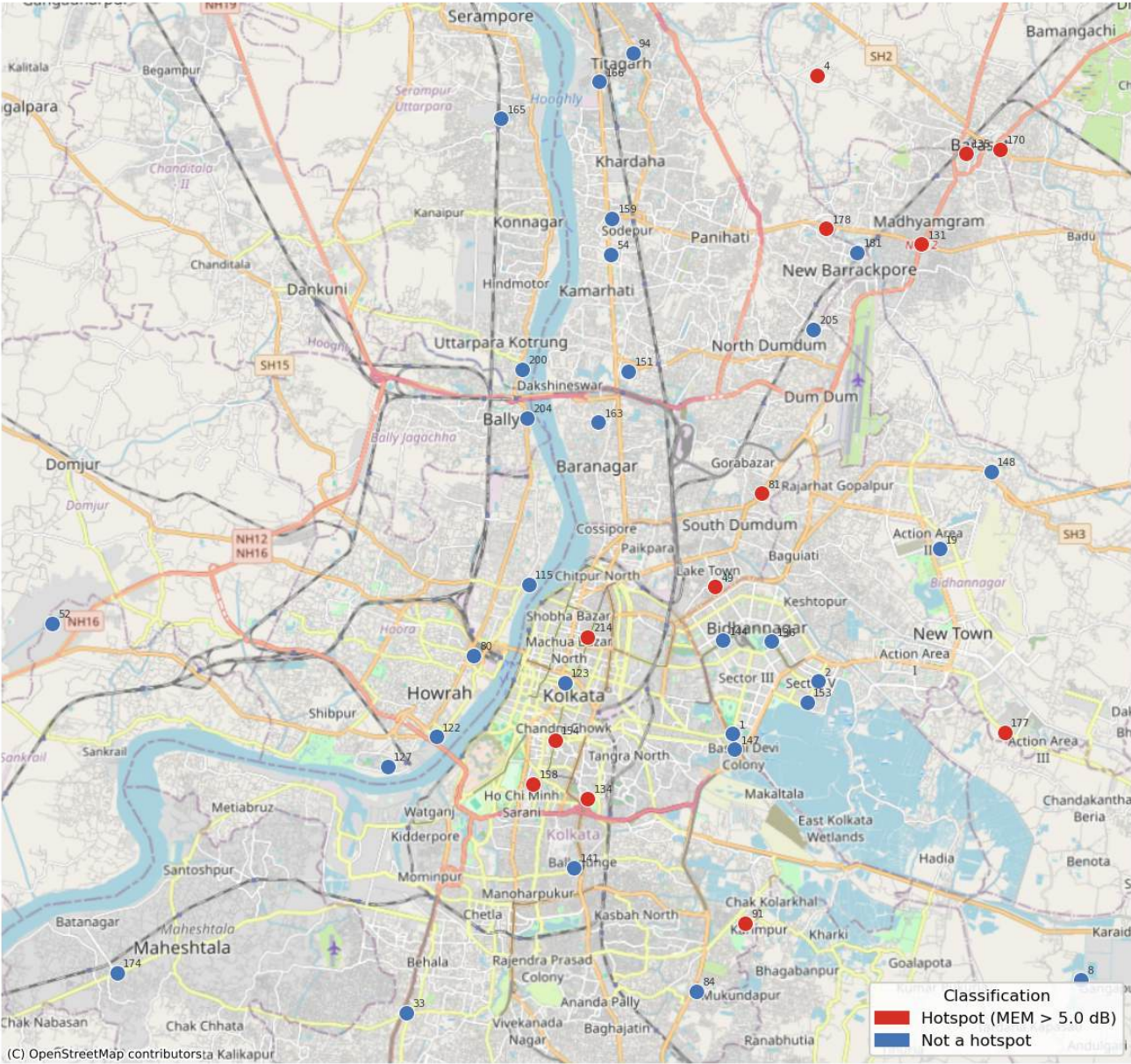


Figure 1

Hotspot Classification (MEM > 5.0 dB)
Diwali 2024 Night period

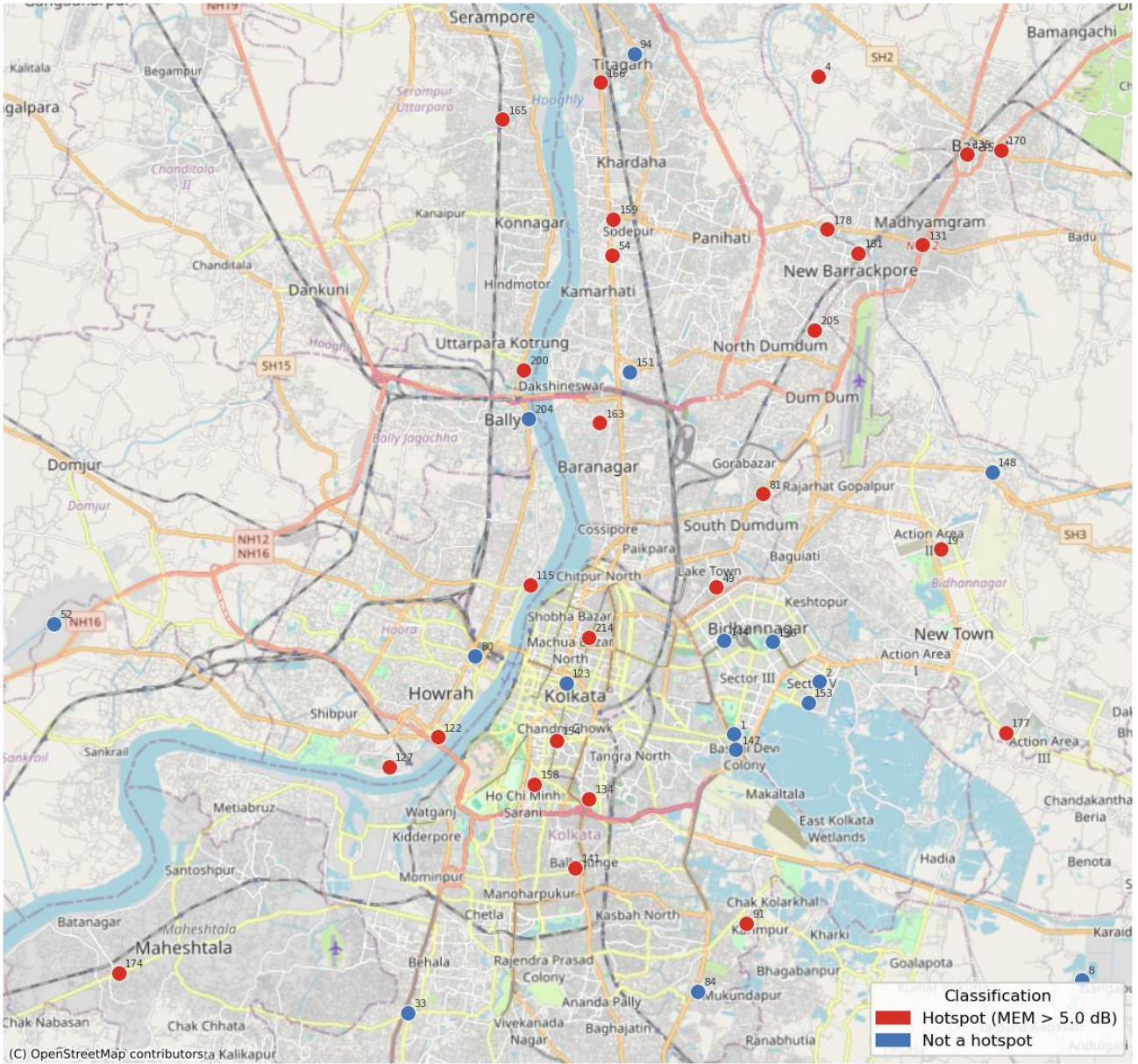


Figure 2

**Hotspot Classification (MEM > 5.0 dB)
Diwali 2025 Day period**

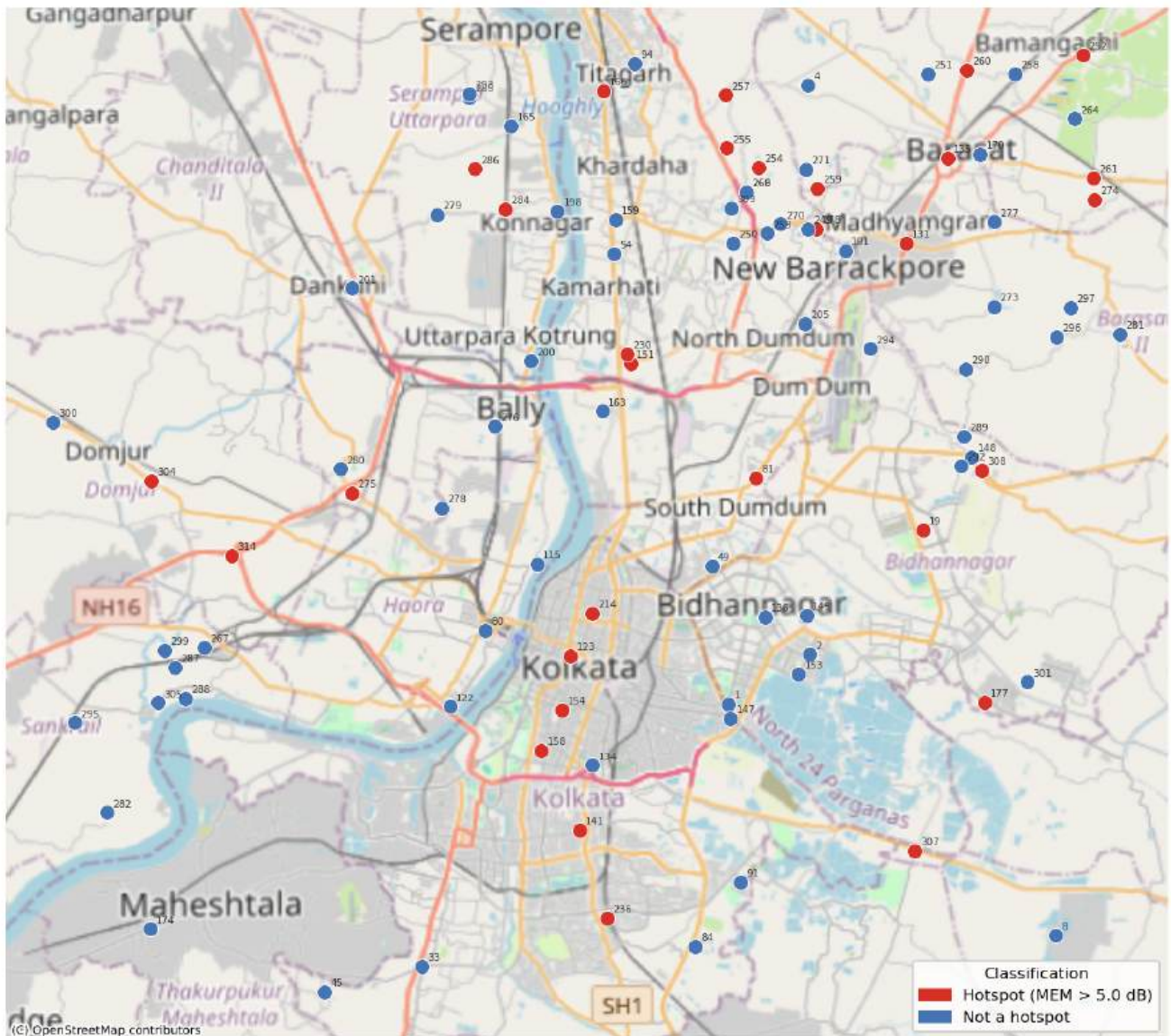


Figure 3

Hotspot Classification (MEM > 5.0 dB)
Diwali 2025 Night period

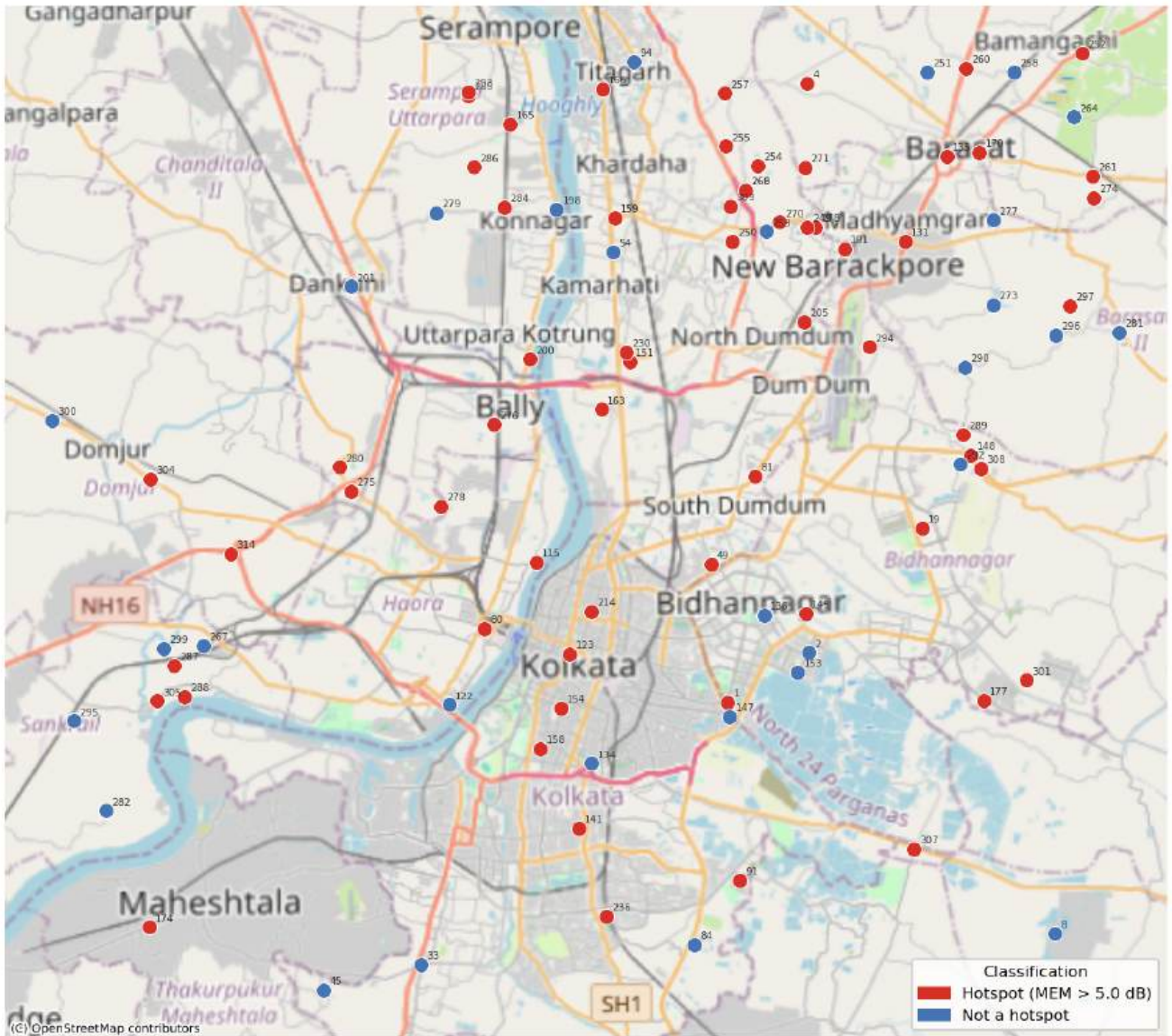


Figure 4

**Hotspot Classification (MEM > 5.0 dB)
Durga Puja 2025 Day period**

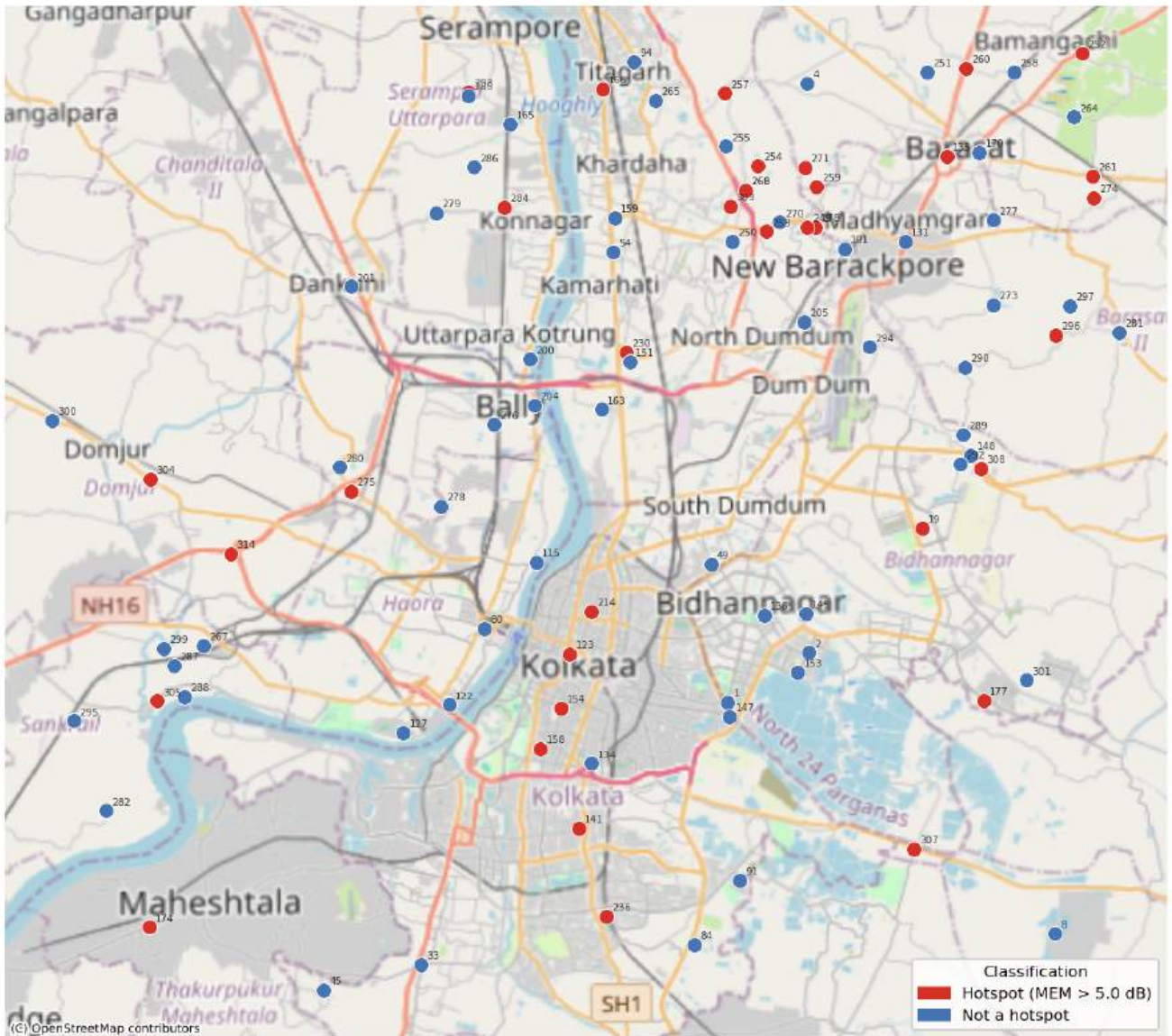


Figure 5

Hotspot Classification (MEM > 5.0 dB) **Durga Puja 2025 Night period**

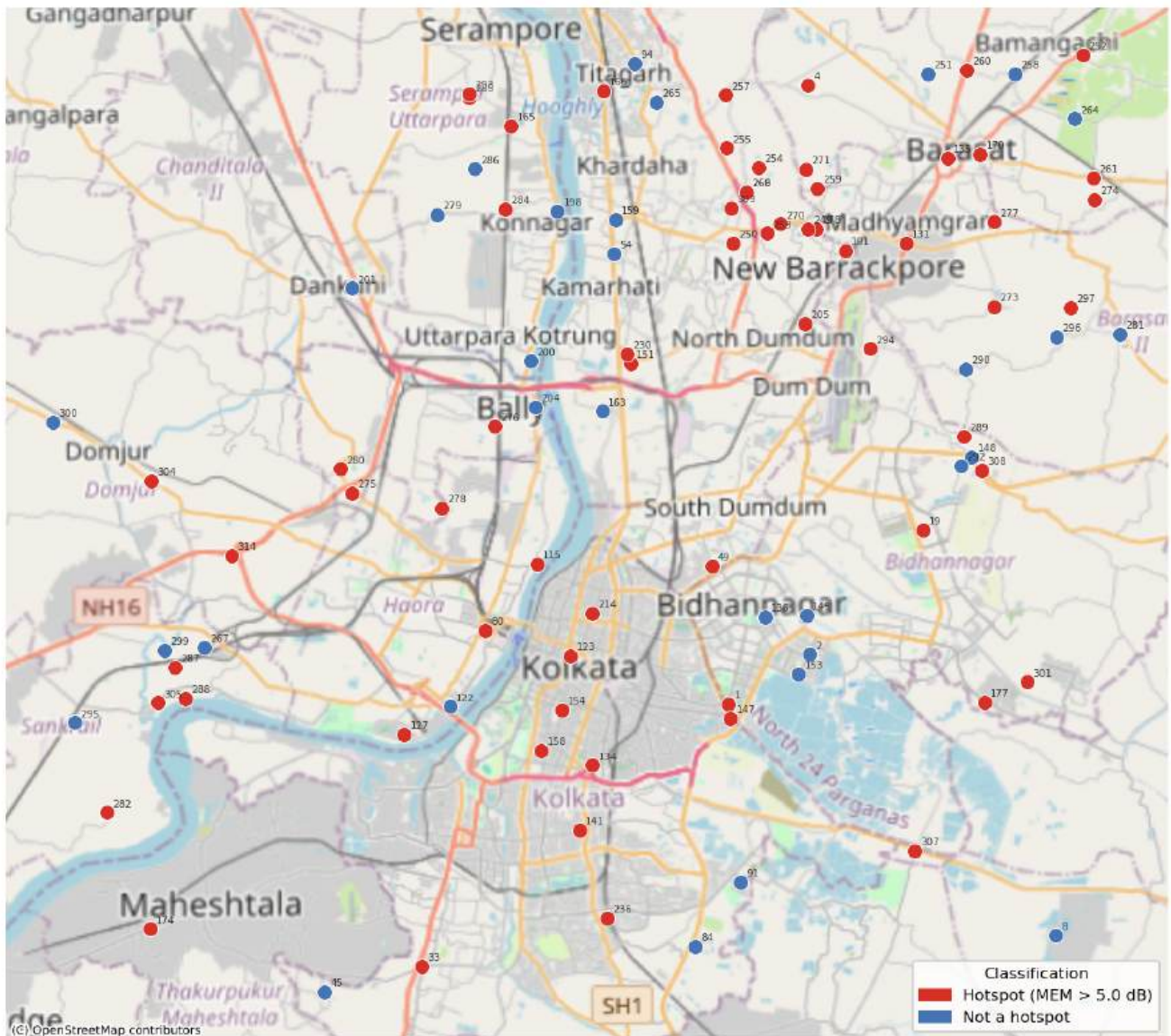


Figure 6